MUHS final-year BPTh (IV BPTh), 2012 syllabus

Community Physiotherapy

Model answers to every 3-mark and 7-mark question asked from Summer 2016 to Summer 2026, written the way to write them in the exam.



Questions	All 240 short-answer slots of the 20 papers (code 51414): 92 distinct 3-mark and 87 distinct 7-mark questions after merging repeats
Arranged	By topic, in syllabus order, with an index by paper at the end
Online version	https://hrushikesh-shinde7.github.io/bpt-study-material/community/

Educational notes for exam preparation. Not medical advice. Every answer is drawn from the full notes, which list the sources for each fact.

How to use the model answers

Every short-answer question from the 20 MUHS papers (Summer 2016 to Summer 2026) is here, once. When a question was repeated, the sessions are listed under it, so you can see which ones come back.

1. **Read the question, close the book, write your own answer** in the time allowed, then compare. Copying without recall does not stick.
2. **Learn the structure** as much as the facts: definition, points, table or diagram, physiotherapy, one closing line.
3. **Practise whole papers.** The index at the back lists each paper's short questions with the page of every answer: sit one paper in 100 minutes, then mark it.

What examiners look for

Question	Length and time	Structure
3 marks (any 5 of 6)	Half a page; about 6 minutes	One-line definition, then 3–5 points, a small table or a labelled sketch; one line of physiotherapy relevance
7 marks (any 5 of 6)	One and a half to two pages; about 14 minutes	Definition or introduction; classification or features; 6–8 headed points or a table with doses; a labelled diagram; precautions; one closing line

Write **doses with units** and **underline the keywords**. A **"Draw"** line tells you which diagram to draw; where the full notes have it, the diagram is printed beside the answer.

> **The paper:** 80 marks in 3 hours. Q1: any 5 of 6 at 3 marks (15). Q2: any 5 of 6 at 7 marks (35). Q3 and Q4: one 15-mark long answer each (30). About 2 minutes a mark. The long answers are covered in the full notes.

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1(a, b) Health, disease and the health care delivery system in India

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3 marks The primary health care setting

Asked: S17 (1c)

Primary health care (Alma-Ata, **1978**): essential health care, made **universally accessible** to individuals and families in the community through their **full participation**, at a cost the community and country can afford; the **first level of contact** with the health system, as close as possible to where people live and work.

- **Principles:** equitable distribution; community participation; intersectoral coordination; appropriate technology; focus on prevention and promotion.
- **Eight elements:** health education; food and nutrition; safe water and sanitation; maternal and child health with family planning; immunisation; control of local endemic diseases; treatment of common diseases and injuries; essential drugs.
- **Where in India:** village (ASHA, anganwadi worker) → sub-centre → PHC, now upgraded as **Ayushman Arogya Mandirs**.

Physiotherapy here means education, screening, antenatal classes, home programmes and CBR.

3 marks WHO definition of health

Asked: S19 (1a)

WHO Constitution (1948): "Health is a state of **complete physical, mental and social well-being** and **not merely the absence of disease or infirmity**."

- **Dimensions:** physical, mental, social; often added spiritual, emotional and vocational.
- **Disease:** a disturbance of the structure or function of body or mind, with signs and symptoms, that departs from health.
- Health is a **continuum**: positive health → ill-health → death.
- **Determinants:** biology and genes; behaviour and lifestyle; environment; socioeconomic conditions; health services.

The physiotherapist works along the whole continuum (promotion, prevention, treatment, rehabilitation), not only with disease.

3 marks How socioeconomic factors affect physiotherapy services in India

Asked: S20 (1c), W24 (1c) (asked 2 times)

- **Cost:** much care is paid **out of pocket**; repeated sessions and lost wages are hard for daily earners.
- **Distribution:** physiotherapists concentrated in **cities and private clinics**; few government posts at CHCs and PHCs.
- **Distance and transport**, especially for disabled and elderly villagers.
- **Awareness and literacy:** rest believed to be the cure; preference for traditional healers.
- **Gender:** women may lack permission or time; preference for female therapists.
- **Home environment:** crowded homes, floor-level living, Indian toilets.
- **Insurance:** limited cover for outpatient rehabilitation (PM-JAY mainly hospital care).

Solutions: CBR; camps; tele-rehabilitation; physiotherapists in Ayushman Arogya Mandirs and CHCs; simple home programmes; community awareness.

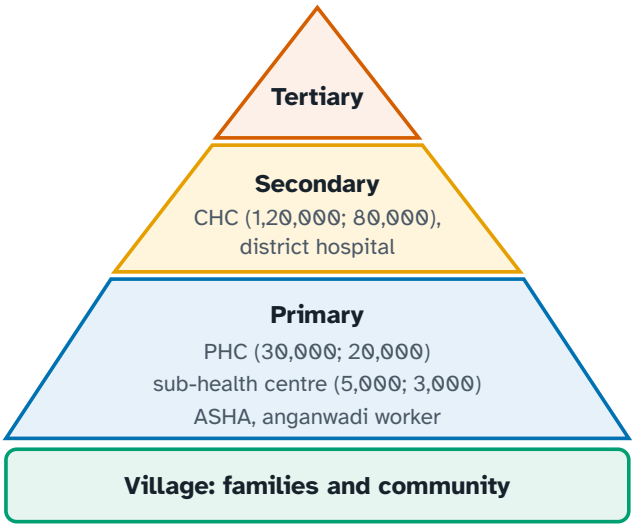
3 marks Health care delivery system in India

Asked: S21 (1f)

Public care in rural India is a **three-tier** pyramid: **primary, secondary, tertiary** (population per centre in plains; hilly and tribal areas in brackets).

Unit	Population, staff
Village	ASHA (about one per 1,000), anganwadi worker
Sub-centre	**5,000** (3,000); ANM, male health worker, community health officer
PHC	**30,000** (20,000); medical officer; 4–6 beds
CHC	**1,20,000** (80,000); **30 beds**; four specialists; first referral unit for four PHCs
District hospital, medical colleges	Specialist and super-specialist care, teaching, research

Three tiers of public health care in rural India



Population served by one centre in plains; second number for hilly, tribal and desert areas (Indian Public Health Standards). Medical colleges and national institutes form the tertiary level.

Sub-centres and PHCs are now **Ayushman Arogya Mandirs** (12 service packages). Urban: urban PHCs, municipal hospitals and a large private sector.

3 marks Levels of health care delivery

Asked: S25 (1d)

- **Primary level:** first contact, close to home: village (ASHA, anganwadi), **sub-centre** (5,000; 3,000) and **PHC** (30,000; 20,000), now **Ayushman Arogya Mandirs**. Health education, maternal and child care, immunisation, common illness, NCD screening.
- **Secondary level: CHC** (1,20,000; 80,000; 30 beds; surgeon, physician, obstetrician, paediatrician) and **district hospital**: specialist and emergency care, X-ray, laboratory, physiotherapy department.
- **Tertiary level:** medical colleges, AIIMS, national institutes: super-specialist care, teaching, research.

Referral moves up; follow-up moves down.

> **Draw:** A pyramid: sub-centre at the base, PHC, CHC, district hospital, medical college at the top, with population norms.

3 marks Role of the physiotherapist in health care delivery systems

Asked: S24 (1e)

Level	Role
Primary	**Health promotion** and fitness; ergonomic and back-care education; antenatal and postnatal classes; **screening** (posture, development, falls risk); training ASHAs and anganwadi workers; home programmes; **CBR**
Secondary	Outpatient and inpatient physiotherapy at **CHCs and district hospitals**: fractures, stroke, chest care, post-operative care, early intervention
Tertiary	Specialised ICU, cardiac, neuro, paediatric, sports and rehabilitation services; teaching and research

The role spans all levels of prevention: primary (fitness, ergonomics), secondary (screening), tertiary (rehabilitation).

7 marks Functioning of primary health centres

Asked: S22 (2d)

The **primary health centre** (PHC) is the first unit with a doctor in the rural three-tier system. It serves **30,000** people in plains (**20,000** in hilly and tribal areas) and is the referral unit for about **six sub-centres**; it refers on to the CHC.

Staff and set-up

- Medical officer, nurses, pharmacist, laboratory technician; **4–6 beds**.
- Now upgraded as an **Ayushman Arogya Mandir** giving comprehensive primary care in **12 service packages**.

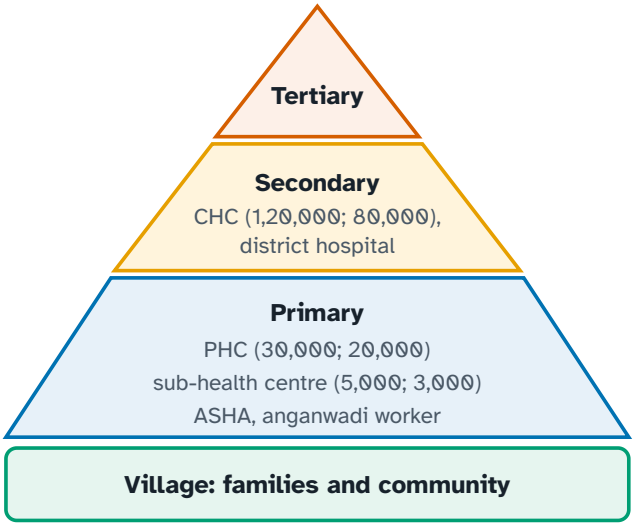
Functions

1. **Medical care:** outpatient and basic inpatient treatment of common illness and injury; **referral**.
2. **Maternal and child health:** antenatal, delivery and postnatal care; newborn care; **family planning**.
3. **Immunisation**.
4. **Disease control:** communicable (tuberculosis, malaria, leprosy); **NCD screening** (hypertension, diabetes, common cancers).
5. **National health programmes** carried out locally.
6. **Health education** and promotion; nutrition.
7. Safe water and environmental **sanitation**.
8. Collection and reporting of **vital statistics** and disease data.
9. Basic **laboratory** services and **essential drugs**.
10. **Training and supervision** of ANMs, health workers and ASHAs.
11. **School health**; elderly, palliative and mental health care (AAM packages).

Place of physiotherapy

- Health education (back care, ergonomics, fitness); antenatal and postnatal classes.
- Screening of posture, development and falls risk; training ASHAs and anganwadi workers.
- Home programmes and **CBR**; referral of complex cases to CHC or district hospital.

Three tiers of public health care in rural India



Population served by one centre in plains; second number for hilly, tribal and desert areas (Indian Public Health Standards). Medical colleges and national institutes form the tertiary level.

Link every function to the eight elements of primary health care (Alma-Ata, 1978).

7 marks Principles and scope of community physiotherapy

Asked: W25 (2d)

Community physiotherapy: the application of physiotherapy to the **whole community**, with a **preventive and promotive** emphasis, to optimise health and function **where people live and work**.

Principles

1. **Prevention and health promotion first**; then treatment and rehabilitation.
2. **Equity and access:** services for rural, poor, elderly, women and disabled people.
3. **Community participation** and ownership.
4. **Holistic, biopsychosocial** approach using the **ICF**.
5. **Teamwork and intersectoral coordination** (health, education, social welfare, industry).
6. **Appropriate, low-cost technology** and local resources.
7. **Education and empowerment** for self-management.
8. **Evidence-based**, measured outcomes; **sustainability**.

Scope

Area	Work
Health promotion	Fitness and obesity programmes in schools, colleges, workplaces
Women's health	Antenatal, postnatal, menopause, pelvic floor
Geriatrics	Fitness, falls prevention, home care
Industrial health	Ergonomics, injury prevention, return to work
Disability	Rehabilitation, **CBR**
NCDs	Prevention and management of diabetes, hypertension, COPD, heart disease
Others	Disaster management; sports and school health; palliative and home care
Professional	Research, policy and advocacy

Where it is delivered

- **Primary level:** villages, anganwadis, sub-centres, PHCs and **Ayushman Arogya Mandirs**: education, screening, training ASHAs and anganwadi workers, home programmes, camps.
- **Secondary and tertiary:** CHCs and district hospitals, with referral up and follow-up back to the community; tele-rehabilitation.

Tie the scope to the levels of prevention: primordial (active schools), primary (fitness, ergonomics), secondary (screening), tertiary (rehabilitation, CBR).

1(c) Physical fitness: components, evaluation and exercise prescription

Full notes p. 11

3 marks Deconditioning effects of bed rest

Asked: W16 (1c)

Prolonged bed rest causes **deconditioning**, a loss of function in many systems:

- **Heart:** higher resting HR, smaller stroke volume, lower VO_2max, **postural hypotension**, **DVT**.
- **Lungs:** basal collapse, retained secretions, pneumonia.
- **Muscles:** antigravity wasting, about **1–1.5% strength loss a day**.
- **Bones and joints:** bone loss, high urine calcium, **contractures**.
- **Skin:** pressure sores; **gut and bladder:** constipation, urinary stasis and stones; **mind:** confusion, low mood, poor balance.

Prevent with early sitting and walking, breathing exercises, ankle pumps, turning and early weight-bearing.

3 marks Precautions when prescribing fitness exercise for a diabetic patient

Asked: S18 (1d)

- **Hypoglycaemia:** check glucose before and after if on **insulin or sulphonylureas**; carry **fast sugar**; do not inject insulin into the exercising limb; lows can come **6–15 hours** later, risk up to **48 hours**.
- **High sugar:** no exercise with high glucose and **ketones**.
- **Feet:** daily inspection, good footwear; cycling or swimming if there is an ulcer.
- **Retinopathy** (severe): avoid vigorous and heavy resistance work, jarring, head-down positions and breath-holding.
- **Autonomic neuropathy:** use RPE, not HR; avoid quick position changes and heat; hydrate.

Exercise at a regular time, often 1–2 hours after a meal, with a partner who knows the signs of hypoglycaemia.

3 marks Principles of aerobic training

Asked: S19 (1e)

- **Overload:** work above the usual level (moderate **40–59% HRR** or vigorous 60–89%).
- **Progression:** increase **duration, then frequency, then intensity**, about 10% a week.
- **Specificity:** rhythmic large-muscle activity builds endurance, not strength.
- **Reversibility:** gains are lost when training stops.
- **Individuality** and **recovery:** dose for age, fitness and disease; rest days avoid overtraining.

Dose as **FITT-VP**: 5 or more days moderate, 30–60 minutes (150 a week); walking, cycling, swimming; with warm-up and cool-down.

Target HR (Karvonen) = resting HR + (maximum HR − resting HR) × intensity.

3 marks FITT principle of aerobic training

Asked: W20 (1c)

FITT (ACSM) sets the dose of aerobic exercise for healthy adults; **FITT-VP** adds volume and progression.

Letter	Aerobic dose
Frequency	**5 or more days** a week moderate, or 3 or more vigorous
Intensity	Moderate **40–59% HRR** (RPE 12–13); vigorous 60–89% (RPE 14–17)
Time	**30–60 min** moderate (**150 a week**) or 20–60 min vigorous (75 a week)
Type	Rhythmic large-muscle activity: walking, cycling, swimming
Volume, progression	About 500–1000 MET-minutes a week; progress gradually

3 marks What is aerobic exercise?

Asked: S21 (1a)

Aerobic exercise is rhythmic, continuous exercise of **large muscle groups** in which energy comes mainly from **oxygen-using pathways**.

- **Intensity and length:** low to moderate, **minutes to hours**.
- **Fuel:** fat and glucose → carbon dioxide and water; **type I** (slow-twitch) fibres.
- **Examples:** brisk walking, jogging, cycling, swimming, dancing.

- **Adaptations:** more capillaries, mitochondria and enzymes; bigger stroke volume; higher **VO_2max**; lower resting HR and BP.

Dose: moderate 40–59% HRR, 30–60 minutes, 5 or more days a week (150 minutes).

3 marks Aerobic vs anaerobic exercise

Asked: W24 (1b)

Feature	Aerobic	Anaerobic
Energy	Oxygen-using pathways	ATP–PC and glycolysis
Intensity, length	Low–moderate; minutes to hours	High; seconds to about 2 min
By-product	CO_2 and water	**Lactate**
Fibres	Type I	Type II
Gains	VO_2max, lower resting HR and BP	Strength, power, speed, muscle size
Examples	Walking, cycling, swimming	Sprinting, weightlifting, jumping

3 marks Adult health-related and skill-related fitness tests

Asked: S26 (1e)

Component	Test
Cardiorespiratory endurance	YMCA 3-minute step test; Rockport 1-mile walk; six-minute walk
Muscular strength	1RM; handgrip dynamometer
Muscular endurance	Push-up; curl-up
Flexibility	Sit-and-reach; back scratch
Body composition	BMI; waist; skinfolds
Agility	Illinois agility run; T-test
Balance	Stork stand; Y-balance
Coordination	Alternate-hand wall toss
Speed	30–50 m sprint
Power	Vertical jump; broad jump
Reaction time	Ruler-drop test

First five rows are **health-related**; the last six are **skill-related**.

7 marks Field tests used in athlete training

Asked: S16 (2b)

A **field test** is a test done away from the laboratory that needs neither extensive training nor expensive equipment (NSCA). A good test is **valid** (measures what it claims) and **reliable** (repeatable). Uses: pre-season screening, setting training loads, tracking progress and deciding **return to play**.

Tests by component

Test (component)	How it is done; what it shows
20 m shuttle run, beep test (aerobic)	Run between lines 20 m apart to beeps; starts at **8.5 km/h**, **+0.5 km/h each minute**; stop when pace is lost (line missed twice). Last stage predicts **VO_2max**
Yo-Yo intermittent recovery (repeated intense running)	2 × 20 m shuttles at rising speed, **10 s** jog between; level 1 starts at 10 km/h; score = total distance. Suits football and team sports
Cooper 12-minute run (aerobic)	Distance covered in 12 min; VO_2max = 22.351 × km − 11.288; correlation **0.897** with treadmill VO_2max
10–40 m sprint (speed)	Timing gates; 0–10 m split = acceleration, later splits = maximal speed
RAST (anaerobic power)	**6 × 35 m** sprints, 10 s rest; power = mass × distance^2 ÷ time^3; fatigue index
Vertical and broad jump (power)	Jump reach minus standing reach, best of 3; broad jump measured to the heels
T-test, Illinois, 505 (change of direction)	T-test: 10 yards forward, shuffle 5 left, 10 right, 5 back, run backwards; reliability 0.98. Illinois (16–19 years) excellent: under **15.2 s** boys, **17.0 s** girls
1RM, handgrip (strength)	Rising single lifts, maximum within **3–5 attempts**; hand dynamometer
Push-ups, curl-ups (endurance)	Maximum without rest; curl-ups to a metronome
Sit-and-reach (flexibility)	Moderately valid for hamstrings, poor for the low back

Order of testing (NSCA)

Non-fatiguing (height, weight, flexibility, skinfolds, vertical jump) → agility → maximal strength and power → sprints → muscular endurance → anaerobic capacity → **aerobic capacity last** or on another day.

Keep the warm-up, surface, footwear, time of day and order the same; full rest between maximal attempts; retest at set points.

Advantages and limits

- **Advantages:** cheap, quick, many athletes at once, sport-specific and realistic.
- **Limits:** less precise than laboratory tests; weather, surface, motivation and pacing affect scores; most give **predicted**, not measured, values; hand-timing error.

T-test and Illinois test change-of-direction speed; true agility adds a reaction to a signal.

7 marks General principles of exercise and its phases

Asked: S16 (2d)

Exercise is planned, structured, repeated activity done to improve or keep **physical fitness**. It works only when it follows the principles of training.

Principles of training

1. **Overload:** the body adapts only when stressed beyond its usual level.
2. **Progression:** increase the load gradually: **duration first, then frequency, then intensity**; about **10% a week**.
3. **Specificity:** the body adapts to the exact type of training (aerobic training builds endurance, not strength).
4. **Reversibility:** gains are lost when training stops ("**use it or lose it**").
5. **Individuality:** prescribe for the person's age, fitness, disease and goals.
6. **Recovery:** adaptation happens during rest; too little rest → **overtraining**.

Dose is written as **FITT-VP**: frequency, intensity, time, type, volume, progression.

Phases of a session

Phase	What it involves
Warm-up (5–10 min)	Light aerobic activity, dynamic movements; raises heart rate, muscle temperature and joint mobility; reduces injury and heart strain
Conditioning (20–60 min)	Main aerobic, resistance or sport work at the prescribed intensity

Phase	What it involves
Cool-down (5–10 min)	Light activity so HR and BP fall gradually; prevents **blood pooling and dizziness**
Stretching (≥10 min)	Static stretches when muscles are warm

Stages of a training programme

- **Initial** (first few weeks): low intensity, building the habit.
- **Improvement** (next few months): steady progression.
- **Maintenance:** keeping the gains long term.

Draw: Phases as a heart-rate curve against time: rise (warm-up), plateau (conditioning), fall (cool-down).

7 marks Deconditioning effects of bed rest

Asked: S16 (2f)

Deconditioning is the loss of function in many body systems caused by prolonged bed rest and inactivity, added to the effects of the illness itself.

Effects by system

System	Effects
Heart and circulation	Higher resting HR; smaller stroke volume; lower **VO_2max**; **postural hypotension** (plasma volume falls); **DVT**
Lungs	Lower lung volumes; basal collapse; retained secretions; **pneumonia**
Muscles	Wasting and weakness of antigravity muscles; about **1–1.5% strength loss a day** of strict bed rest
Bones and joints	Bone loss, high urine calcium; **contractures** and stiffness
Skin	**Pressure sores** over sacrum, heels, hips
Metabolic	Insulin resistance; protein loss
Kidney, bladder, gut	Urinary stasis, infection and stones; **constipation**, poor appetite
Nervous system and mind	Poor balance, confusion, low mood, poor sleep, sensory deprivation

Prevention and physiotherapy

- **Avoid unnecessary bed rest**; sit out and walk early.
- **Breathing exercises** and positioning for the chest.
- **Ankle pumps** and active leg exercise against DVT.
- Active and resisted exercise for strength; ROM against contractures.
- **Positioning and turning** for the skin.
- **Standing and weight-bearing** as early as possible for bone, circulation and balance.

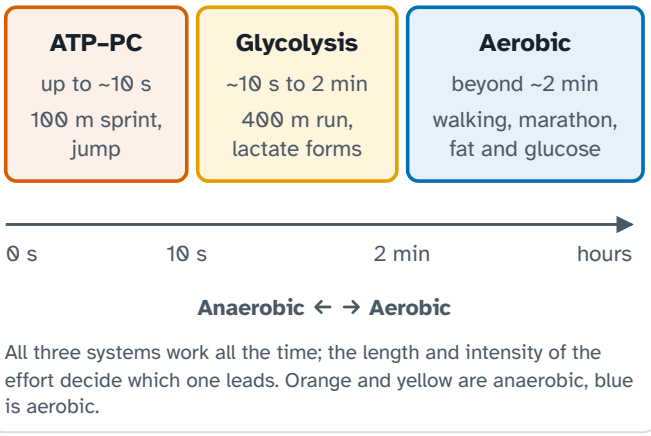
Avoid unnecessary bed rest: get the patient sitting out, standing and walking as early as possible.

7 marks Energy systems

Asked: W16 (2f)

ATP is the only direct fuel for contraction, but stored ATP lasts only a few seconds of maximal work. Three systems rebuild it. **All three work all the time**; the **length and intensity** of the effort decide which one leads.

Which energy system leads, by length of effort



All three systems work all the time; the length and intensity of the effort decide which one leads. Orange and yellow are anaerobic, blue is aerobic.

System	How ATP is made	Rate and use
Phosphagen (ATP–PC)	**Creatine phosphate** + ADP → ATP (creatine kinase); no oxygen	**Fastest, smallest store**; all-out efforts of seconds: sprint start, jump, lift
Fast glycolysis	Glucose or glycogen → pyruvate → **lactate**; net **2 ATP** (3 from glycogen)	Fast, medium store; about 30 s to 2 min: **400 m**; acid adds to fatigue
Oxidative	Carbohydrate and fat burnt in **mitochondria** (Krebs cycle, electron transport) → CO_2 + water; about **30–32 ATP** per glucose	**Slowest, largest store**; rest and efforts over 3 min: walking, **marathon**

Duration and main system (NSCA)

0–6 s phosphagen; **6–30 s** phosphagen and fast glycolysis; **30 s–2 min** fast glycolysis; **2–3 min** fast glycolysis and oxidative; **over 3 min** oxidative.

Key terms

- **Rate against capacity:** the faster a system makes ATP, the less it can make in total.
- **Respiratory exchange ratio:** about 0.7 on fat, 1.0 on carbohydrate.
- **Lactate threshold:** about **50–60% VO_2max** untrained, 70–80% trained.
- **Oxygen deficit** at the start of exercise; **EPOC** afterwards rebuilds creatine phosphate, clears lactate and returns temperature and heart rate to rest.

Training implications (NSCA work : rest)

- Phosphagen: 5–10 s near-maximal bouts, **1:12 to 1:20**.
- Fast glycolysis: 15–30 s bouts, **1:3 to 1:5**.
- Oxidative: bouts over 3 min at lower intensity, **1:1 to 1:3**, or **continuous training** (more capillaries and mitochondria, higher VO_2max).

Draw: A time line 0 s → 6 s → 30 s → 2 min → 3 min → hours, with the phosphagen, glycolysis and oxidative bands and one sport under each.

7 marks Hazards of a sedentary lifestyle

Asked: W18 (2a)

A sedentary lifestyle joins two separate risks. **Sedentary behaviour** is any waking behaviour at **1.5 METs or less** while sitting, reclining or lying. **Physical inactivity** is not meeting the physical activity recommendations (Tremblay et al., 2017). A person can meet the guidelines and still sit all day.

Size of the problem

- World: **31%** of adults insufficiently active (2022); India about **49%** (women 57%, men 42%).
- ICMR-INDIAB: **54.4%** inactive; towns 65% against villages 50%.
- Insufficient activity: **20–30% higher risk of death** (WHO); 9% of premature deaths worldwide (Lee et al., 2012).

Hazards by system

System	Hazards
Heart	Coronary heart disease, stroke, **hypertension**; cardiovascular death hazard ratio 1.18 with high sitting
Metabolism	**Type 2 diabetes (hazard ratio 1.91)**, obesity, high triglycerides, low HDL, metabolic syndrome; idle muscle loses lipoprotein lipase activity
Cancer	Colon, breast, endometrium; cancer death hazard ratio 1.17
Musculoskeletal	**Sarcopenia**, **osteoporosis**, stiffness, back and neck pain from static posture
Mind	**Depression** (relative risk 1.25), anxiety, poor sleep, lower cognition
Fitness	Low VO_2max and strength; falls and dependence in old age

High sitting raises all-cause death (hazard ratio **1.24**; Biswas et al., 2015). Risk climbs above **6–8 h a day** of sitting or 3–4 h of television. About **60–75 min a day** of moderate activity offsets long sitting (Ekelund et al., 2016).

Physiotherapy prescription

- **150–300 min a week** of moderate activity plus strengthening on **2 days** (WHO, 2020); sit less and replace sitting with any activity, even light.
- **Break up sitting:** a **2-minute walk every 20 minutes** cut the glucose rise after a drink by 24–30% (Dunstan et al., 2012).
- **Desk workers:** build to **2 h, then 4 h a day** of standing and light walking at work; sit-stand desk; avoid static standing (Buckley et al., 2015).
- Habits: stand for calls, take the stairs, walking meetings, active commuting; step counter and reminders; workplace and school programmes; ergonomic advice.

Physical activity is primordial and primary prevention of most non-communicable diseases.

7 marks Aerobic and anaerobic exercise with examples; MET

Asked: W22 (2e)

Definitions

- **Aerobic exercise:** rhythmic, continuous exercise of large muscle groups in which energy comes mainly from **oxygen-using pathways**.
- **Anaerobic exercise:** short, intense exercise in which energy comes mainly from **ATP–PC and glycolysis** without enough oxygen.

Comparison

Feature	Aerobic	Anaerobic
Intensity, length	Low to moderate; minutes to hours	High; seconds to about 2 min
Lactate threshold	Mostly below it, so it can be kept up	Above it; lactate and acid build up
Fuel, by-product	Fat and glucose; CO_2 and water	Creatine phosphate and glucose; **lactate**
Fibres	Type I (slow-twitch)	Type II (fast-twitch)
Adaptations	More capillaries and mitochondria; bigger stroke volume; higher **VO_2max**; lower resting HR and BP	Strength, power, speed, muscle size, lactate tolerance
Examples	**Brisk walking, jogging, cycling, swimming, dancing**	**Sprinting, weightlifting, jumping, high-intensity intervals**

MET (metabolic equivalent)

- The energy used **sitting quietly at rest**: **3.5 ml O_2/kg/min** (about 1 kcal/kg/hour).
- Bands: sedentary **1.0–1.5**; light **1.6–2.9**; moderate **3.0–5.9**; vigorous **6.0 or more**.
- Values (Compendium, 2024): yoga 2.3; slow walking 3.0; brisk walking **4.8**; cycling 6.8; running at about 10 km/h **9.3**.
- **Energy:** kcal/min = METs × 3.5 × kg ÷ 200; a 60 kg person walking briskly uses about 5 kcal a minute.
- **Volume:** MET-minutes = METs × minutes; ACSM advises **500–1000 a week**.
- **Uses:** grading and prescribing intensity, weekly volume, comparing activities, weight control, choosing activities in cardiac rehabilitation.
- **Limit:** measured rest averages about 2.6 ml/kg/min, so METs are estimates; check with HR or RPE.

Most daily activity mixes both: all energy systems work together, and the length and intensity of effort decide which leads.

7 marks Define aerobic exercise; explain circuit training

Asked: S25 (2a)

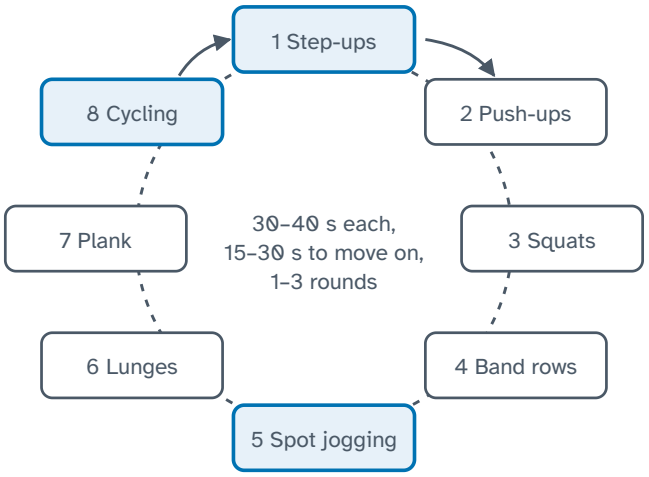
Aerobic exercise

Rhythmic, continuous exercise of **large muscle groups** in which energy comes mainly from **oxygen-using pathways**; low to moderate intensity for minutes to hours (brisk walking, cycling, swimming). It raises **VO_2max** and lowers resting HR and BP.

Circuit training: definition

A series of exercises (**stations**) done one after another with **little rest**, working different muscle groups and adding aerobic stations; the whole set is repeated. Developed by **Morgan and Adamson** at Leeds in 1953.

A circuit of eight stations



Blue stations are aerobic; the others alternate upper body, lower body and trunk so each muscle group rests while another works.

Method

- About **10–15 stations**, alternating upper body, lower body, trunk and aerobic work.
- **30–40 s** or **12–15 repetitions** at each, at **40–60% 1RM** for a weight circuit.
- **15–30 s** rest while moving on; **1–3 circuits**, about 30 minutes; warm-up and cool-down.
- Progress by adding a round, more work time, less rest or more load.

Types

- **Circuit weight training:** light weights, many repetitions.
- **Aerobic circuit:** aerobic bouts between stations, or heart rate kept at 70–80% of maximum.
- **High-intensity circuit:** 12 body-weight exercises, 30 s each, 10 s change.
- **Circuit class therapy** for patients; sport-specific circuits.

Effects

- Muscular endurance and strength; VO_2max up about **5% (men) and 8% (women)** in early studies, and **11%** when heart rate was kept at 70–80% of maximum.
- After stroke, circuit class therapy improved six-minute walk distance by about **61 m** (Cochrane, 2017).

Advantages and disadvantages

- **Advantages:** several fitness components in one short session; suits **groups** (schools, workplaces, **cardiac rehabilitation phase III**, older adults with chair stations); little equipment; enjoyable.
- **Disadvantages:** smaller strength gains than heavy training; needs space and supervision; technique slips with fatigue.

Precautions: technique first; **avoid breath-holding**; adapt stations for joint problems; monitor **HR or RPE** in patients.

> **Draw:** A ring of eight stations with clockwise arrows: step-ups, push-ups, squats, band rows, spot jogging, lunges, plank, cycling.

7 marks Effects of exercise in a person with hypertension

Asked: W25 (2b)

Hypertension affects **35.5%** of Indian adults (ICMR-INDIAB, 2023). Exercise lowers blood pressure both after a single session and with training, and is a first-line lifestyle treatment.

Acute effects

- During aerobic exercise **systolic BP rises** with workload while diastolic stays level or falls slightly; heavy isometric holds and breath-holding raise both sharply.
- After one session BP stays lower for up to **22 hours**: **post-exercise hypotension** (Pescatello et al., 2004).

Chronic effects

- Resting BP falls most in hypertension: about **5–8 mmHg** (2–4 in prehypertension, 1–2 in normal BP).
- Network meta-analysis (Edwards et al., 2023): aerobic **4.5/2.5**, dynamic resistance 4.6/3.0, combined 6.0/2.5, interval training 4.1/2.5, **isometric 8.2/4.0 mmHg**.
- Fewer cardiovascular events; better fitness and mood; **less need for drugs**.

Mechanisms

- Lower **sympathetic activity**: plasma noradrenaline down 29%, renin activity down 20%.
- Lower **vascular resistance** (7%); better **endothelial function**; less arterial stiffness with aerobic training.
- **Weight loss**; better **insulin sensitivity**.

Prescription (ACSM)

- **Aerobic: 5–7 days a week**, 40–59% heart rate reserve (RPE 12–13), **30 min or more** a day; walking, cycling, swimming.
- **Resistance: 2–3 days a week**, 60–70% 1RM (40–50% in older people and beginners), 2–4 sets of 8–12.
- Flexibility 2–3 days a week.

Precautions

- Do not start if resting BP is above **200/110**; stop a test above **250/115**.
- **Avoid breath-holding** (Valsalva) and maximal straining.
- **Long warm-up and cool-down**: vasodilators and alpha-blockers cause dizziness after exercise.
- Use **RPE with beta-blockers**; watch dehydration and low potassium with **diuretics**.

Classic answers banned weights; current evidence shows resistance and isometric training also lower BP, provided there is no breath-holding.

7 marks Principles of exercise prescription

Asked: W25 (2f)

Exercise prescription is a planned, individual dose of exercise (FITT-VP) written to improve or maintain health and fitness safely and to meet the person's goals.

Principles of training

- **Overload**; **progression** (duration → frequency → intensity, about 10% a week); **specificity**; **reversibility**; **individuality**; **recovery**.

FITT-VP for healthy adults (ACSM)

Type	Dose
Aerobic	Moderate **40–59% HRR** (RPE 12–13) 30–60 min, ≥5 days (**150 min a week**); or vigorous 60–89% (RPE 14–17), ≥3 days (75 min a week); walking, cycling, swimming
Resistance	**2–3 days a week**; **60–70% 1RM** for beginners (40–50% older or very unfit); **2–4 sets of 8–12**; multi-joint, opposing groups balanced
Flexibility	2–3 days a week; hold **10–30 s** (30–60 s older adults), total about 60 s per stretch
Neuromotor	Balance and agility 2–3 days a week, especially older adults
Volume, progression	About 500–1000 MET-minutes a week; raise load when the top of the repetition range is easy

Karvonen: target HR = resting HR + (max HR – resting HR) × intensity.

Steps and safety

1. **Screen:** history, risk factors, medicines, red flags; medical clearance when needed.
2. **Test fitness** and set goals with the person.
3. Write the FITT-VP for aerobic, resistance, flexibility and balance work.
4. Teach **warm-up, cool-down**, self-monitoring (**talk test, RPE, pulse**) and warning signs.
5. **Review and retest every 4–6 weeks**; adjust for adherence and enjoyment.

Adapt for disease: glucose checks in diabetes, no Valsalva and long cool-down in hypertension.

1(c) Obesity in children and adults: body composition, BMI and weight management

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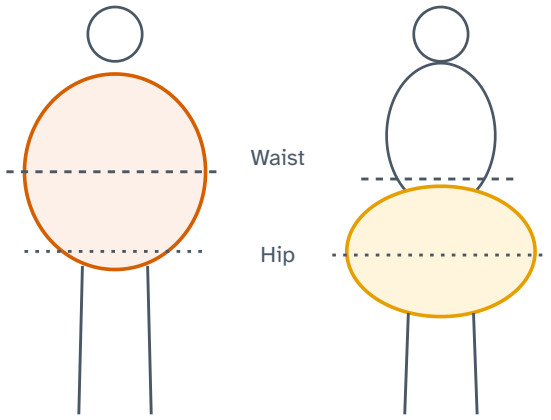
3 marks Define obesity; list its types

Asked: W16 (1d), S25 (1f) (asked 2 times)

Obesity (WHO): **abnormal or excessive fat accumulation that may impair health**. Screened by **BMI** (kg/m²) and **waist**; confirmed by body fat where possible.

Basis	Types
Fat distribution	**Android** (apple, central; men; higher risk of diabetes and heart disease); **gynoid** (pear, hips and thighs; women)
Fat cells	**Hyperplastic** (more cells; childhood onset; harder to reverse); **hypertrophic** (bigger cells; adult onset)
Cause	**Primary** (intake more than use); **secondary** (hypothyroidism, Cushing's, PCOS, steroids, Prader–Willi)
Severity	By BMI grade

Apple and pear: where the fat sits, and where to measure



Waist (long dashes): midway between the lowest rib and the top of the iliac crest, at the end of a normal breath out. Hip (short dashes): the widest part of the buttocks. Abdominal fat (orange) carries more risk than hip and thigh fat (yellow).

3 marks Grades of obesity and their clinical significance

Asked: W17 (1a)

Grade	WHO (kg/m²)	Asian Indian 2009
Normal	18.5–24.9	18.5–22.9
Overweight	25–29.9	23–24.9
Obese	Class I 30–34.9; II 35–39.9; III ≥40	≥25

Revised Indian definition (2025): any BMI **over 23** is obesity; grades I 23–24.9, II 25–27.5, III 27.6–32.4, IV ≥32.5.

Significance: risk of diabetes, hypertension, heart disease and death **rises with each grade**; Asian Indians carry more fat at the same BMI, so cut-offs are lower; grade guides treatment (lifestyle for all; drugs in higher grades; bariatric surgery for severe obesity with disease).

3 marks Body mass index and its uses

Asked: W22 (1f), S23 (1d) (asked 2 times)

BMI = weight (kg) ÷ height (m)². Example: 68 kg, 1.60 m → 68 ÷ 2.56 = **26.6 kg/m²**.

- **WHO:** normal **18.5–24.9**; overweight **25–29.9**; obese **≥30**.
- **Asian Indian (2009):** normal up to **22.9**; overweight **23–24.9**; obese **≥25** (2025: over 23 is obesity).
- **Uses:** simple, cheap **screening** of individuals and populations; grading obesity; **tracking change**; deciding treatment.
- **Limits:** does not separate fat from muscle (a muscular athlete scores high); says nothing about **fat distribution**; needs age charts in children.

Add waist circumference: a normal BMI with a large waist is still at risk.

3 marks Fat mass and fat-free mass

Asked: S24 (1c)

Body weight = fat mass + fat-free mass; body fat % = fat mass ÷ body weight × 100.

- **Fat mass:** all body fat: **essential fat** (nerves, bone marrow, organs; more in women) and **storage fat** under the skin and around organs.

- **Fat-free mass:** muscle, bone, organs and water (about **three-quarters water**); it sets the **resting metabolic rate**.
- **Measured by** skinfolds, BIA, DEXA, underwater weighing.

Weight loss should reduce fat mass and keep fat-free mass: resistance training, enough protein, no crash diets.

7 marks Methods of measuring body composition

Asked: S16 (2e), W16 (2c) (asked 2 times)

Body composition divides body weight into **fat mass** (essential and storage fat) and **fat-free mass** (muscle, bone, organs, water). **Body fat % = fat mass ÷ body weight × 100**. It shows what BMI cannot: whether excess weight is fat or muscle.

Method	Principle	Comment
BMI, waist, hip, waist-to-hip, waist-to-height	Tape and scale	Field methods; waist-to-hip risk at **≥0.90 men, ≥0.85 women**; waist-to-height **over 0.5**
Skinfold thickness	**Calipers** at biceps, triceps, subscapular, suprailiac; equations give density and % fat	Cheap, portable; needs a trained tester; less accurate in severe obesity
Bioelectrical impedance (BIA)	Small current passes faster through **water-rich lean tissue** than fat	Quick; affected by hydration, food and exercise before the test
Underwater (hydrostatic) weighing	**Archimedes' principle**: fat floats, lean tissue sinks → body density	The old **reference method**; needs a tank and full breath-out
Air displacement (Bod Pod)	Body volume from displaced air	Accurate, comfortable, costly
DEXA	Two low-dose X-ray energies separate fat, lean tissue and bone	Accurate; regional fat and **bone density**; costly
CT and MRI	Images of fat areas	Measure **visceral fat**; research use

Choosing a method

- **Community and clinic:** BMI, waist, skinfolds, BIA (cheap, portable).
- **Laboratory:** hydrostatic weighing, Bod Pod, DEXA, CT and MRI (accurate, costly).
- Repeat with the **same method and time of day** to track change.

Obesity by body fat (2025 Indian definition): over **25.5% in men** and over **38% in women**. Weight loss should cut fat mass and keep fat-free mass.

7 marks Set-point theory; three causes of childhood obesity

Asked: S20 (2a)

Set-point theory

- The brain (**hypothalamus**, using signals such as **leptin** from fat) defends a **"set" body weight**, like a **thermostat**.
- When a person diets, **hunger rises** and **metabolism falls**, so weight returns towards the old level.
- This explains **regain after crash diets** ("yo-yo" dieting): rapid loss is mostly water and muscle, and weight returns, often with more fat.
- **Application:** slow weight loss (about **0.5 kg a week**) with **regular exercise** and lasting habits is thought to help the body settle at a lower weight; target 5–10% of body weight over 3–6 months.

Childhood obesity

Measured by **BMI-for-age on IAP 2015 charts** (5–18 years): overweight from the adult-equivalent **23** line, obesity from the **27** line.

Three causes

1. **Diet:** fast food, sugary drinks and snacks, large portions. *Prevention:* no junk food in canteens; fruit and home food in tiffin; water bottles.
2. **Inactivity and screen time:** too little play and sport; short sleep. *Prevention:* at least **60 minutes a day** of moderate to vigorous activity (WHO); limit recreational screen time; regular sleep.
3. **Family and environment:** parents with obesity and family eating habits; urban living without safe play space. *Prevention:* the whole family changes habits together; active travel to school.

Other causes: academic pressure and emotional eating; early-life factors (birth weight, mother's diabetes in pregnancy, formula feeding); rarely endocrine or genetic disease.

Obesity in childhood continues into adult life; praise effort, not weight.

7 marks Obesity and its effects on adult health

Asked: W20 (2f)

Obesity (WHO): abnormal or excessive fat accumulation that may impair health. In Asian Indians any BMI **over 23 kg/m²** counts as obesity (2025); abdominal obesity is a waist of **≥90 cm in men, ≥80 cm in women**.

Causes: energy intake more than use (high-calorie food, sugary drinks); sedentary work and screen time; poor sleep and stress; genetics; pregnancy without return to activity; menopause; drugs (steroids); hypothyroidism, PCOS; ageing.

Effects on adult health

System	Effects
Metabolic	**Insulin resistance, type 2 diabetes**, high cholesterol, **metabolic syndrome**, fatty liver, gallstones, gout
Heart and vessels	**Hypertension**, coronary heart disease, heart failure, stroke, varicose veins, DVT
Lungs	Breathlessness on effort, **obstructive sleep apnoea**, obesity hypoventilation
Joints	**Knee and hip osteoarthritis**, low back pain, flat feet, plantar heel pain
Women	PCOS, infertility, gestational diabetes, pre-eclampsia, **stress incontinence**
Cancer	Breast (after menopause), colon, endometrial
Mind and life	Low self-esteem, depression, stigma, poor fitness and quality of life, **shorter life**

Role of physiotherapy

- Assess BMI, waist, body fat and fitness; screen BP, sugar and joints.
- **Aerobic** at least 5 days a week, 40–59% HRR, building to **more than 250 minutes a week**; **resistance** 2–3 days to keep fat-free mass; low impact for painful knees.
- Lifestyle: dietitian-led **500 kcal a day deficit**; steps, sleep, food diary; **no crash diets**.

Target 5–10% of body weight over 3–6 months; losing more than 5% within about 20 weeks improved disability in knee osteoarthritis.

7 marks Causes of childhood obesity

Asked: W21 (2a), W24 (2a) (asked 2 times)

Childhood obesity is judged by **BMI-for-age on the IAP 2015 charts** (5–18 years): **overweight** from the line equal to an adult BMI of **23**, **obesity** from the line equal to **27**. Under 5 years, WHO standards are used.

Causes (with prevention)

1. **Unhealthy food:** fast food, **sugary drinks** and snacks; large portions. → No junk food or sugary drinks in canteens; fruit and home food in tiffin; water bottles.
2. **Physical inactivity:** too little play and sport. → At least **60 minutes a day** of moderate to vigorous activity, vigorous and muscle- and bone-strengthening on **3 days a week** (WHO); daily PE periods.
3. **Screen time:** TV, mobile and games replace play. → Limit recreational screen time.
4. **Short sleep.** → Regular sleep routine.
5. **Family factors:** parents with obesity, family eating habits. → Whole family changes habits together; family meals.
6. **Urban environment:** no safe play space. → Active travel to school; school sport.
7. **Academic pressure and emotional eating.** → Fun, non-competitive games; no blame or teasing.
8. **Early-life factors:** low or high birth weight, mother's diabetes in pregnancy, formula feeding.
9. **Rarely endocrine or genetic disease** (secondary obesity).

Why it matters

- Early type 2 diabetes, high BP, fatty liver, sleep apnoea; knee pain, flat feet, knock knees, slipped capital femoral epiphysis; teasing, low self-esteem; obesity **continues into adult life**.

For a 5-cause question, pick diet, inactivity, screen time, sleep and family habits, and add one line of prevention for each.

7 marks Grades of obesity and their clinical significance

Asked: S24 (2d)

Obesity is graded by **BMI = weight (kg) ÷ height (m)²**, with waist and body fat added.

WHO and Asian Indian grades

Category	WHO (kg/m²)	Asian Indian, 2009
Underweight	<18.5	<18.5
Normal	18.5–24.9	18.5–22.9
Overweight	25–29.9	23–24.9
Obese	**Class I 30–34.9; II 35–39.9; III ≥40**	**≥25**

Revised Indian definition (2025)

- Any BMI **over 23** is obesity; no separate "overweight" step.
- **Stage 1:** extra fat without effects on organ function or daily activities.
- **Stage 2:** BMI over 23 plus a large waist (**≥90 cm men, ≥80 cm women**) or waist-to-height **over 0.5**, plus limits in daily activities (knee pain, breathlessness) or an obesity-related disease (type 2 diabetes).
- **BMI grades:** I 23–24.9; II 25–27.5; III 27.6–32.4; IV ≥32.5. Body fat **over 25.5% (men), over 38% (women)** is obesity.

Clinical significance

- Risk of **diabetes, hypertension, heart disease and death rises with each grade**.
- At the same BMI, **Asian Indians carry more body and abdominal fat**, so cut-offs are lower.
- **Grade guides treatment:** lifestyle for all; drugs and closer medical care in higher grades or with disease; **bariatric surgery** for severe obesity with disease.
- **Waist adds what BMI misses:** normal BMI with a large waist is still at risk.
- **Physiotherapy:** higher grades need screening for disease first; low-impact exercise if joints hurt; target 5–10% loss over 3–6 months.

Always show the BMI working in a case, then grade it by both WHO and Indian cut-offs.

7 marks Anthropometric assessment in obesity

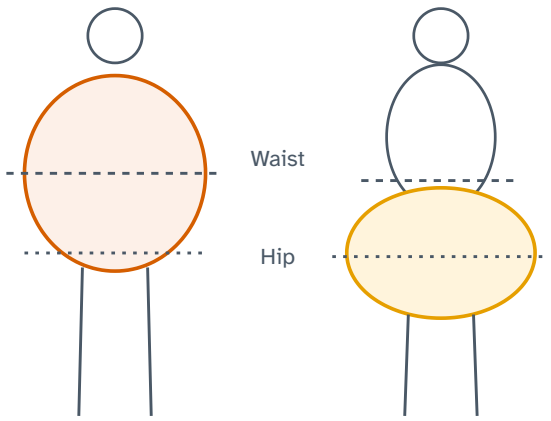
Asked: S26 (2c)

Anthropometry is the measurement of body size and proportions with simple tools (scale, stadiometer, tape, calipers). It is the cheap field method of judging obesity and fat distribution.

Measurements

1. **Weight** in light clothes, no shoes, on a calibrated scale; **height** with a stadiometer.
2. **BMI** = weight (kg) ÷ height (m)²; graded by WHO and Asian Indian cut-offs (obesity **over 23** in Indians, 2025).
3. **Waist circumference:** midway between the **lowest rib and the top of the iliac crest**, at the end of a normal breath out; abdominal obesity **≥90 cm men, ≥80 cm women**.
4. **Hip circumference:** widest part of the buttocks.
5. **Waist-to-hip ratio:** risk at **≥0.90 men, ≥0.85 women** (WHO). **Waist-to-height ratio:** risk **over 0.5**.
6. **Skinfolds** with calipers at biceps, triceps, subscapular and suprailiac sites → body density and % fat.
7. **Mid-upper-arm** and **neck circumference** (neck for sleep apnoea risk).
8. **Children:** BMI-for-age on **IAP charts** (adult-equivalent 23 and 27 lines).

Apple and pear: where the fat sits, and where to measure



Waist (long dashes): midway between the lowest rib and the top of the iliac crest, at the end of a normal breath out. Hip (short dashes): the widest part of the buttocks. Abdominal fat (orange) carries more risk than hip and thigh fat (yellow).

Standard technique

- Same **time of day**, same side, **mean of two readings**; repeat **monthly**.

Interpretation

- Apple (android) shape with a large waist carries more risk than pear (gynoid).
- BMI cannot separate fat from muscle, so add waist and skinfolds.

> **Draw:** Apple and pear outlines with the waist line (between lowest rib and iliac crest) and the hip line (widest part of buttocks).

Part 2: Women's health

2(a–c) Women in India: social issues, legal rights and health promotion

Full notes p. 37

7 marks Strategies of health promotion for rural women

Asked: S21 (2d)

Health promotion for rural women works through village workers and the places where women already gather. Their main problems are **anaemia** (57% of women aged 15–49, NFHS-5), back pain, pelvic floor problems, pregnancy care, chulha smoke and diabetes, with little time and limited mobility.

1. Know the community

- Meet the **sarpanch, ASHA, ANM, anganwadi worker** and **self-help group** leaders; learn the main problems and the women's **daily timetable**.

2. Use existing platforms

- **Anganwadis**, **Village Health, Sanitation and Nutrition Days**, self-help group meetings, mahila mandals, PHC and **Ayushman Arogya Mandir** clinics, temples and schools.

3. Choose the message

- Nutrition and **iron**; **antenatal and postnatal exercise**; **pelvic floor exercise**.
- **Back care**: safe ways to lift water pots, squat and carry; osteoporosis prevention.
- Physical activity (walking groups, yoga); clean cooking fuel; toilet use; menstrual hygiene; screening for BP, diabetes and cancer.

4. Use the right methods

- **Local language**, pictures, **demonstrations and practice**, songs and role plays; short sessions at convenient times; **women educators**; home visits for those who cannot come.

5. Involve the family

- Men, **mothers-in-law** and elders, so women get permission and time.

6. Link to schemes and services

- **JSY, PMMVY, PMSMA, Ujjwala**, pensions, PHC referral.

7. Train local women

- ASHAs and self-help group leaders continue the programme (**sustainability**).

8. Evaluate

- Attendance; **knowledge before and after**; behaviour change (exercise, iron tablets, institutional delivery); feedback.

Example: hold antenatal exercise sessions at the PMSMA clinic on the 9th of the month, where pregnant women already gather.

7 marks Social issues affecting the physical function of women in India

Asked: W21 (2f), W24 (2f) (asked 2 times)

Women's health in India is shaped by social conditions: **57%** of women aged 15–49 are anaemic (NFHS-5); about **1 in 4** women aged 20–24 married before 18; **29%** of ever-married women reported spousal violence; female literacy **65%** against 82% for men (Census 2011).

Issues and their effects on the body

Social issue	Effect on body and function
Gender discrimination, son preference	Girls fed less and later, less medical care: **stunting, anaemia, weak bones**
Early marriage, repeated pregnancies	Obstructed labour, fistula, **prolapse, incontinence**, anaemia, low bone mass, maternal deaths
Poverty and malnutrition	Anaemia, fatigue, low muscle strength, **osteoporosis**, poor healing

Social issue	Effect on body and function
Heavy domestic and farm work; double burden	**Back and neck pain** from squatting, stooping and **head-loading**; knee osteoarthritis; no time to rest or exercise
Chulha smoke	Chronic bronchitis, **COPD**, eye problems
No toilet	Holding urine and stool, **urinary infections, pelvic floor problems**, unsafe night trips
Low literacy and awareness	Late care; little knowledge of exercise, posture, pelvic floor health
Restricted mobility and decision-making	Need permission or escort; **cannot attend repeated physiotherapy**
Domestic violence, dowry	Injuries, chronic pain, depression, anxiety
Cultural limits on exercise and sunlight	Inactivity, obesity, **vitamin D deficiency**
Menstrual and widowhood stigma	Infections; isolation and poverty in old age

Role of the physiotherapist

- **Education** in local language at anganwadis and self-help groups: back care, safe lifting and carrying, posture.
- **Pelvic floor training**; antenatal and postnatal exercise; walking groups and yoga for bone health.
- Home programmes and **home visits** for women who cannot travel; women therapists.
- Link to schemes: **JSY, JSSK, PMMVY, Ujjwala** (LPG), **Swachh Bharat** (toilets), **Anaemia Mukt Bharat**.

Draw: Flow: no toilet → holding urine → UTI and pelvic floor problems → pelvic floor training and education.

2(d, e) Pregnancy: physiological changes, discomforts and antenatal care

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3 marks Diastasis recti

Asked: S17 (1a)

Diastasis recti: separation of the two **rectus abdominis** muscles along the **linea alba**, from stretching in pregnancy; **60%** of first-time mothers at 6 weeks after birth (Sperstad et al., 2016).

- **Effects:** abdominal doming, poor trunk support, back and pelvic pain; rarely hernia.
- **Test:** crook lying head lift; finger-widths at the umbilicus and **4.5 cm above and below**; **two finger-widths or more** = diastasis.
- **Management:** roll to the side to get up; avoid straining; **transversus abdominis with pelvic floor**; curl-ups only when the abdomen stays flat; splinting with crossed hands; surgery rarely.

3 marks Exercises contraindicated during pregnancy

Asked: S17 (1f), S22 (1e) (asked 2 times)

- **Contact sports:** boxing, hockey, kabaddi.
- **High fall risk:** horse riding, skiing, gymnastics, off-road cycling.
- **Scuba diving, sky diving**; **hot yoga** or hot Pilates; unaccustomed exercise at high altitude (ACOG, 2020).
- **Prolonged supine exercise** after the first trimester (supine hypotension).
- **Double leg raises and full sit-ups** (strain the abdominal wall).
- **Heavy lifting with breath-holding**; **ballistic** and end-range stretching (lax ligaments); deep squats and lunges with pelvic girdle pain; high-impact jumping late in pregnancy.

No exercise at all with absolute contraindications, for example placenta praevia after 26 weeks or pre-eclampsia.

3 marks Six physiological changes in pregnancy

Asked: W17 (1b)

1. **Hormones:** progesterone, oestrogen and **relaxin** rise → ligament laxity, smooth muscle relaxation.
2. **Heart and blood:** blood volume up **40–50%**; cardiac output up; resting HR up **10–20 beats**; dilutional anaemia; clots more easily.
3. **Supine position:** from about **20 weeks** the uterus presses on the inferior vena cava → **supine hypotension**.
4. **Breathing:** diaphragm rises; tidal volume and minute ventilation up; FRC falls.
5. **Musculoskeletal:** weight gain; **centre of gravity forward**; abdominals stretch (diastasis); pelvic floor loaded; feet flatten.
6. **Gut, kidneys, metabolism:** heartburn, constipation; frequency and leakage; BMR and heat production up.

3 marks Early bird classes

Asked: W18 (1b)

Early bird classes are antenatal education sessions in the **first trimester**, before the main antenatal classes (from about 28 weeks).

- **Content:** changes of pregnancy; **nutrition**; **posture and back care**; **safe exercise**; **pelvic floor** training; discomforts of early pregnancy (nausea, fatigue); **warning signs**.
- **Purpose:** build **healthy habits early**; prevent problems; reduce anxiety.
- Later antenatal classes then prepare her for labour.

Run them where women already gather: anganwadi, PHC or the PMSMA clinic.

3 marks Physiotherapy management of hyperemesis gravidarum

Asked: W19 (1a)

Hyperemesis gravidarum: severe nausea and vomiting of pregnancy with **dehydration, weight loss and electrolyte imbalance**, often needing hospital care; medical treatment is fluids and anti-emetics.

- **Relaxation and slow breathing** to ease nausea and anxiety.
- **Acupressure wristbands at P6** (some women find them helpful).

- Gentle **bed and chair exercises** and **ankle pumps** to prevent deconditioning and clots while in bed.
- **Early walking** as she improves; energy conservation.
- Posture and positioning; **emotional support**.

3 marks Musculoskeletal dysfunctions in pregnancy

Asked: S21 (1b)

Caused by **relaxin** (lax ligaments), weight gain, **centre of gravity forward** and stretched abdominals.

- **Low back pain:** posture, pelvic tilts, cat–camel, side-lying with a pillow between the knees, belt.
- **Pelvic girdle pain** (sacroiliac, symphysis pubis): pelvic belt; knees together when turning; sit to dress.
- **Diastasis recti:** transversus with pelvic floor; roll to the side.
- **Carpal tunnel syndrome** (fluid): night wrist splint in neutral, nerve gliding.
- **Leg cramps**; heel, foot and knee pain; rib pain; De Quervain's, coccyx pain, meralgia paraesthetica.

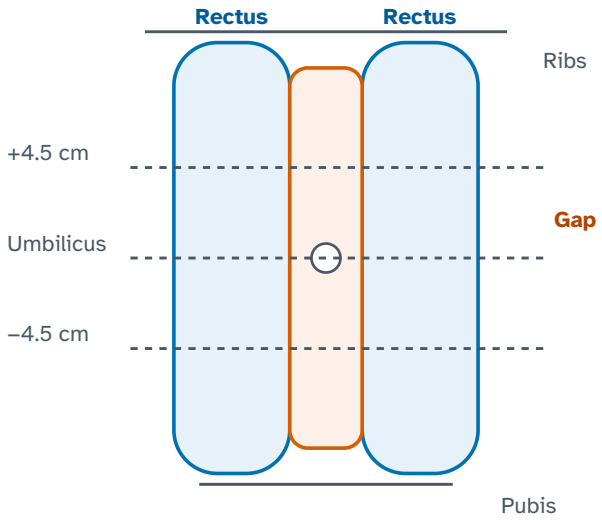
Weak, loaded pelvic floor also gives urinary leakage: daily pelvic floor training.

3 marks Assessment of diastasis recti

Asked: W24 (1a)

1. **Crook lying**; she lifts head and shoulders slowly until the shoulder blades just clear the bed.
2. Fingers across the midline at **three levels**: umbilicus, **4.5 cm above**, **4.5 cm below**; count **finger-widths** between the muscle edges (or calipers, ultrasound).
3. **Two finger-widths or more** at any level = diastasis.
4. Note **depth and tension** of the linea alba (firm, or fingers sink) and **doming**.
5. Record pain, back and pelvic symptoms, **pelvic floor** function, lifting and getting up.

Testing for diastasis recti: three levels



Two finger-widths or more at any level counts as diastasis (the definition Sperstad et al. used). Recti in blue, the widened linea alba in orange.

7 marks Diastasis recti: evaluation and management

Asked: W16 (2d), S22 (2f), S24 (2e) (asked 3 times)

Diastasis recti: separation of the two **rectus abdominis** muscles along the **linea alba**, from stretching in pregnancy. In first-time mothers it is present in **33% at 21 weeks** of pregnancy, **60% at 6 weeks** after birth, 45% at 6 months and 33% at 12 months (Sperstad et al., 2016).

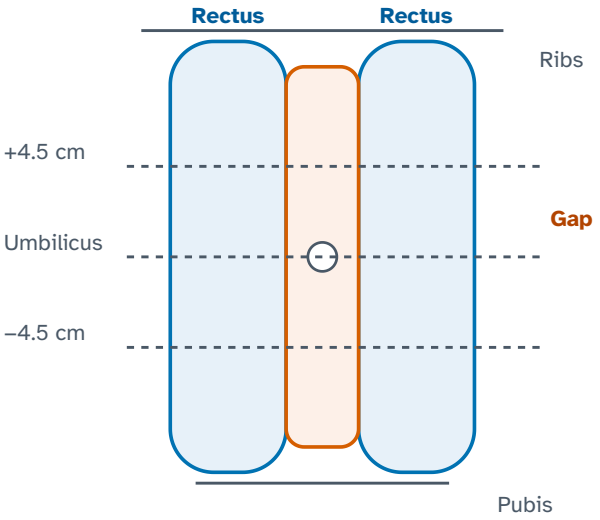
Risk factors and effects

- **Usually listed:** multiple or large babies, many pregnancies, weak abdominals, obesity, heavy lifting (the Sperstad study found no clear risk factor).
- **Effects:** abdominal **doming** or bulging, poor trunk support, possibly back and pelvic pain, cosmetic worry; rarely hernia.

Evaluation

- **Crook lying**; she lifts head and shoulders slowly until the shoulder blades just clear the bed.
- Fingers across the midline at **three levels**: at the umbilicus, **4.5 cm above** and **4.5 cm below**; count finger-widths between the muscle edges (or calipers, ultrasound). **Two finger-widths or more** at any level = diastasis.
- Note **depth and tension** of the linea alba (firm, or fingers sink) and doming.
- Record pain, back and pelvic symptoms, pelvic floor function and function (lifting, getting up).

Testing for diastasis recti: three levels



Two finger-widths or more at any level counts as diastasis (the definition Sperstad et al. used). Recti in blue, the widened linea alba in orange.

Management in pregnancy

- **Roll to the side** to get up from lying; avoid straining and breath-holding.
- **Transversus abdominis and pelvic floor** activation; good posture; avoid exercises that make the abdomen dome.

Management after birth

- **Transversus drawing-in with pelvic floor**; pelvic tilts; side-lying and four-point exercises.
- Progress to **head lifts and curl-ups only when she can keep the abdomen flat** (pre-activate transversus first); later, loaded functional training.
- **Splinting** the abdomen with crossed hands or a binder when coughing and getting up in the early days.
- Correct lifting; constipation care; weight management.

- **Surgical repair** only if a large gap with hernia or major symptoms persists after long rehabilitation.

Draw: Front view of the recti with the widened linea alba and fingers placed at +4.5 cm, umbilicus and –4.5 cm.

7 marks Guidelines for women exercising in pregnancy

Asked: W18 (2b)

Benefits

Less **gestational diabetes**, **pre-eclampsia** and excessive weight gain; better mood and sleep; less back pain; better fitness for labour and recovery. Activity is **not associated with miscarriage or preterm birth** (Canadian guideline, 2018).

Dose

- At least **150 minutes a week** of moderate activity over at least **3 days**, ideally daily; aerobic plus resistance (Canadian guideline; ACOG). Women active before pregnancy can continue their usual activity.
- **Intensity:** moderate by the **talk test** (can talk, not sing) or **Borg 12–14**, because **heart rate is altered** in pregnancy.
- **Good choices:** walking, stationary cycling, swimming and water exercise, low-impact aerobics, prenatal yoga and Pilates (not hot), light resistance, **daily pelvic floor training**.

Precautions

- Warm-up and cool-down; drink water; **avoid heat** and humidity.
- **Avoid lying flat for long after the first trimester** (side-lying or inclined).
- Avoid breath-holding and heavy straining; avoid end-range and bouncing stretches (lax ligaments); good footwear; eat enough.

Avoid (ACOG, 2020)

- Contact sports; high fall-risk activities; scuba and sky diving; hot yoga; unaccustomed exercise at high altitude.

Absolute contraindications

- Significant heart disease; restrictive lung disease; incompetent cervix or cerclage; multiple pregnancy at risk of preterm labour; persistent bleeding in 2nd or 3rd trimester; **placenta praevia after 26 weeks**; preterm labour; ruptured membranes; **pre-eclampsia** or pregnancy-induced hypertension; severe anaemia.

Stop exercising if

- **Vaginal bleeding**; regular painful contractions; leaking fluid; breathlessness before starting; dizziness; headache; chest pain; muscle weakness affecting balance; **calf pain or swelling**.

Screen first and work with the obstetrician; high-risk pregnancies need tailored or reduced activity.

7 marks Physiological changes in pregnancy

Asked: S20 (2f), W20 (2b) (asked 2 times)

Hormones

Progesterone, oestrogen and relaxin rise → **ligament laxity** and smooth muscle relaxation (veins, bowel, ureters).

Changes by system

System	Change	Meaning for PT
Heart and blood	Blood volume up **40–50%**; cardiac output up; resting HR up **10–20 beats**; BP dips mid-pregnancy; dilutional anaemia; clots more easily	HR a poor guide: use **talk test or Borg**; clot risk
Supine position	From about **20 weeks** uterus presses on the **inferior vena cava**	**Supine hypotension**: side-lying or wedge under right hip
Breathing	Diaphragm rises, rib cage widens; tidal volume and minute ventilation up; FRC falls	Mild breathlessness normal; breathing exercises; upright positions
Musculoskeletal	Weight gain; **centre of gravity forward**; posture changes; lax pelvic and spinal ligaments; abdominals stretch and may separate; pelvic floor loaded; feet flatten	Back and **pelvic girdle pain**, **diastasis recti**, leakage, balance changes, foot pain
Kidneys and bladder	Filtration rises; uterus presses on the bladder	Frequency, urgency, stress leakage
Gut	Slower stomach and bowel; relaxed lower oesophageal sphincter	Nausea, heartburn, constipation
Metabolism	BMR up; insulin resistance later; heat production up	Avoid overheating; gestational diabetes risk
Fluid and skin	Fluid retention; stretch marks	Ankle swelling, **carpal tunnel syndrome**, varicose veins

Weight gain (Institute of Medicine, 2009)

- Underweight **12.5–18 kg**; normal BMI **11.5–16 kg**; overweight **7–11.5 kg**; obese **5–9 kg**.

Draw: Posture in pregnancy: centre of gravity forward, changed lumbar curve, knees locked; side-lying rest with pillows under the abdomen and between the knees.

7 marks Aims and principles of obstetric physiotherapy

Asked: W21 (2d), W24 (2d) (asked 2 times)

Obstetric physiotherapy is the care of the woman through pregnancy, labour and the postnatal period, in antenatal and postnatal classes and individual treatment.

Aims

1. Help her **adapt to the physical changes** of pregnancy; prevent or treat discomforts (back and pelvic girdle pain, cramps, swelling).
2. Keep her **fit and active safely**; control weight gain.
3. **Protect** the back, pelvis, abdomen and **pelvic floor**.
4. **Prepare for labour:** knowledge, positions, **relaxation, breathing**, less fear.
5. Help **recovery after birth** and the return to fitness.
6. **Prevent long-term problems:** incontinence, prolapse, back pain, diastasis.

Principles

1. **Assess first**; screen for contraindications and red flags; work with the **obstetrician**.
2. **Moderate, regular exercise** with precautions (150 minutes a week; talk test or Borg 12–14; no long supine lying after the first trimester); **progress with the trimester**.

3. **Education and self-management:** posture, body mechanics, rest positions.
4. Include the **partner and family**; respect beliefs; **group classes** for support.
5. **Individualise:** high-risk pregnancies need tailored or reduced activity.

Delivery

- **Early bird class** (first trimester): changes, posture, safe exercise, pelvic floor, warning signs.
- **Antenatal classes** from about 28 weeks: labour preparation, relaxation and breathing.
- **Postnatal:** early pelvic floor and abdominal exercise, feeding posture, back care.

Know the stop signs: bleeding, painful contractions, leaking fluid, dizziness, chest pain, calf pain or swelling.

7 marks Antenatal programme: goals and structure of the class

Asked: S25 (2e)

Antenatal classes are group sessions usually from about **28 weeks**, **weekly for 6–8 sessions**, in groups of **8–12**, with the partner invited to some. An **early bird class** in the first trimester comes before them.

Goals

- **Knowledge** of pregnancy, labour and the baby.
- **Fitness and posture**; relief of discomforts.
- **Pelvic floor training**.
- **Relaxation, breathing and positions** for labour.
- Postnatal recovery and baby care.
- **Confidence and less fear**.

Structure of a session (about 90 minutes)

Part	Content
Welcome (10 min)	Welcome and questions
Talk (20 min)	One topic
Exercise (30 min)	Warm-up, low-impact aerobic, strengthening, **pelvic floor**, stretching
Relaxation and breathing (20 min)	Practice for labour
Summary (10 min)	Recap and handout

Topics across the course

1. Body changes and posture.
2. Exercise and discomforts.
3. Pelvic floor.
4. Signs of labour and when to go to hospital.
5. Stages of labour and pain relief.
6. Labour positions and breathing (partner session).
7. Caesarean section.
8. Postnatal care, feeding and baby handling.

Safety in class

- Screen each woman for **contraindications** and red flags first; work with the obstetrician.
- Moderate intensity by the **talk test or Borg 12–14**; no long supine lying; drink water; avoid heat.
- Teach the **stop signs**: bleeding, painful contractions, leaking fluid, dizziness, calf pain or swelling.

2(e) Labour and postnatal care

Full notes p. 52

3 marks Complications of caesarean section

Asked: S17 (1e)

Caesarean section is delivery of the baby through an abdominal and uterine incision, done as a planned (elective) operation, for example for **cephalopelvic disproportion**, or as an emergency.

- **Early:** bleeding; infection of the wound or uterus; **deep vein thrombosis** and pulmonary embolism; **chest infection and atelectasis** after anaesthesia; urinary retention or bladder injury; paralytic ileus and wind pain; spinal headache.
- **Late:** wound breakdown; **incisional hernia**; adhesions and chronic pain; delayed mobility and breastfeeding.
- **Later pregnancies: scar rupture** and abnormal placenta.
- **Baby:** fast breathing for a short time after birth.

Physiotherapy prevents many of these: deep breathing, **supported huffing** with a pillow over the wound, ankle pumps and early walking.

3 marks Positions in labour

Asked: W19 (1e)

Upright and mobile positions help labour progress, ease pain and reduce fear.

- **First stage:** walking; standing and **leaning forward** on the partner or bed; sitting on a **birthing ball**; kneeling forward over a pillow or ball; **all-fours** (eases backache when the baby faces forward); side-lying for rest or with an epidural.
- Upright and mobile positions shortened the first stage by about **1 hour 22 minutes** and reduced caesarean and epidural use (Lawrence et al., 2013).
- **Second stage:** the position **she chooses**: kneeling, squatting with support, all-fours, side-lying.
- **Avoid lying flat on the back:** it presses on the great vessels and narrows the pelvic outlet (WHO, 2018).

3 marks Ergonomic advice for a lactating mother

Asked: S21 (1d)

Feeding takes hours a day, so a slouched posture loads the neck and back.

- **Bring the baby to the breast**, not the breast to the baby: a **pillow on the lap**.
- Chair with back support (cushion behind the low back), **feet flat or on a low stool**, shoulders relaxed, arm supported.
- Avoid long neck bending; stretch neck and upper back after feeds.
- Change holds: cradle, cross-cradle, **football hold** (after caesarean), **side-lying** (night, sore perineum).
- Keep **wrists straight**; support the head with the forearm.
- Change nappies at **waist height**; lift the baby close with knees bent; do not carry on one hip for long.

3 marks Postpartum blues

Asked: S22 (1f)

Postpartum (baby) blues: short-lived tearfulness, mood swings, irritability, anxiety and poor sleep in about **half or more** of new mothers, starting **2–3 days** after birth and settling within **2 weeks**.

- **Cause:** sudden fall in hormones, sleep loss and adjustment to the baby.
- **Management:** reassurance that it is common; rest and sleep; family help with the baby; talking about feelings; gentle walks and relaxation.
- **Refer:** symptoms beyond 2 weeks or severe suggest **postnatal depression**: screen with the **Edinburgh Postnatal Depression Scale**. Confusion or hallucinations mean **postpartum psychosis**, an emergency.

3 marks Breast engorgement and its management

Asked: S24 (1a)

Breast engorgement: painful, swollen, hard breasts, usually **2–5 days** after birth as milk comes in, from **milk stasis** plus swelling of the breast tissue.

- **Frequent, effective feeding** on demand (**8–12 times a day**) with a correct latch.
- **Hand-express** a little to soften the areola before latching.
- **Cold packs** between feeds for swelling.
- Gentle **lymphatic drainage strokes towards the armpit**; avoid deep or forceful massage.
- Supportive, non-tight bra; pain relief as prescribed.
- **Refer** if fever or a red, hot area appears (**mastitis**).

3 marks Three goals of postnatal care

Asked: W25 (1a)

Puerperium: the **6 weeks** after delivery while the body returns towards its non-pregnant state.

1. **Prevent complications:** chest problems after anaesthesia, **deep vein thrombosis**, urinary retention, constipation (deep breathing, ankle pumps, early walking).
2. **Restore the pelvic floor and abdominal wall:** pelvic floor contractions **from the first day**; transversus drawing-in; check for **diastasis recti**.
3. **Teach posture and body mechanics** for feeding, lifting and carrying the baby, and relieve pain of the perineum, back, wound and breasts.

Also: support emotional well-being and rest; gradual return to fitness and work with a lifelong pelvic floor habit.

7 marks Physiotherapy management of pain during labour

Asked: S17 (2b)

Labour pain comes with contractions that open the cervix in the first stage and with stretching of the pelvic floor and perineum in the second. Pain and anxiety cause **fast breathing and over-breathing** and raise cardiac output and BP.

Aims: reduce fear and pain; help her stay in control and mobile; help labour progress; save energy; keep the baby well oxygenated.

Methods

1. **Antenatal education:** what happens in labour; positions, breathing and relaxation practised beforehand; partner's role. Less fear means less pain.
2. **Positions and mobility:** walking, leaning forward on the partner or bed, **birthing ball**, kneeling forward, **all-fours** (eases backache when the baby faces forward), side-lying for rest.
3. **Breathing:** slow, relaxed breathing in early labour; lighter, slightly faster at the peak; **"pant-pant-blow"** in transition; avoid over-breathing.
4. **Relaxation: Mitchell's physiological relaxation** or progressive relaxation; music.
5. **Massage and warmth:** firm **sacral pressure** or massage during contractions; warm packs on the back or lower abdomen.
6. **TENS:** two pairs of electrodes on the back, one either side of the spine at **T10–L1** and one at **S2–S4**; low background current with a **boost button** during contractions; start early; not in water.
7. **Warm water:** shower or bath in the first stage.
8. **Continuous support** by a partner or companion: reduces fear and the need for drugs.
9. **With medical analgesia** (epidural, Entonox, opioids given by doctors): adjust positions to side-lying or supported sitting.

Evidence: WHO (2018) recommends relaxation and massage for women who want them. TENS changes pain scores little, but most women would use it again and no harms were reported.

> **Draw:** Back view with TENS electrode pairs at T10–L1 and S2–S4 on either side of the spine.

7 marks Role of breathing exercises in labour

Asked: S19 (2a)

In labour **oxygen use rises**, and pain causes **fast breathing and over-breathing** (tingling, dizziness). Breathing patterns are taught in **antenatal classes** with relaxation, so the woman can use them in labour.

Why breathing helps

- **Focuses attention** away from pain and keeps her in control.
- Aids **relaxation** and saves energy.
- Keeps the **oxygen supply** to mother and baby.
- Prevents exhausting **breath-holding** and stops her pushing too early.

Breathing by stage

Stage	Breathing
Early first stage	**Slow, relaxed breathing**: a sigh out at the start, slow breaths in through the nose and out through the mouth, a sigh at the end
Strong contractions	**Lighter, slightly faster** breathing at the peak, back to slow breathing as it fades
Transition	Urge to push before full dilatation: **"pant-pant-blow"** to avoid pushing too early
Second stage	Push with the **natural urge**, breathing out or with short breath-holds, not long forced breath-holding
Crowning	**Pant** as the head crowns so the perineum stretches slowly

Stage	Breathing
After birth	Deep breathing with ankle pumps and pelvic floor contractions

Precautions

- **Avoid over-breathing:** if she feels tingling or dizziness, slow down and breathe into **cupped hands**.
- Practise before labour with the partner, who coaches during contractions.

Combine breathing with upright positions, relaxation and sacral massage for the best effect.

7 marks Physiotherapy during labour

Asked: S23 (2a)

Labour is the process by which contractions open the cervix and the baby is born, followed by the placenta. Most of the physiotherapy is taught in **antenatal classes** (what happens, positions, breathing, relaxation, partner's role) and used on the day.

Aims: reduce fear and pain; keep her in control and mobile; help labour progress; save energy; **protect the perineum**; keep the baby well oxygenated.

Stages of labour and what the physiotherapist does

First stage
Contractions open the cervix to 10 cm (active from 5 cm)
PT: walk, upright, breathing, relaxation, massage, TENS

↓

Second stage
Full dilatation to the birth of the baby
PT: position of her choice, push with the urge

↓

Third stage
Birth of the baby to delivery of the placenta

↓

Fourth stage and puerperium
First hours, then 6 weeks of recovery
PT: breathing, ankle pumps, pelvic floor, feeding posture

The active first stage usually lasts up to 12 hours in a first labour and up to 10 hours in later labours (WHO, 2018).

First stage (cervix opens to 10 cm; active phase from 5 cm)

- **Upright, mobile positions:** walking, leaning forward, birthing ball, kneeling forward, all-fours; side-lying for rest or with an epidural. They shortened the first stage by about **1 hour 22 minutes**.
- **Breathing:** slow and relaxed, lighter at the peak; avoid over-breathing (cupped hands if tingling).
- **Relaxation** (Mitchell's), **sacral massage**, warm packs, warm shower.
- **TENS:** electrodes at **T10–L1** and **S2–S4**, boost button during contractions; not in water.

Transition
Urge to push before full dilatation: **"pant-pant-blow"**.

Second stage (full dilatation to birth)

- Position of **her choice** (kneeling, supported squatting, all-fours, side-lying); **avoid lying flat on the back**.
- Push with the **natural urge**, breathing out or short breath-holds; **pant as the head crowns** so the perineum stretches slowly.

Fourth stage and puerperium
Deep breathing, **ankle pumps**, gentle **pelvic floor contractions**, feeding posture.

Continuous support by a partner or companion reduces fear and the need for drugs; with an epidural, adjust positions to side-lying or supported sitting.

2(d) Menopause and osteoporosis

Full notes p. 63

3 marks Prevention of osteoporosis with ageing

Asked: W18 (1c)

Osteoporosis (WHO, 1994): a systemic skeletal disease with **low bone mass** and **microarchitectural deterioration** of bone, so bones are fragile and fracture easily.

- **Build peak bone mass** in childhood and youth: sport, jumping, running, enough calcium and protein.
- **Exercise throughout life:** weight-bearing and impact activity, **progressive resistance training**, balance training.
- **Nutrition:** calcium from milk, curd, **ragi**, sesame, green leafy vegetables; **vitamin D** from sunlight (supplements if deficient); enough protein.
- **Lifestyle:** stop smoking; limit alcohol and caffeine; keep a healthy weight.
- **Prevent falls:** balance training, home safety, vision and drug review.
- **Screen and treat:** DEXA for women at risk; medicines (bisphosphonates) when fracture risk is high.

3 marks Risk (predisposing) factors for osteoporosis

Asked: W22 (1c), S26 (1f) (asked 2 times)

Osteoporosis: low bone mass and microarchitectural deterioration of bone with increased fragility and fracture risk (WHO).

Group	Factors
Cannot change	**Female sex**, **older age**, **early menopause**, family history of hip fracture, small thin build, **previous fragility fracture**, Asian or white ethnicity
Can change	**Physical inactivity**, low body weight, low **calcium and vitamin D** (little sunlight), smoking, alcohol, excess caffeine and salt
Diseases and drugs	Long-term **steroids**, rheumatoid arthritis, thyroid or parathyroid excess, diabetes, malabsorption, kidney disease, **immobility**

Physiotherapy targets the changeable factors: resistance, impact and balance exercise, sunlight and diet advice.

3 marks Types of osteoporosis

Asked: S23 (1b)

Osteoporosis: low bone mass with microarchitectural deterioration; diagnosed by DEXA **T-score –2.5 or below**.

Type	Features
Primary type I (postmenopausal)	Women about **50–70**; oestrogen loss; mainly **trabecular** bone; **vertebral crush** and **distal radius (Colles')** fractures
Primary type II (senile)	Men and women over about **70**; trabecular and cortical bone; **hip** and vertebral fractures
Idiopathic (juvenile)	Children and young adults, no known cause; rare
Secondary	Steroids, thyroid and parathyroid excess, rheumatoid arthritis, malabsorption, kidney or liver disease, alcohol, long immobilisation, drugs (anticonvulsants, heparin)
Regional	Disuse after paralysis or a plaster; complex regional pain syndrome

7 marks Importance of physiotherapy in postmenopausal women

Asked: S17 (2c), S22 (2b) (asked 2 times)

Menopause is the permanent end of menstruation from loss of ovarian follicle activity, diagnosed after **12 months without a period**; Indian women reach it at about **46 years**. **Postmenopause** is the years after it.

Changes that make physiotherapy necessary

- **Oestrogen and progesterone fall**; inhibin falls, so **FSH rises markedly** and LH rises.
- Rapid **bone loss** in the first years (osteoporosis, fractures of wrist, spine, hip); loss of **muscle mass and strength**.
- Weaker **pelvic floor** and tissues (stress incontinence, urgency, prolapse); worse cholesterol, abdominal fat, higher BP and heart risk; hot flushes, poor sleep, mood swings.

Role of physiotherapy

Problem	Physiotherapy
Bone loss	**Progressive resistance** and **impact (weight-bearing)** exercise 2–3 days a week; posture and back extensor training
Falls and fractures	**Balance and strength training** (tandem and single-leg stance, tai chi); home safety; footwear
Pelvic floor weakness	**Pelvic floor muscle training**, bladder training, continence advice
Weight, heart and diabetes risk	**Aerobic exercise 150 minutes a week**, resistance training, weight management
Joint and back pain	Exercise, posture correction, pain relief, ergonomic advice
Hot flushes, sleep, mood	Regular exercise, relaxation and paced breathing, yoga, sleep routine; layered clothing, cool rooms
Knowledge	Education on bone health, **calcium and vitamin D**, sunlight, no smoking or alcohol, screening

Precautions

- Screen for osteoporosis and fracture: red flags are **sudden severe back pain, height loss**, night pain; refer for DEXA and X-ray.
- **Spine-sparing movement:** avoid repeated or loaded forward bending and twisting (sit-ups, toe touches); teach the **hip hinge** and log-roll out of bed. Impact work only when fracture is excluded and joints allow.

Aerobic exercise alone is not enough for bone: the programme must include resistance and balance training (Too Fit to Fracture, 2014).

7 marks Physiological variations associated with menopause

Asked: S23 (2e)

Menopause: the permanent end of menstruation from loss of ovarian follicle activity, diagnosed after **12 months without a period**. Average age in Indian studies is about **46 years** (about 51 in Western women).

Stages

- **Perimenopause:** years of irregular cycles and symptoms before, and the first year after, the last period.
- **Postmenopause:** the years after. **Premature menopause:** before 40. **Surgical menopause:** after removal of both ovaries (sudden symptoms).

Hormonal changes

Oestrogen (oestradiol) and progesterone fall; inhibin falls, so **FSH rises markedly** and LH rises; androgens become relatively higher.

Effects of low oestrogen by system

System	Change	Physiotherapy
Vasomotor	**Hot flushes**, night sweats, palpitations, poor sleep	Regular exercise, paced breathing, cool rooms
Mind	Mood swings, irritability, anxiety, low mood, poor concentration	Aerobic exercise, yoga, relaxation
Bone	**Rapid bone loss** in the first years; osteoporosis; wrist, spine, hip fractures	Resistance and impact exercise
Muscles and joints	Loss of muscle mass and strength; joint aches, stiffness	Strengthening, posture work
Genitourinary	Vaginal dryness, urgency, frequency, infections, **stress incontinence**, **prolapse**	Pelvic floor muscle training, bladder training
Heart and metabolism	Worse cholesterol, abdominal fat, insulin resistance, higher BP and heart risk	Aerobic exercise 150 minutes a week
Skin and others	Thin dry skin, hair changes, weight gain	Weight management

Medical treatment (hormone therapy, vaginal oestrogen, bone-protecting drugs) is the doctor's decision; exercise and education are the physiotherapist's part.

2(f) Urogenital dysfunction: incontinence, prolapse and the pelvic floor

Full notes p. 70

3 marks Muscles of the pelvic floor

Asked: S16 (1e)

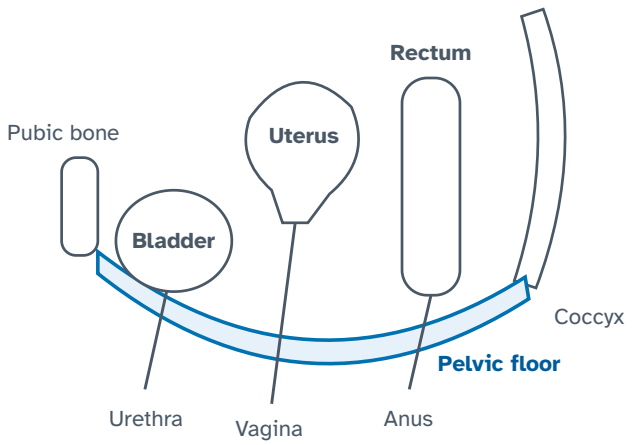
The **pelvic floor** is a muscular sling from the pubis to the coccyx, pierced by the urethra, vagina and anal canal.

Layer	Muscles
Pelvic diaphragm (deep)	**Levator ani**: pubococcygeus (with **puborectalis**), iliococcygeus; and **coccygeus**
Deep perineal pouch	Deep transverse perineal, **external urethral sphincter**
Superficial perineal pouch	Bulbospongiosus, ischiocavernosus, superficial transverse perineal
Others	**External anal sphincter**; the **perineal body** is their central tendon

Nerve: pudendal (S2–S4) and direct S3–S4 branches to levator ani. Mostly **slow-twitch** fibres for support, with fast-twitch fibres for coughs and lifts.

Functions: support pelvic organs; continence; relaxation for voiding; sexual function; spinal and pelvic stability with the diaphragm, transversus and multifidus.

The pelvic floor as a sling, side view



The muscles (blue) form a sling from pubis to coccyx. Three tubes pass through it, so one muscle controls the bladder, the vagina and the bowel.

3 marks Continence-promoting advice

Asked: S18 (1f)

Simple lifestyle advice that reduces load on the bladder and pelvic floor; given with pelvic floor training.

- **Drink normally**, about **1.5–2 litres a day**; cutting fluids concentrates urine and irritates the bladder.
- Reduce **caffeine**, fizzy drinks and alcohol.
- Avoid **constipation and straining**: fibre, fluids, a footstool on the toilet.
- **Lose weight** if overweight (BMI over 30); stop smoking; treat chronic cough.
- Sit fully on the toilet, relax and empty completely; do not hover; no "just in case" visits.
- Do **the knack** before lifting, coughing or sneezing; lift with the load close.
- Keep up **pelvic floor exercises for life**, especially in pregnancy, after birth and around menopause.

3 marks Interferential therapy in stress urinary incontinence

Asked: W18 (1d)

Principle: two **medium-frequency currents** cross in the pelvis and produce a low-frequency **beat** deep in the pelvic floor, with little skin discomfort. It contracts the muscle **without a vaginal probe**, aiming to restore reflex activity and urethral closure.

- **Electrodes:** bipolar, under the **ischial tuberosities** and over the **anterior perineum** (Laycock and Green); or four electrodes, two suprapubic and two by the ischial tuberosities, crosswise.
- **Trial settings:** carrier about **4000 Hz**; beat **0–100 Hz**; **15 minutes**, **3 times a week** for 15 sessions.
- **Evidence:** small trials only; NICE advises it only if she **cannot contract** the pelvic floor, always **with supervised training**.
- **Avoid in:** pregnancy, infection, bleeding, pelvic cancer, pacemaker.

3 marks Deferment techniques for urge incontinence

Asked: W19 (1f)

Urgency incontinence: leakage with or just after a sudden urge (overactive detrusor). Deferment helps the woman hold on until she reaches the toilet.

- **Stop** and stand still or **sit down**; pressure on the perineum helps.
- Do several **quick strong pelvic floor contractions**.
- **Breathe slowly** and relax.
- **Distract the mind** (count backwards); curl the toes or cross the legs.
- When the urge fades, **walk calmly** to the toilet.

Used within **bladder training** (at least 6 weeks): lengthen the gap between voids by about **15 minutes** at a time towards **3–4 hours**; cut caffeine.

3 marks PERFECT method of pelvic floor muscle grading

Asked: W20 (1b)

Scheme by **Laycock and Jerwood (2001)**, assessed by digital vaginal examination with consent.

Letter	What is measured
P: Power	Maximal contraction on the **Modified Oxford scale 0–5** (0 none, 1 flicker, 2 weak, 3 moderate with lift, 4 good, 5 strong)
E: Endurance	Seconds a maximal contraction is held (up to **10**)
R: Repetitions	Number of those holds repeated (up to **10**)

Letter	What is measured
F: Fast	Number of quick **1-second** contractions (up to **10**)
ECT	**Every Contraction Timed**: results become her exercise programme

P3 E5 R4 F6: start with 5-second holds × 4, then 6 quick contractions.

3 marks Define uterine prolapse and its types

Asked: W21 (1d)

Pelvic organ prolapse: descent of the front wall of the vagina, the back wall, the **uterus (cervix)** or the vault after hysterectomy. Uterine prolapse is descent of the uterus.

Compartment	Type
Anterior	**Cystocele**, urethrocele, cystourethrocele
Middle	**Uterine (uterovaginal)**; vault prolapse
Posterior	**Rectocele**, enterocele

Degrees: first, cervix descends within the vagina; **second**, cervix comes out through the vaginal opening on straining; **third (procidentia)**, whole uterus outside.

3 marks Indications for hysterectomy

Asked: S23 (1f)

Hysterectomy: surgical removal of the uterus; total, subtotal (cervix left) or radical (Wertheim's); abdominal, vaginal or laparoscopic.

- **Fibroids** with heavy bleeding or pressure (the commonest).
- **Abnormal uterine bleeding** not controlled by drugs.
- **Adenomyosis** and **endometriosis**.
- **Uterovaginal prolapse**.
- **Cancer** of the uterus, cervix or ovary.
- Chronic pelvic pain from **pelvic inflammatory disease**.
- Life-threatening **bleeding after childbirth**.

PT after surgery: breathing and supported coughing, leg exercises, early walking, gentle pelvic floor work, no heavy lifting for about 6 weeks.

3 marks Kegel's exercises

Asked: S24 (1f)

Pelvic floor muscle training described by the gynaecologist **Arnold Kegel (1948)**; supervised training for **at least 3 months** is first-line for stress and mixed incontinence (NICE).

- **Teach:** "**squeeze and lift** as if stopping wind and the flow of urine", without holding the breath or squeezing the buttocks. Stopping the urine stream is only for finding the muscle once.
- **Dose:** slow holds up to **10 seconds × 10** and up to **10 quick** 1-second contractions; at least **8 contractions 3 times a day**, set from PERFECT.
- **Progress:** lying → sitting → standing → lifting, stairs, jumping.
- **The knack:** contract just before and during a cough, sneeze or lift.

Uses: stress, urge and mixed incontinence; prolapse; pregnancy and after birth.

3 marks Tests for assessing incontinence in women

Asked: S25 (1b)

After a history of when, how much and how often she leaks, the physiotherapist uses:

- **Bladder diary** (frequency–volume chart) for at least **3 days**.
- **ICIQ-UI short form**: severity and bother.
- **Pad test:** in the **1-hour test** a gain of **1 g or more** is positive.
- **Cough stress test:** leakage at the moment of a cough with a full bladder.
- **Q-tip test:** swab swings **more than 30°** on straining in urethral hypermobility.
- **Digital vaginal examination: PERFECT** and Modified Oxford scale; perineometer or EMG.
- **Medical:** urine test, post-void residual by ultrasound, urodynamics before surgery.

7 marks Types of stress incontinence and their physiotherapy

Asked: S19 (2f)

Stress urinary incontinence (SUI) (ICS): involuntary leakage on effort or exertion, or on sneezing or coughing. **Causes:** weak pelvic floor and urethral support after **childbirth**, menopause, chronic cough, obesity, heavy lifting.

Types by severity (Stamey grades)

- **Grade 1:** leakage with coughing, sneezing or lifting.
- **Grade 2:** with smaller rises in pressure: walking, standing up, climbing stairs.
- **Grade 3:** leakage regardless of activity or position.

Types by mechanism

- **Urethral hypermobility:** bladder neck and urethra drop on effort because support is weak (Q-tip swings more than 30°).
- **Intrinsic sphincter deficiency:** urethral closure itself is weak (after surgery, severe atrophy).

Physiotherapy

- **Assess:** bladder diary, pad test, cough stress test, **PERFECT**.
- **Pelvic floor muscle training**, supervised for **at least 3 months**: first-line for all types. Slow holds up to **10 seconds × 10**, up to **10 fast** contractions; at least **8 contractions 3 times a day**; lying → sitting → standing → during lifting and stairs.
- **The knack:** strong contraction just before and during a cough, sneeze or lift (key for grade 1).
- **Biofeedback** (perineometer, EMG) or **electrical stimulation** (about **50 Hz**) for a very weak or unaware muscle; **vaginal cones**.
- **Lifestyle:** weight loss, treat cough and constipation, safe lifting.
- **Grade 3 or sphincter deficiency:** refer to the urogynaecologist (surgery, bulking agents), with PT before and after.

Trained women were about **eight times** more likely to report cure (Dumoulin et al., 2018). Review with diary, pad test and PERFECT every 4–6 weeks.

7 marks Physiotherapy management of uterine prolapse

Asked: S20 (2d)

Uterine prolapse: descent of the uterus (cervix) into or out of the vagina. **Causes:** childbirth (many births, long labour, big baby, instruments, pushing before full dilatation), loss of oestrogen after menopause, chronic rise in abdominal pressure (cough, constipation, heavy lifting, obesity), ageing.

Symptoms: something coming down; **dragging or heaviness worse by evening**; low backache; leakage or incomplete emptying; difficulty emptying the bowel.

Assessment

Symptoms, obstetric and bowel history, menopause; inspection on straining; **PERFECT**; bladder diary.

Physiotherapy

1. **Pelvic floor muscle training**, individually taught and supervised, for **at least 16 weeks**: first option for symptomatic prolapse (NICE); reduced symptoms in the **POPPY trial**.
2. **Dose** from PERFECT: slow holds up to 10 seconds × 10, up to 10 fast; at least 8 contractions 3 times a day; lying → sitting → standing → functional tasks.
3. **The knack** before coughing, sneezing, lifting.
4. **Constipation:** fibre, fluids; footstool with knees above hips, lean forward, breathe out, do not strain.
5. **Lifestyle:** weight loss; treat chronic cough; safe lifting with the load close.
6. **Modify straining activities:** heavy lifting, breath-held sit-ups, high-impact jumping.
7. **Pessary** (fitted by a doctor) for women unfit for or not wanting surgery; PT continues.
8. **Surgery** (vaginal hysterectomy with repair, colporrhaphy) for large prolapse: before and after, breathing and supported coughing, leg exercises, early walking, gentle pelvic floor and deep abdominal work, roll to the side, **no heavy lifting for about 6 weeks**.

Training improved prolapse by one POP-Q stage in **19%** of women against **8%** of controls (Brækken et al., 2010).

7 marks Types and degrees of urogenital (uterine) prolapse and their management

Asked: S21 (2c), W22 (2a) (asked 2 times)

Pelvic organ prolapse: descent of one or more of the front wall of the vagina, the back wall, the uterus (cervix) or the vault after hysterectomy.

Types (by compartment)

- **Anterior:** cystocele (bladder), urethrocele, cystourethrocele.
- **Middle (apical): uterine (uterovaginal)**; vault prolapse after hysterectomy.
- **Posterior:** rectocele (rectum), enterocele (small bowel in the pouch of Douglas).

Degrees (grades) of uterine prolapse

Degree	What is seen	POP-Q
First	Cervix descends but stays **inside the vagina**	Stage I: more than 1 cm above the hymen
Second	Cervix comes **out through the vaginal opening on straining**	Stage II: within 1 cm of the hymen
Third (procidentia)	**Whole uterus outside** the vaginal opening	Stage III: more than 1 cm below; IV: complete eversion

Causes: childbirth, menopause, chronic raised abdominal pressure, ageing. **Symptoms:** dragging, heaviness by evening, backache, bladder and bowel problems.

Management

- **First and second degree (stage I–II):** supervised **pelvic floor muscle training for at least 16 weeks** (NICE; POPPY trial); at least 8 contractions 3 times a day from the PERFECT result; the knack.
- **Lifestyle:** weight loss, treat constipation and cough, safe lifting; modify straining (heavy lifting, breath-held sit-ups, jumping).
- **Pessary** by the gynaecologist if surgery is unsuitable or not wanted.
- **Large prolapse:** surgery (vaginal hysterectomy with repair, colporrhaphy) with PT before and after: breathing, leg exercises, early walking, gentle pelvic floor work, no heavy lifting for about 6 weeks.

Draw: Degrees of uterine prolapse: three small side views with the cervix inside the vagina, at the opening, and the uterus outside.

7 marks Types of uterine prolapse and the effects of exercise

Asked: W25 (2e)

Pelvic organ prolapse: descent of the vaginal walls, the **uterus (cervix)** or the vault after hysterectomy, usually after childbirth, menopause or a chronic rise in abdominal pressure.

Types

- **By compartment:** anterior (cystocele, urethrocele); middle (**uterovaginal**, vault); posterior (rectocele, enterocele).
- **By degree: first**, cervix descends within the vagina; **second**, cervix comes out through the opening on straining; **third (procidentia)**, whole uterus outside (POP-Q stages I–IV).

How exercise helps (mechanisms)

1. **Timing (the knack):** contracting before a cough, lift or strain narrows the **levator hiatus** and lifts the floor, so the organs are not pushed down.
2. **Strength training:** months of near-maximal contractions make the muscle **thicker and stiffer**, so the floor sits higher at rest. On ultrasound, trained women had a smaller hiatus and a **higher bladder and rectum** (Brækken 2010).

Effects shown in trials

- **Stage improves:** one POP-Q stage better in **19%** of trained women vs **8%** of controls (Brækken 2010).
- **Fewer symptoms:** less dragging and heaviness at 12 months in the **POPPY trial** (447 women, 5 physiotherapy visits over 16 weeks).
- **Indian trial:** one-to-one training beat a self-help manual for symptoms and quality of life (Kashyap 2013, Chandigarh).
- **Reviews:** first-line for stage I–III, with no serious side effects; needs supervision (Bø 2022). Long-term benefit is uncertain, and training added around surgery did not improve the result.

- **Also:** less leakage and better support of the spine and pelvis.

Prescription
Supervised training for at least **16 weeks** for stage 1–2 (NICE): **3 sets of 8–12** near-maximal contractions **3–4 times a week**, then maintenance; the knack; weight loss, treat constipation and cough. Modify, do not stop, straining activities (heavy lifting, breath-held sit-ups, jumping).

Large prolapse needs a pessary or surgery; exercise continues alongside.

Part 3: Geriatrics

3(a–c) Senior citizens in India: policies, NGOs, living arrangements and communication

Full notes p. 85

3 marks Policies that facilitate an elderly person

Asked: W18 (1e)

A **senior citizen** in India is a person aged **60 or over**.

- **National Policy on Older Persons (1999):** financial and food security, health care, shelter, welfare, protection from abuse.
- **Maintenance and Welfare of Parents and Senior Citizens Act (2007):** children must maintain parents; tribunals (up to **₹10,000 a month**); abandonment punishable.
- **NPHCE (2010):** geriatric clinics and wards; **rehabilitation and physiotherapy at CHCs**; home-based care.
- **AB PM-JAY for 70 and over (2024): ₹5 lakh a year** hospital cover.
- **IGNOAPS** pension for elders below the poverty line; **Rashtriya Vayoshri Yojana** free assistive devices; Elderline **14567**.

3 marks Role of NGOs in elderly care in India

Asked: W19 (1b)

Examples: HelpAge India, Dignity Foundation, Agewell Foundation, Nightingales Medical Trust, Silver Innings, local religious and community trusts.

- **Health:** mobile medical units and camps (eye, hearing, **physiotherapy**).
- **Care:** old age homes, **day care centres**, senior citizens' clubs.
- **Protection:** helplines; protection from **abuse**.
- **Money:** livelihood and pension help; sponsorship of destitute elders.
- **Advocacy:** awareness and policy; **caregiver training**; disaster relief.
- **Linking** elders to government schemes (NGOs get grants under Atal Vayo Abhyuday Yojana).

7 marks National policies that facilitate the elderly

Asked: S18 (2f)

A **senior citizen** in India is a person aged **60 or over** (Maintenance and Welfare of Parents and Senior Citizens Act, 2007). India had **104 million** elderly in the 2011 Census (**8.6%**), projected to reach about **227 million (14.9%) by 2036**. Most live in villages, many are widows and poor, and most worked without pensions, so state support is essential.

Policy or scheme	Main points
National Policy on Older Persons (1999)	State support for **financial and food security**, health care and nutrition, shelter, education, welfare, and **protection** against abuse and exploitation
Maintenance and Welfare of Parents and Senior Citizens Act (2007)	Children and heirs **must maintain** parents; **maintenance tribunals** (allowance up to **₹10,000 a month**); old age homes for indigent elders in districts; medical care; **abandoning** a senior citizen is punishable
National Programme for Health Care of the Elderly (NPHCE, 2010)	Geriatric clinics and wards at district hospitals; **rehabilitation units and physiotherapy at CHCs**; home-based care through PHCs and sub-centres; regional geriatric centres and national centres of ageing
Ayushman Arogya Mandirs	Elderly and palliative care in the primary health care package
AB PM-JAY for 70 and over (2024)	**₹5 lakh a year** hospital cover for every person aged 70 and over, whatever the income (**Ayushman Vay Vandana card**)
Indira Gandhi National Old Age Pension Scheme	Monthly **pension** for elderly below the poverty line, topped up by states
Rashtriya Vayoshri Yojana (2017)	**Free assistive devices** (sticks, walkers, wheelchairs, hearing aids, spectacles, dentures) for poor elders, through camps
Atal Vayo Abhyuday Yojana	Grants to NGOs for old age homes, day care centres, mobile medical units; **Elderline 14567**
Other benefits	Higher income-tax exemption, savings schemes with higher interest, priority counters, reserved bus seats

Role of the physiotherapist

- Rehabilitation and physiotherapy in **NPHCE units** at CHCs and district hospitals; home-based care.
- Assess and prescribe aids at **Vayoshri camps**; link elders to pensions, PM-JAY and NGOs.

Start with the definition and numbers, then write the policy, the Act and NPHCE first: these carry the marks.

7 marks Communication barriers in the elderly (geriatric population)

Asked: S20 (2c), W21 (2b), W24 (2b) (asked 3 times)

Communication barriers are factors in the older person, the professional or the setting that stop a message being given or understood. Good communication matters for **safety**, adherence to the exercise programme and **dignity**.

Barrier	Examples	What to do
Hearing	**Presbycusis** (high tones lost first), difficulty in noise	Face the person; quiet room; speak slowly, clearly, in a **lower pitch**; do not shout; hearing aid in and working

Barrier	Examples	What to do
Vision	Poor near vision, cataract, glare	Good light without glare; glasses on; **large-print, high-contrast** handouts
Cognition	Slow processing, poor memory, dementia, delirium	Short sentences, **one instruction at a time**, repeat, demonstrate, picture reminders, involve family
Speech and language	Aphasia or dysarthria after stroke, loose dentures	Yes/no questions, gestures, picture boards, patience
Emotional	Depression, anxiety, fear, embarrassment	Build trust, listen, respect, privacy
Physical	Pain, fatigue, breathlessness	Short sessions, comfortable position
Cultural and educational	Different language, **low literacy**, beliefs about illness	Local language, simple words, familiar examples
Professional	Jargon, hurry, talking to the relative instead of the patient, **"elderspeak"**, ageist attitudes	Address the person directly; allow time; **teach-back** (ask them to repeat in their own words)
Environment	Noise, poor lighting, interruptions	Quiet, well-lit room; sit at the **same level**

When explaining a home programme, demonstrate it, give a large-print handout and check with teach-back.

3(d) Theories of ageing

Full notes p. 92

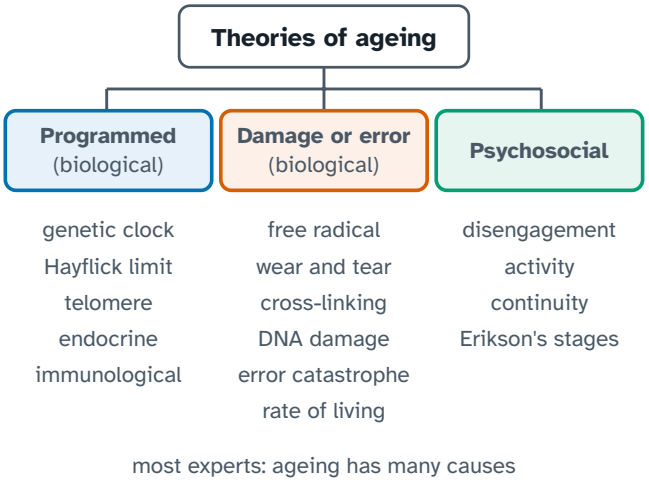
3 marks Enumerate (three) theories of ageing

Asked: S22 (1c), W22 (1b) (asked 2 times)

Theories fall into three groups: **programmed** (genetic clock, Hayflick limit, telomere, endocrine, immunological), **damage** (free radical, wear and tear, cross-linking, DNA damage, error catastrophe, rate of living) and **psychosocial** (disengagement, activity, continuity, Erikson).

1. **Hayflick limit / telomere** (programmed): cells divide about **40–60** times and stop as **telomeres shorten**.
2. **Free radical** (damage; Harman, 1956): **reactive oxygen species** damage DNA, proteins and membranes.
3. **Activity** (psychosocial; Havighurst): staying **active and involved** brings successful ageing.

Theories of ageing: the classification tree



Programmed theories say ageing follows a built-in plan; damage theories say it is the sum of accumulated harm; psychosocial theories explain how people adapt to growing old.

7 marks Define ageing; mention types of ageing; enumerate genetic theories of ageing

Asked: W16 (2a)

Definition

Ageing is the progressive, time-related decline in the structure and function of the body that **reduces the ability to adapt to stress** and raises the risk of disease and death. It is **universal, progressive, intrinsic** (from within) and **harmful**.

Types of ageing

Type	Meaning
Chronological	Age in years since birth
Biological (physiological)	Age of the body's systems; two people of 70 may differ greatly
Psychological	Changes in memory, learning, personality, coping
Social	Changing roles: retirement, grandparenthood, widowhood
Functional	What the person can do compared with others of the same age
Primary and secondary	**Primary**: changes due to age itself. **Secondary**: changes added by disease, inactivity, lifestyle

Genetic (programmed) theories

1. **Genetic clock:** lifespan is set by genes that switch on and off in sequence; each species has a typical maximum lifespan; progeria shows early ageing from one gene.
2. **Hayflick limit (cellular senescence):** normal cells divide a limited number of times, then stop; human fibroblasts stop after about **40–60 divisions** (Hayflick and Moorhead, 1961).
3. **Telomere theory:** telomeres shorten at each division; when too short, the cell stops dividing; **telomerase** restores them in germ and cancer cells.
4. **Endocrine (neuroendocrine):** a clock in the **hypothalamus and pituitary** drives ageing through hormones (fall in growth hormone, sex hormones, melatonin).
5. **Immunological:** programmed decline of immunity (**thymus shrinks**), so infections, cancer and autoimmunity rise.

Physiotherapy aims at **successful ageing** (Rowe and Kahn): low disease and disability, high physical and cognitive function, active engagement with life. Much decline is secondary and can be changed.

7 marks Theories of ageing: enumerate and explain any three

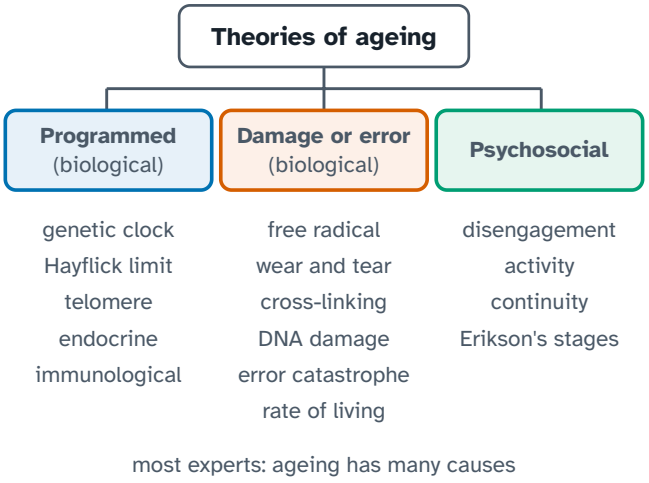
Asked: S17 (2e), S18 (2e), W19 (2d), W20 (2c), S24 (2c) (asked 5 times)

Ageing is the progressive, time-related decline in body structure and function that reduces the ability to adapt to stress; it is universal, progressive, intrinsic and harmful.

Classification

- **Programmed (genetic):** genetic clock, Hayflick limit, telomere, endocrine, immunological.
- **Damage or error (stochastic):** free radical, wear and tear, cross-linking, DNA damage, error catastrophe, rate of living.
- **Psychosocial:** disengagement, activity, continuity, Erikson's eighth stage.

Theories of ageing: the classification tree



Programmed theories say ageing follows a built-in plan; damage theories say it is the sum of accumulated harm; psychosocial theories explain how people adapt to growing old.

1. Telomere theory and Hayflick limit (programmed)
Normal cells divide a limited number of times and stop: human fibroblasts after about **40–60 divisions** (Hayflick, 1961). **Telomeres**, the protective ends of chromosomes, shorten at each division; when too short, the cell stops dividing. Senescent cells build up in old tissues.

2. Free radical theory (damage; Harman, 1956)
Reactive oxygen species made during normal metabolism damage DNA, proteins and membranes; damaged **mitochondria** make more radicals. Evidence: oxidative damage and **lipofuscin** rise with age. PT: regular moderate exercise **increases antioxidant defences**.

3. Activity theory (psychosocial; Havighurst, 1961)
Staying **physically and socially active** and replacing lost roles leads to successful ageing and life satisfaction. Basis of **exercise groups**, day centres and volunteering. The opposite view, **disengagement theory**, is criticised because withdrawal harms health.

Meaning for physiotherapy
No single theory explains ageing; much decline is **secondary** (inactivity, disease), which exercise can change: stretching counters **cross-linking** stiffness, strengthening and weight control reduce joint wear, and group exercise supports activity and continuity.

> **Draw:** The classification tree: ageing theories branching into programmed, damage and psychosocial, with all names.

7 marks Theories of ageing: enumerate and describe any one

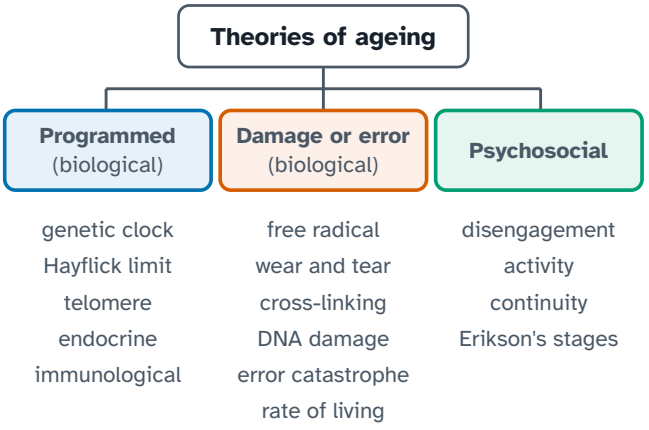
Asked: S21 (2b)

Ageing: the progressive, time-related decline in body structure and function that reduces adaptation to stress and raises the risk of disease and death.

Theories

- **Programmed:** genetic clock; Hayflick limit; telomere; endocrine; immunological.
- **Damage: free radical**; wear and tear; cross-linking; DNA damage; error catastrophe; rate of living.
- **Psychosocial:** disengagement; activity; continuity; Erikson's eighth stage.

Theories of ageing: the classification tree

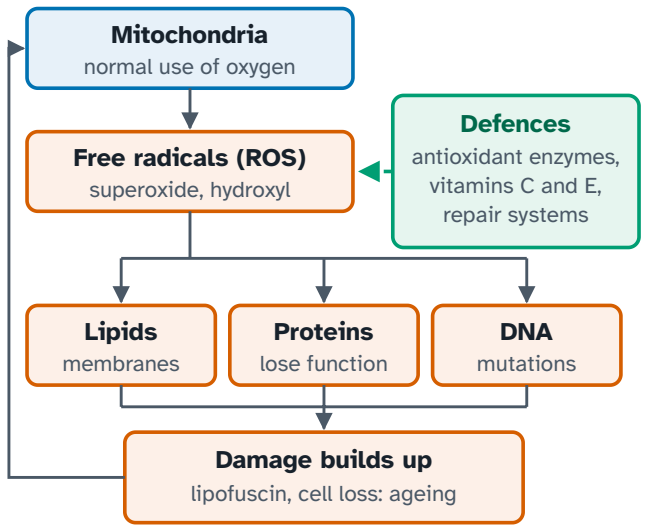


Programmed theories say ageing follows a built-in plan; damage theories say it is the sum of accumulated harm; psychosocial theories explain how people adapt to growing old.

Free radical theory (Harman, 1956)

- **Free radical:** a molecule with an **unpaired electron**, so highly reactive. Main **reactive oxygen species**: superoxide, hydrogen peroxide, hydroxyl radical.
- **Source: normal metabolism**: electrons leak from the respiratory chain in the **mitochondria** (mitochondrial theory, Harman 1972).
- **Damage:** lipid peroxidation of **membranes**; oxidised, non-working **proteins**; **DNA** mutations.
- **Vicious circle:** damaged mitochondria leak more radicals.
- **Defences: superoxide dismutase, catalase, glutathione peroxidase**; vitamins C and E; repair. Ageing follows when damage outruns defence.
- **Evidence for:** oxidative damage rises with age; **lipofuscin** (age pigment) accumulates.
- **Evidence against:** antioxidant pills do not prolong life (beta-carotene, vitamin A and E slightly raised mortality in 68 trials); naked mole-rats live over 28 years despite high oxidative damage.

Free radical theory: the vicious circle



Loop: damaged mitochondria leak more radicals

Orange boxes are the damage, green is the body's defence. Regular exercise strengthens the defences; antioxidant pills do not.

Meaning for physiotherapy

Regular **moderate exercise** briefly raises radicals but trains the body to make more of its own **antioxidant enzymes** (hormesis); high-dose antioxidant pills can blunt training gains. Advise exercise, fruit and vegetables and no smoking. Much decline in old age is **secondary** (inactivity, disease), which physiotherapy can change.

Ageing has many causes; today they are grouped as interacting **hallmarks of ageing**, so name one limitation of any theory you describe. The telomere theory (Hayflick limit, TTAGGG caps, telomerase) is an equally good choice.

> **Draw:** The classification tree; beside it, the free radical circle: mitochondria → radicals → lipids, proteins, DNA → ageing, with defences blocking the radicals.

3(e) Physiology of ageing and the effects of exercise

Full notes p. 101

3 marks Visual changes due to ageing

Asked: S16 (1d)

With ageing the eye lets in less light and focuses and adapts less well; these changes add to falls risk.

- **Presbyopia:** cannot focus on near objects.
- **Smaller pupil and yellower lens:** less light reaches the retina; colours change.
- Poor **dark adaptation** and **glare** tolerance.
- Lower **contrast sensitivity** and **depth perception**; narrower visual field.
- **Cataract**, **glaucoma** and **macular degeneration** more common.

Effect: misjudges steps and kerbs; falls at night; reading difficulty. **Advice:** good lighting without glare, **contrast edges on steps**, regular eye checks.

3 marks Musculoskeletal changes due to ageing

Asked: W16 (1e), S25 (1e) (asked 2 times)

Ageing reduces the reserve of muscle, bone and connective tissue; much of the loss is **secondary** (inactivity) and responds to exercise.

Tissue	Change and effect
Muscle	**Sarcopenia**: loss of mass, mainly **type II** fibres; fewer motor units; strength and power fall faster than mass: slow sit-to-stand, falls
Bone	Loss of mass and quality (fast after menopause): **osteoporosis**, hip, wrist and spine fractures
Cartilage	Less water and proteoglycan, thinner: **osteoarthritis**
Tendons, ligaments	Stiffer collagen with **cross-links**: reduced ROM, slow healing
Discs	Dehydrate, lose height: stiff spine, spondylosis
Posture and gait	Forward head, **kyphosis**, flexed hips and knees; slower, shorter steps, wider base

3 marks Six cardiopulmonary (cardiorespiratory) changes in the elderly

Asked: S19 (1c), S20 (1f), W24 (1f) (asked 3 times)

With age the heart and lungs keep normal resting function but **lose reserve**, which shows with exercise or illness.

Change	Result
Stiffer arteries	Raised **systolic BP** and pulse pressure
Lower maximum heart rate	Lower maximum cardiac output and **VO_2max**
Weaker baroreflex	**Postural hypotension**
Stiffer chest wall, weaker muscles	More work of breathing
Loss of lung recoil	Lower **FEV_1** and vital capacity, higher **residual volume**
Weaker cough and clearance	More chest infections

Use the Borg scale and a longer warm-up; rise slowly from lying.

3 marks Physiological changes in the respiratory system with ageing

Asked: S26 (1c)

The lungs lose reserve with age; it shows in illness, after surgery and on effort.

- **Chest wall stiffer** (costal cartilage calcification, kyphosis); **respiratory muscles weaker**.
- Loss of **elastic recoil**; small airways close earlier in expiration.
- **FEV_1 and vital capacity fall**; **residual volume rises**; total lung capacity changes little.
- More ventilation–perfusion mismatch: resting **PaO_2 falls** slowly.
- Weaker **cough**, slower mucociliary clearance; blunted response to low O_2 and high CO_2.

PT: breathing exercises, early mobilisation and airway clearance in illness.

7 marks Age-related changes in the nervous system

Asked: S17 (2a), S26 (2d) (asked 2 times)

Ageing is the gradual decline in the body's reserve and its ability to adapt to stress. In the nervous system it shows as **slower sensing, processing and responding**.

Central nervous system

- **Brain volume falls**, more in the **frontal lobes and hippocampus**; loss of neurons in some areas, of synapses and dendrites; white matter changes.
- Fall in neurotransmitters: **dopamine** (slower movement), **acetylcholine** (memory).
- **Cognition:** slower processing speed and **reaction time**; weaker short-term and working memory and divided attention; knowledge, vocabulary and judgement kept.
- **Motor control:** slower, less smooth movement; poorer fine coordination; slower **anticipatory postural adjustments**.
- **Sleep:** lighter, broken; earlier waking.

Peripheral nervous system

- Loss of motor and sensory fibres; **slower nerve conduction**; **ankle jerks** reduced or absent.
- Reduced **vibration and position sense**, most in the feet.

Autonomic nervous system

- Weaker **baroreflex**: dizziness on standing, **postural hypotension**.
- Poor temperature control; bladder urgency and nocturia.

Special senses

Presbyopia, glare and contrast problems; **presbycusis**; loss of vestibular hair cells (unsteady on turning); reduced taste and smell.

Effect and physiotherapy

Late, weak balance reactions, so **falls** rise with age. PT: **balance**, strength, **dual-task** and aerobic training; rise slowly from lying; good lighting.

Contrast primary ageing (everyone) with secondary ageing (disease, inactivity), which exercise can change.

7 marks Age-related changes in the neurological system and the effects of exercise

Asked: W25 (2c)

Ageing reduces reserve and the ability to adapt to stress; in the nervous system it makes sensing, processing and responding slower.

Changes

- **Brain:** volume loss (frontal lobes, hippocampus); fewer synapses; white matter change; less **dopamine** and **acetylcholine**.
- **Cognition:** slower processing and **reaction time**; weaker working memory and divided attention.
- **Motor control:** slower, less coordinated movement; slower anticipatory postural adjustments.
- **Peripheral:** fewer fibres, **slower conduction**, reduced ankle jerks and vibration and position sense in the feet; fewer motor units.
- **Autonomic:** weaker baroreflex (**postural hypotension**), poor temperature control, bladder urgency.
- **Senses:** presbyopia, presbycusis, vestibular loss. Result: late, weak balance reactions and **falls**.

Effects of exercise

- **Strength and power:** resistance training works at any age; residents aged about **90** gained **174%** in strength after **8 weeks** (Fiatarone, 1990). Early gains are **neural** (better recruitment and firing), later gains add muscle size.
- **Nervous system:** faster reaction time; better coordination and balance reactions.
- **Falls:** exercise reduced the rate of falls by **23%**; balance and functional exercise worked best (Sherrington, 2019).
- **Brain:** aerobic exercise improves blood flow to the brain, **memory and mood**, and lowers the risk of cognitive decline.

Dose (WHO, 2020) and precautions

150–300 minutes moderate aerobic a week; strengthening on **2 or more** days; balance and strength on **3 or more** days. Rise slowly (postural hypotension); avoid breath-holding; supervise balance work; Borg scale with beta-blockers.

7 marks Age-related changes in the respiratory system

Asked: S22 (2a)

Ageing is the gradual loss of the body's **reserve**. An older person breathes normally at rest, but the loss shows on effort, in **pneumonia, after surgery and with bed rest**. Smoking, inactivity and disease add **secondary** loss.

Changes

Area	Change	Effect
Chest wall	**Stiffer**: costal cartilage calcification, kyphosis; rib cage compliance about **30% lower**	Less expansion; more work of breathing
Lung tissue	Loss of **elastic recoil**; total compliance falls	Small airways close earlier
Muscles	Diaphragm strength about **25% lower** at 65–75	Breathlessness on effort; weak cough
Volumes	TLC same; **residual volume up about 50%** (20 to 70); VC down; **FEV_1 down 20–30 ml a year**	Air trapping; fixed 0.70 ratio over-diagnoses COPD
Airway closure	Closing volume rises; airways close in normal breathing at about **44 lying, 65 sitting**	Poorer oxygenation lying flat
Gas exchange	Alveolar area **75 → 60 m^2** (30 to 70); more V/Q mismatch; **$PaO_2 \approx 109 - 0.43 \times age$**; $PaCO_2$ unchanged	Less oxygen margin in illness
Control	Response to **hypoxia down about 50%**, to hypercapnia 40–50%; less awareness of bronchospasm	Late warning of hypoxia and asthma attacks
Defences	Slower **mucociliary clearance**; less sensitive, weaker **cough**; poorer swallow	**Silent aspiration** (71% of older pneumonia patients), chest infections

Effect on function

Breathlessness at lower workloads, lower exercise tolerance, more pneumonia and slower recovery after surgery.

Physiotherapy implications

- **Pre-operative breathing exercises** and **early mobilisation**: lung complications are 2–3 times more likely at 60–79.
- **Upright positioning**: PaO_2 about 5 mmHg higher sitting than lying.
- **Inspiratory muscle training**: maximal inspiratory pressure up about 26 cmH_2O.
- **Aerobic training**: VO_2max up about 16%; posture and chest mobility work against kyphosis.
- **Airway clearance** in infection; avoid long bed rest; stop smoking; vaccination advice.
- Monitor **oxygen saturation and Borg**, not breathlessness alone.

Viva point: residual volume is the lung volume that **rises** with age.

7 marks Age-related changes in the sensorimotor system

Asked: S23 (2f)

The sensorimotor system senses the body and surroundings, processes the information and produces movement. With ageing all three steps slow: **slower sensing, slower processing, slower and weaker responses**.

Sensory changes

- **Touch, vibration, proprioception:** fewer receptors and sensory fibres; reduced position and vibration sense, **most in the feet**: poor balance on uneven ground and in the dark.
- **Vision:** presbyopia, less light reaches the retina, poor dark adaptation, lower contrast and depth perception: misjudges steps.
- **Vestibular:** loss of hair cells and nerve fibres: unsteady on turning.
- **Hearing:** presbycusis; slower nerve conduction overall.

Central processing

- Brain volume loss, fewer synapses, less dopamine.
- Slower **processing speed and reaction time**; weaker divided attention (dual tasks).

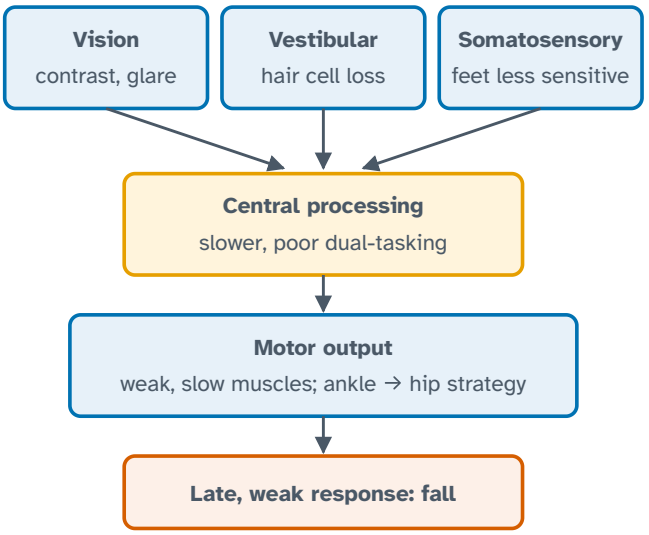
Motor changes

- **Sarcopenia** with loss of **type II fibres** and **fewer motor units**: strength and power fall faster than mass.
- Slower, less smooth movement; poorer fine coordination; slower **anticipatory postural adjustments**.
- Gait: slower, shorter steps, wider base, more time on both feet.

Result

Balance reactions are **late and weak**: reduced ankle range and strength shift control from the **ankle strategy to the hip strategy**, the corrective step is too slow or too short, and walking while talking (**dual-tasking**) worsens gait. A slip or trip becomes a fall, so **falls rise with age**.

Balance as a system, and where ageing affects it



Each part of the balance system declines with age (see sensorimotor changes); disease and drugs add to the decline.

Physiotherapy

Balance training (cut falls by 23%), resistance and power training, dual-task practice, good lighting and footwear.

3(f) Geriatric evaluation and the role of physiotherapy

Full notes p. 110

3 marks Adverse effects of immobility in the elderly

Asked: S23 (1c)

Older people lose function quickly in bed and **regain strength more slowly** than the young.

- **Muscle and function:** rapid loss of strength and walking ability (**hospital-associated disability**); contractures.
- **Heart and lungs: postural hypotension** and falls on first standing; **DVT**; pneumonia.
- **Bone and skin:** bone loss; **pressure sores**.
- **Bladder and bowel:** incontinence, retention, constipation, faecal impaction.
- **Mind: delirium**, depression, fear of falling.
- **Metabolic:** insulin resistance, poor appetite, weight loss.

Prevent: sit out and walk early, exercise in bed, turning, toileting schedule.

7 marks Geriatric assessment scales

Asked: W17 (2b)

Comprehensive geriatric assessment is a multidimensional, team-based assessment of an older person's medical, functional, psychological and social needs. Standard scales make it **objective**, show **risk** (falls, frailty), set goals and measure change.

Scale	What it measures	Scoring and cut-off
Katz Index of ADL	Bathing, dressing, toileting, transferring, continence, feeding	**0–6**; 6 independent, 2 or less very dependent
Barthel Index	Ten basic ADL and mobility items	**0–100**; higher more independent
Lawton IADL	Telephone, shopping, food, housekeeping, laundry, transport, medicines, finances	**0–8**; higher more independent
MMSE	Orientation, registration, attention, recall, language, copying	**0–30**; **23 or less** suggests cognitive impairment

Scale	What it measures	Scoring and cut-off
Geriatric Depression Scale (15)	Depressive symptoms, yes/no	More than **5** suggests depression
Timed Up and Go	Rise, walk 3 m, turn, return, sit	**13.5 s or more**: raised falls risk
Berg Balance Scale	14 balance tasks	**0–56**; below about **45**: falls risk
Tinetti POMA	Balance and gait	**0–28**; below **19** high falls risk
SPPB	Standing balance, 4 m gait speed, five chair stands	**0–12**; low scores predict disability
Gait speed	Usual speed over 4–10 m	Below about **0.8 m/s**: poor function, frailty
Fried frailty phenotype	Weight loss, exhaustion, weak grip, slow walking, low activity	**3 or more** frail; 1–2 pre-frail
FES-I	Concern about falling in 16 activities	**16–64**; higher more fear

Fitness is measured with the **Senior Fitness Test** (chair stand, arm curl, 6-minute walk, sit-and-reach, back scratch, 8-foot up-and-go).

Group the scales by domain (ADL, IADL, cognition, mood, balance and falls, frailty) and give one cut-off for each.

7 marks Principles of geriatric rehabilitation

Asked: W18 (2d)

Geriatric rehabilitation aims to restore and keep the older person's **function and independence** in the tasks that matter to them, despite multiple illnesses and reduced reserve.

Principles

1. **Comprehensive assessment:** include cognition, mood, home and carers; **treat the person, not the age**.
2. **Function is the goal:** independence in meaningful tasks, goals set with the person and family.
3. **Teamwork:** doctor, nurse, physiotherapist, occupational therapist, social worker, family.
4. **Mobilise early;** avoid unnecessary bed rest (loss of strength, postural hypotension, pressure sores, delirium).
5. **Start low, go slow, but progress:** older muscle still adapts to overload; allow longer recovery.
6. **Prevent the next problem:** falls, pressure sores, contractures, deconditioning, incontinence.
7. **Adapt the environment and use aids:** walking aids, raised toilet seat, rails, lighting.
8. **Consider multiple illnesses and drugs:** BP, sugar, heart rate response, pain, sensory loss.
9. **Simple instructions**, repetition, written or pictorial home programmes; **involve carers**.
10. **Dignity and choice;** motivation through meaningful goals.
11. **Continuity:** link to community programmes, senior citizens' groups and follow-up.

Role of the physiotherapist

Health promotion (exercise groups, walking clubs, fall-prevention classes); screening; treatment of OA, back pain, stroke, Parkinson's, fractures, COPD, incontinence; home visits, aids and carer training; palliative care.

Remember the order: assess → set functional goals → treat early and progressively → prevent → adapt → follow up.

7 marks Functional assessment of the elderly

Asked: W18 (2f), W19 (2f) (asked 2 times)

Functional assessment measures what an older person can do in daily life. It is the core of **comprehensive geriatric assessment**, because function predicts **independence, care needs and outcome** better than diagnosis alone.

Domains and tools

Domain	What to assess	Tools
History	Illnesses, **drugs** (sedatives, BP drugs), **falls in the last year**, continence, sleep, weight change, the person's goals	Interview, family report
Physical	BP lying and standing, vision, hearing, feet and footwear, pain, ROM, grip, sensation	Postural BP, dynamometer, monofilament, Snellen chart
Basic ADL	Feeding, bathing, dressing, toileting, transfers, continence, walking	**Katz** (6 items), **Barthel** (0–100), FIM
Instrumental ADL	Telephone, shopping, cooking, housework, transport, medicines, money	**Lawton IADL** (8 items)
Mobility and balance	Transfers, gait, stairs, balance, falls risk	**TUG**, Berg, Tinetti, gait speed, SPPB
Cognition	Memory, orientation, attention	**MMSE**, Mini-Cog, MoCA
Mood	Depression, loneliness, fear of falling	**GDS**, FES-I
Nutrition and frailty	Weight loss, muscle mass, exhaustion	MNA, **Fried phenotype**, Clinical Frailty Scale
Social and home	Who lives with them, carers, money, floor level, stairs, **toilet type**, lighting, hazards	Home visit and checklist

Use of the findings

- Set **functional goals** with the person and family.
- Identify **risk**: falls (TUG 13.5 s or more), frailty, cognitive impairment (MMSE 23 or less).
- Decide on aids, home changes, referral and level of care; **re-test** to measure progress.

ADL are basic self-care tasks; IADL are the more complex tasks needed to live alone.

7 marks Components of fitness assessment in the geriatric population

Asked: S25 (2c)

Functional fitness is the physical capacity to do normal daily activities safely and independently without undue fatigue. Its assessment gives a **baseline**, finds weak components, guides the exercise prescription and shows progress.

Screen first

BP, heart disease, **recent falls**, pain, drugs; stop if **chest pain, dizziness or unusual breathlessness**.

Senior Fitness Test (Rikli and Jones)

For community-living adults aged **60–94**; norms from 7,183 people; reliability 0.80–0.98.

Component	Test and procedure
Lower-body strength	**30-second chair stand**: seat **43 cm (17 in)**, arms crossed; number of full stands
Upper-body strength	**30-second arm curl**, seated: **5 lb (2.27 kg)** women, **8 lb (3.63 kg)** men; number of curls

Component	Test and procedure
Aerobic endurance	**6-minute walk** round a **50-yard (45.7 m)** course; or **2-minute step**, knee to midway between kneecap and iliac crest
Lower-body flexibility	**Chair sit-and-reach**: one leg straight, ankle at 90°; fingertips to toe in inches (minus if short)
Upper-body flexibility	**Back scratch**: gap or overlap between middle fingers
Agility, dynamic balance	**8-foot up-and-go**: rise, walk round a cone **2.44 m** away, sit; seconds
Body composition	Height, weight, **BMI**

Interpretation

- **Normal range** = middle 50% for age and sex; for example chair stand **12–17** (women 60–64) and **14–19** (men 60–64).
- **Criterion standards (2013):** the score needed to stay independent to about 90; for example women aged 60–64 need **15 chair stands** and **625 yards**; at 90–94 both sexes need 9 stands, 400 yards and an up-and-go of 8.0 s.
- Older risk zones: fewer than **8 chair stands**, under **350 yards**, up-and-go over **9 s**.

Add

- **Grip strength**; **single-leg stance**; **gait speed** (below about 0.8 m/s: frailty); **TUG** (13.5 s or more: falls risk); SPPB.

Use

- Compare each score with the normal range and the standard; train the **weakest components** first.
- **Re-test** to show change and motivate the person.

The chair stand tests lower-body strength, not endurance: a common MCQ.

3(f) Falls in the elderly

Full notes p. 119

3 marks Risk factors for falls in the elderly

Asked: W20 (1f)

Fall (WHO): an event which results in a person coming to rest inadvertently on the ground, floor or other lower level. About **28–35%** of people aged 65 and over fall each year; a **previous fall** is the strongest predictor.

Group	Risk factors
Intrinsic	**Leg weakness**, arthritis, foot problems; poor balance, slow or shuffling gait, previous falls; **poor vision**, neuropathy, vestibular loss; **postural hypotension**, stroke, Parkinson's, dementia, diabetes, nocturia; **fear of falling**
Extrinsic	Poor lighting, slippery floors, loose rugs, stairs without rails, **Indian toilets** without grab bars; **four or more drugs**, sedatives, BP drugs; loose slippers, faulty walking aids

Physiotherapy targets them with balance and strength training, home modification and education.

3 marks Extrinsic risk factors for falls in the elderly

Asked: W25 (1d)

Extrinsic factors are outside the person: the environment, medicines, footwear and aids. They are the easiest to change.

- **Lighting:** poor or dim light, no night light.
- **Floors:** slippery or wet floors, **loose rugs**, clutter, cords.
- **Stairs** without rails; **low beds and chairs**.
- **Indian (squat) toilets**, bathrooms without grab bars.
- **Outdoors:** uneven roads, animals.
- **Medicines: four or more drugs**; sedatives, sleeping pills, antidepressants, antipsychotics, BP drugs, diuretics.
- **Footwear and aids:** loose slippers, high heels, worn soles; wrong-height or faulty walking aids.

Remedy: home safety check, medication review by the doctor, well-fitting shoes, aids checked and fitted.

7 marks Causes of falls in the elderly; environmental modifications for prevention

Asked: W22 (2c)

Fall (WHO): an event which results in a person coming to rest inadvertently on the ground or floor or other lower level. About **28–35%** of people over 65 fall each year (**32–42%** over 70); results include fractures, head injury, a long lie, fear of falling and loss of independence.

Causes

- **Intrinsic:** leg weakness, arthritis, foot problems; poor balance and slow gait; poor vision, neuropathy, vestibular loss; **postural hypotension**, stroke, Parkinson's, dementia, depression, diabetes, urinary urgency and nocturia, acute illness; **fear of falling**, cognitive impairment.
- **Extrinsic:** environmental hazards (below); **medicines** (four or more drugs, sedatives, antidepressants, antipsychotics, BP drugs, diuretics); poor footwear and faulty walking aids.

Environmental modifications

Area	Modification
Lighting	Bright, even light; switches at room entrances and by the bed; **night lights to the toilet**
Floors	Remove **loose rugs**, cords, clutter; non-slip mats; wipe spills; **contrasting colour on step edges**
Stairs	**Rails on both sides**; good light at top and bottom; nothing left on steps
Bathroom and toilet	**Grab bars**; western toilet or **raised commode** over the Indian toilet; non-slip floor; bathing stool; hand shower
Bedroom	Bed and chair at **knee height**; firm chair with armrests; urinal or commode at night
Kitchen and storage	Common items between waist and shoulder height; no climbing on stools
Outdoors	Even paths, rails at entrances; avoid wet or dark roads

Evidence: home safety assessment and modification reduced the rate of falls (rate ratio **0.81**), more in people at high risk (Gillespie et al., 2012).

Add footwear (firm, well-fitting, thin non-slip soles), a phone or alarm within reach, and balance and strength exercise, the most effective single measure.

> **Draw:** A safe bathroom: grab bars, raised seat over the toilet, non-slip floor, bathing stool.

Part 4: Concepts of rehabilitation

4(a) Disability: definitions, models, types, evaluation and prevention

Full notes p. 127

3 marks Define impairment, disability and handicap

Asked: W16 (1a), W25 (1f) (asked 2 times)

Defined by the WHO in the **ICIDH** (International Classification of Impairments, Disabilities and Handicaps, **1980**).

- **Impairment** (**organ level**): any loss or abnormality of psychological, physiological or anatomical structure or function.
- **Disability** (**person level**): any restriction or lack, resulting from an impairment, of ability to perform an activity in the manner or within the range considered normal for a human being.
- **Handicap** (**society level**): a disadvantage for a given individual, resulting from an impairment or a disability, that limits or prevents the fulfilment of a role that is normal for that individual, depending on age, sex, and social and cultural factors.

Example: polio → paralysed leg (impairment) → cannot walk far (disability) → cannot farm or go to school (handicap).

The ICIDH was replaced by the **ICF (2001)**, in which disability is the umbrella term for impairments, activity limitations and participation restrictions.

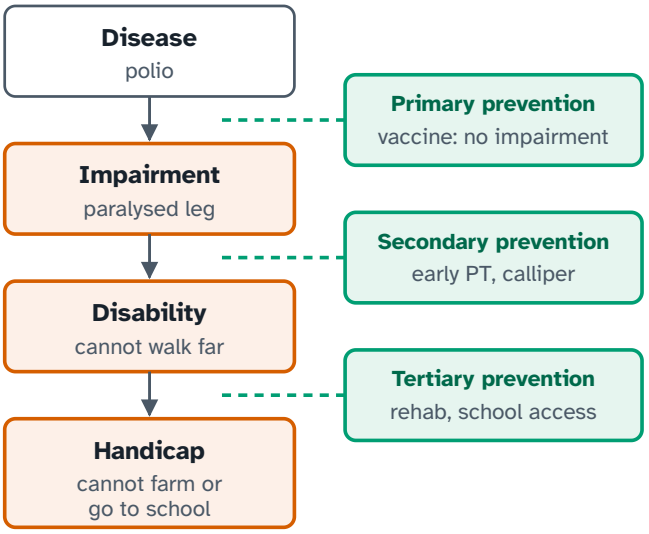
3 marks Full form of ICIDH (asked as "ICFDH")

Asked: W17 (1c)

ICIDH: International Classification of Impairments, Disabilities and Handicaps, published by the **WHO in 1980**.

- Describes the consequences of disease as a one-way chain: **disease → impairment → disability → handicap**.
- **Impairment:** loss or abnormality of structure or function (organ level).
- **Disability:** restriction or lack of ability to perform an activity in the normal manner (person level).
- **Handicap:** a disadvantage that limits or prevents a role normal for that individual (society level).
- Replaced by the **ICF (2001)**: body functions and structures, activities and participation, with environmental and personal factors, all interacting **both ways**.

The ICIDH chain and the three levels of prevention



Each level of prevention breaks one link in the chain (green dashed lines). The ICF later replaced this one-way chain with two-way interactions and added environmental and personal factors.

3 marks Models of disability

Asked: W17 (1d)

Each model explains disability differently and so leads to a different response.

Model	Sees disability as	Response
Medical (individual)	A problem of the person, caused by disease or injury	Cure or treat the impairment; professionals decide
Social	Created by **barriers in society** (steps, attitudes, laws)	Remove barriers: ramps, inclusive schools, laws
Biopsychosocial (ICF)	Interaction of health condition, body, activity, participation and context	Treat the person and change the environment together
Rights-based	A human rights issue	UN Convention (2006); **RPWD Act 2016**
Charity	A tragedy needing pity and help	Donations and care; criticised for creating dependence
Nagi	Pathology → impairment → functional limitation → disability	Used in rehabilitation research

Contrast the medical and social models first; the ICF model joins the two.

3 marks Define disability and classify its types

Asked: S19 (1f), W20 (1a) (asked 2 times)

Disability (ICIDH, WHO 1980): any restriction or lack, resulting from an impairment, of ability to perform an activity in the manner or within the range considered normal. In the **ICF** it is the umbrella term for impairments, activity limitations and participation restrictions.

21 specified disabilities (RPWD Act 2016), in groups:

- **Locomotor:** locomotor disability, leprosy-cured, cerebral palsy, dwarfism, muscular dystrophy, acid attack victims.
- **Sensory:** blindness, low vision, deaf and hard of hearing, speech and language disability.
- **Intellectual:** intellectual disability, specific learning disabilities, autism spectrum disorder; **mental illness**.
- **Chronic neurological:** multiple sclerosis, Parkinson's disease; **blood disorders:** haemophilia, thalassaemia, sickle cell disease; **multiple** disabilities including deaf-blindness.

Other ways: congenital or acquired; temporary or permanent; visible or invisible; mild, moderate, severe, profound. **Benchmark disability: 40% or more**.

3 marks Primary and secondary disability

Asked: S16 (1c)

Feature	Primary disability	Secondary disability
Meaning	Loss of function caused **directly** by the disease or injury	Further problems that develop **later** because of the primary disability, inactivity or poor care
Examples	Paraplegia after spinal cord injury; hemiplegia after stroke	Contractures, pressure sores, obesity, osteoporosis, pain, urinary infections, depression, deconditioning
Can it be prevented?	Only by preventing the disease or injury itself	**Largely preventable** by good care

Physiotherapy role: preventing secondary disability is a major job of the physiotherapist: positioning, exercise, education and early mobility.

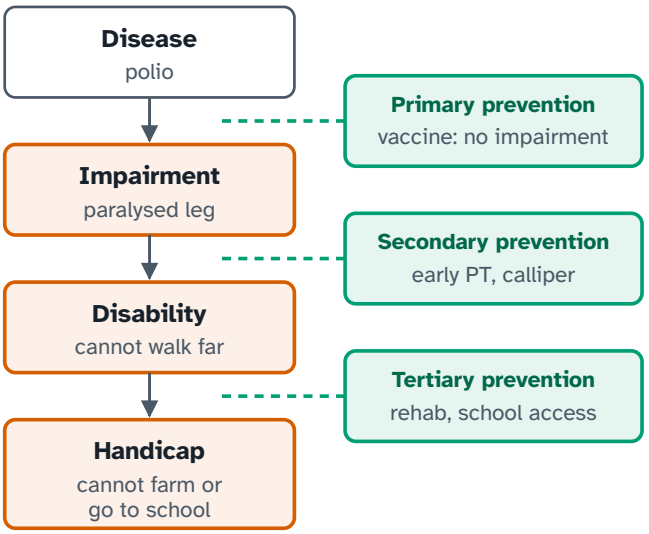
3 marks Levels of disability prevention

Asked: W16 (1b), W22 (1a), W25 (1e) (asked 3 times)

Three levels (WHO, 1981); each breaks one link in the chain **disease → impairment → disability → handicap**.

Level	Aim	Examples
Primary	Prevent the **impairment** occurring	Immunisation (polio, measles, rubella); iodine, vitamin A, folic acid; antenatal care; helmets, seat belts; genetic counselling
Secondary	Stop an impairment becoming a **disability**	Screening (RBSK); early treatment; early physiotherapy and splinting; early surgery (clubfoot, cataract)
Tertiary	Stop a disability becoming a **handicap**	Rehabilitation; aids and appliances; inclusive education; accessibility; employment; rights; CBR

The ICIDH chain and the three levels of prevention



Each level of prevention breaks one link in the chain (green dashed lines). The ICF later replaced this one-way chain with two-way interactions and added environmental and personal factors.

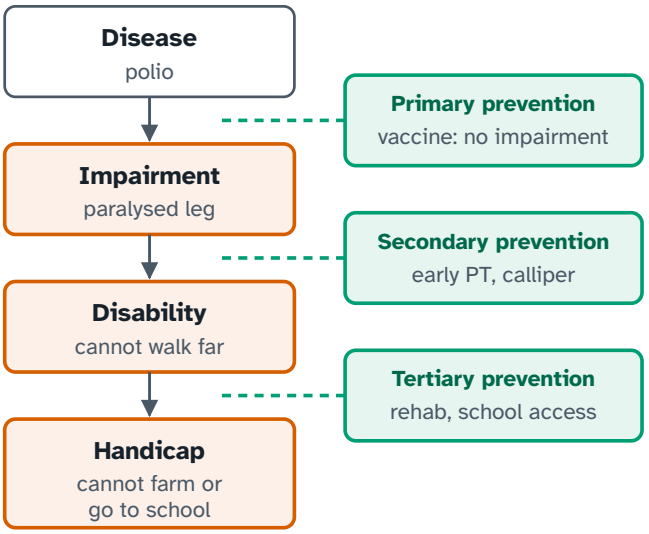
Polio example: vaccine (primary); early physiotherapy and calliper (secondary); rehabilitation and school access (tertiary).

7 marks Levels of disability prevention

Asked: W20 (2a), S26 (2f) (asked 2 times)

The WHO report on disability prevention and rehabilitation (**1981**) describes three levels of prevention based on the **ICIDH chain**: disease → **impairment** (organ level) → **disability** (person level) → **handicap** (society level). Each level breaks one link of the chain.

The ICIDH chain and the three levels of prevention



Each level of prevention breaks one link in the chain (green dashed lines). The ICF later replaced this one-way chain with two-way interactions and added environmental and personal factors.

1. Primary prevention (first level)
Aim: prevent the impairment from occurring.

- **Immunisation**: polio, measles, rubella.
- **Nutrition**: iodine, vitamin A, folic acid, iron.
- **Antenatal care** and safe delivery; avoiding consanguineous marriage; genetic counselling.

- Road and workplace safety: helmets, seat belts.
- Control of leprosy and diabetes; healthy lifestyle and exercise; health education.

2. Secondary prevention (second level)
Aim: stop an impairment becoming a disability by **early detection and treatment**.

- Newborn and developmental screening (**RBSK**); hearing and vision screening; screening for diabetic foot.
- Early treatment of infections and injuries.
- **Early physiotherapy**, positioning and splinting.
- Early surgery (clubfoot, cataract).

3. Tertiary prevention (third level)
Aim: stop a disability becoming a handicap.

- **Rehabilitation**: medical, vocational, social; aids and appliances.
- Prevention of **secondary conditions**.
- Inclusive education; accessible buildings and transport; employment.
- Legal rights and schemes; community awareness; **CBR**.

Role of the physiotherapist

Level	Physiotherapy action
Primary	Health education, safety advice, exercise for a healthy lifestyle
Secondary	Early detection, early therapy, positioning and splinting
Tertiary	Rehabilitation, aids and their training, inclusion, CBR

Polio example: vaccine (primary) → early physiotherapy and calliper (secondary) → rehabilitation and school access (tertiary).

7 marks Types of disability; evaluation of permanent physical impairment of the upper extremity

Asked: S23 (2d)

Types of disability

- **RPWD Act 2016: 21 specified disabilities:** locomotor (locomotor disability, leprosy-cured, cerebral palsy, dwarfism, muscular dystrophy, acid attack victims); sensory (blindness, low vision, deaf and hard of hearing, speech and language); intellectual (intellectual disability, specific learning disabilities, autism); mental illness; chronic neurological (multiple sclerosis, Parkinson's disease); blood disorders (haemophilia, thalassaemia, sickle cell disease); multiple disabilities including deaf-blindness.
- Other ways: congenital or acquired; temporary or permanent; visible or invisible; mild, moderate, severe, profound; primary or secondary.
- **Benchmark disability: 40% or more**, certified by a medical authority.

Evaluation of permanent impairment of the upper extremity
Done when the condition has reached **maximum improvement** from treatment; based on **measured function**, not opinion. The upper limb has an **arm component** and a **hand component**, each worth **90%**.

Component	What is measured	Value
Arm: range of motion	Shoulder, elbow, wrist, each **30%**; mean % loss of range × 0.3, added across joints	Up to 90%
Arm: muscle strength	MRC grades 0–5 as % loss; mean loss around each joint × 0.3	Up to 90%
Arm: coordinated activities	Ten set activities, **9% each**	Up to 90%
Hand: prehension	Opposition 8%; lateral pinch 5%; cylindrical 6%; spherical 6%; hook grasp 5%	30%
Hand: sensation	Complete loss: thumb ray 9%, index 6%, middle, ring and little 5% each	30%
Hand: strength	Grip 20%, pinch 10%	30%

Combining formula: a + b(90 – a) ÷ 90 (a higher, b lower value). Arm values are combined by the formula; hand values are added; then arm and hand are combined by the formula. Example: arm 27% and hand 64% → 64 + 27 × 26 ÷ 90 = **71.8%**.

Extra weightage up to 10% for persistent pain, infection, deformity, contracture, dominant limb (4%) or shortening (2% for each inch beyond the first); the total **never exceeds 100%**.

A result of 40% or more is a benchmark disability, needed for most reservations and benefits.

4(b, d) Rehabilitation: types, IBR vs CBR and the rehabilitation team

Full notes p. 133

3 marks Define rehabilitation

Asked: S20 (1d)

Rehabilitation (WHO): "a set of interventions designed to **optimize functioning** and **reduce disability** in individuals with health conditions in interaction with their environment".

Older WHO wording (still quoted in Indian textbooks): all measures that reduce the impact of disabling and handicapping conditions and enable disabled people to achieve **social integration**.

- **Aim:** the highest possible independence in daily life, school, work and community, with the least restriction.
- **Kinds:** medical, vocational, social, educational, psychological.
- **By place:** institution-based (IBR); outreach; community-based (CBR).

3 marks Role of the occupational therapist in the rehabilitation team

Asked: S26 (1d)

The occupational therapist helps the person do the **daily activities** of life, school and work as independently as possible.

- Assesses and trains **ADL**: feeding, dressing, bathing, toileting, cooking, using money and transport.
- **Hand function**: fine motor training, hand splints, sensory re-education.
- **Adaptive devices** (built-up spoons, button hooks, reachers, sock aids) and home, school and workplace modification.
- **Cognitive and perceptual retraining** (memory, attention, neglect) after brain injury and stroke.
- Pre-vocational and work skills; wheelchair seating; energy conservation; play and sensory integration in children.

3 marks Role of the medical social worker in rehabilitation

Asked: W20 (1e)

The medical social worker deals with the **social and economic** side of disability.

- **Psychosocial assessment**: family, income, housing, education, work, community support.
- **Counselling** the person and family to accept the disability and take part in rehabilitation.
- **Linking to benefits**: disability certificate and **UDID** card, pensions, travel concessions, free aids (**ADIP**), scholarships, loans for self-employment.
- Arranging financial help through hospital funds, government schemes and NGOs.
- **Discharge planning**, return to school or work, follow-up; **advocacy** against discrimination and abuse.

3 marks Role of the social worker in CBR

Asked: S26 (1b)

In CBR the social worker works in the village with the person, family and community.

- **Psychosocial assessment** of the family: income, housing, education, work, support.
- **Community survey** to identify persons with disability.
- **Awareness meetings** to reduce stigma.
- Forming **self-help groups** and **CBR committees**.
- Linking families to **schemes** (certificate, UDID, pensions, ADIP aids), livelihood and education.
- **Mobilising local resources**; counselling the family; advocacy against discrimination.

3 marks Vocational rehabilitation

Asked: W19 (1c), S21 (1e) (asked 2 times)

Purpose (ILO Convention 159, 1983): "to enable a disabled person to **secure, retain and advance in suitable employment** and thereby to further such person's integration or reintegration into society."

1. **Vocational evaluation:** physical capacity (functional capacity evaluation), education, skills, interests, aptitude, work history, local job market.
2. **Guidance and counselling:** a realistic goal matching ability and interest.
3. **Training:** industrial training institutes, Vocational Rehabilitation Centres for Handicapped, NGOs; work conditioning and hardening.
4. **Placement:** old job with changes, open employment (**4%** reservation, RPWD Act 2016), sheltered workshop, supported or self-employment (NDFDC loans).
5. **Follow-up** to keep the job.

3 marks What is a non-government organisation (NGO)? Give examples

Asked: W17 (1e)

NGO: a **voluntary, non-profit** organisation run **independently of the government**, registered as a society or trust.

Examples:

- **ADAPT** (formerly the Spastics Society of India): cerebral palsy.
- **Indian Red Cross Society**.
- **HelpAge India**: older people.
- **Bhagwan Mahaveer Viklang Sahayata Samiti** (Jaipur foot).
- Amar Seva Sangam, Mobility India, CBM.

NGOs give services where government services do not reach, train CBR workers, and advocate for rights.

3 marks Role of NGOs in community welfare, CBR and the rehabilitation team

Asked: S16 (1a), S23 (1a), S25 (1c) (asked 3 times)

NGO: a voluntary, non-profit organisation independent of the government (for example ADAPT, Indian Red Cross, BMVSS).

- **Surveys and early identification**; awareness to reduce stigma.
- **Services where government does not reach:** therapy, camps, aids and appliances, special and inclusive schools.
- **Training** CBR workers, parents and volunteers.
- **Vocational training**, self-help groups, livelihood.
- **Advocacy** for rights and accessibility; linking people to government schemes.
- Fund-raising, research, disaster relief; piloting programmes that government later adopts.

3 marks Spastics Society of India and its functions

Asked: S18 (1a)

Started on **2 October 1972** in **Mumbai** by **Dr Mithu Alur** to provide education and treatment for children with **cerebral palsy**; now called **ADAPT** (Able Disabled All People Together).

Functions:

- Special and **inclusive education**; early intervention.
- **Therapy:** physiotherapy, occupational and speech therapy.
- Training courses for teachers and therapists.
- Vocational training and placement of young adults.
- Advocacy for inclusive education; outreach in slums and villages; research and publications.

3 marks Functions of the Red Cross Society

Asked: S20 (1a), W21 (1f) (asked 2 times)

Founded on **Henry Dunant's** idea after the battle of Solferino (International Committee, **Geneva, 1863**). The **Indian Red Cross Society** was set up by an **Act of 1920**; the **President of India** is its president.

Functions:

- **Disaster** preparedness and relief.
- **Blood donation** and blood banks; **first aid** training; ambulance and hospital services.
- Health education; maternal and child welfare; care of the sick and disabled.
- Help for war victims; tracing missing family members.
- **Junior and Youth Red Cross** in schools and colleges.

Seven principles: humanity, impartiality, neutrality, independence, voluntary service, unity, universality.

7 marks Differences between institution-based (IBR) and community-based rehabilitation (CBR)

Asked: W16 (2b), W21 (2c), S24 (2f), W24 (2c), S25 (2f) (asked 5 times)

IBR: rehabilitation in hospitals, rehabilitation centres and special schools with specialist staff and equipment; the person comes to the centre.

CBR (ILO, UNESCO and WHO, 2004): "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities", carried out by people with disabilities, their families and communities with government and non-government services.

Feature	IBR	CBR
Place	Hospital, rehabilitation centre, special school, mostly in cities	Home, village, school, workplace
Who delivers	Specialists: doctors, therapists, prosthetists, psychologists	CBR workers, family, ASHA and anganwadi workers, teachers, volunteers, supported by professionals
Approach	**Medical model**: treats the impairment	**Social and rights model**: health, education, livelihood, social life, empowerment
Decisions	By professionals; the person is a patient	With the person and family, who are partners
Cost	**High** (buildings, equipment, travel and stay)	**Low**; local materials, existing services
Coverage	Few people, mainly urban	Many, including rural and poor families
Technology	Advanced equipment, investigations, surgery	Simple, local, appropriate (bamboo or wooden aids, home adaptations)
Carry-over	Skills learnt in an artificial setting	Skills learnt at home and in the village
Continuity	Ends at discharge; follow-up hard	Ongoing, sustained by the community
Inclusion	May separate the person from family	Keeps the person in the family; changes attitudes
Best for	Acute and complex care, prostheses, intensive therapy, staff training	Long-term needs, common disabilities, prevention, school and work inclusion
Limits	Expensive, crowded, far away, dependency	Limited specialist skill; depends on training and community commitment

They are not rivals: CBR handles most needs locally and **refers** complex cases to institutions; institutions train CBR workers and give specialist back-up; **outreach** programmes link the two.

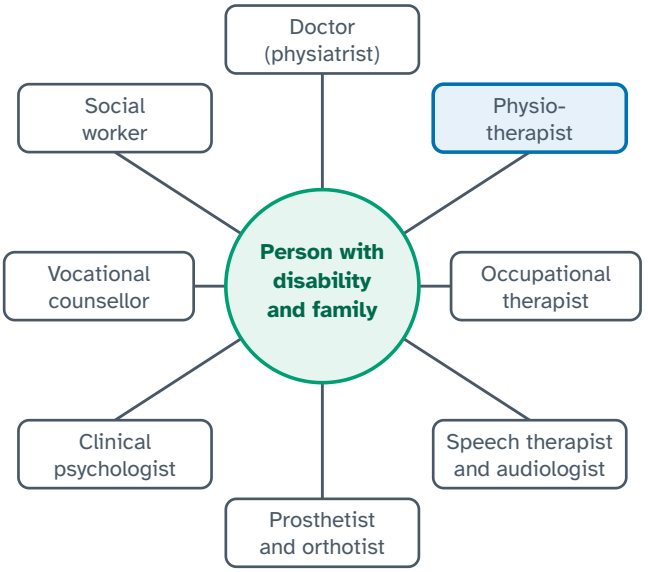
> **Draw:** IBR, outreach and CBR as three circles from the institution to the village.

7 marks Role of the medical social worker in the rehabilitation team

Asked: S21 (2e)

Rehabilitation is a **team effort** around the person and family. The medical social worker is the member who handles the **social, family and economic** effects of disability and links the person to resources outside the hospital.

The rehabilitation team



The person and family (green) are members of the team and set the goals; the physiotherapist is shown in blue. Add the nurse, the special teacher and, in CBR, the CBR worker and community.

1. Psychosocial assessment
Family structure, income, housing, education, work and community support; identifies problems that block rehabilitation.

2. Counselling
Helps the person and family **accept the disability** and take part in rehabilitation; supports the family through the change in roles.

3. Linking to benefits
Disability certificate and **UDID card**; pensions; travel concessions; free aids and appliances (**ADIP**); scholarships; loans for self-employment.

4. Financial help
Through hospital funds, government schemes and **NGOs**.

5. Discharge planning and return to the community
Home and community resources; return to school or work; follow-up after discharge.

6. Advocacy
Against discrimination and abuse; for the person's rights.

7. In community-based rehabilitation
Community survey to identify persons with disability; awareness meetings; forming **self-help groups** and **CBR**

committees; linking families to schemes, livelihood and education; mobilising local resources.

8. Teamwork
Shares findings in team meetings so the team sets **common goals** (interdisciplinary model) and avoids conflicting advice.

The social worker turns medical recovery into social integration, which is the final aim of rehabilitation.

7 marks Significance of vocational guidance and counselling in rehabilitation

Asked: W17 (2e), S23 (2c) (asked 2 times)

Vocational rehabilitation (ILO Convention 159, 1983) aims "to enable a disabled person to **secure, retain and advance in suitable employment**" and so to integrate into society. **Vocational guidance and counselling** helps the person choose a **realistic work goal** that matches ability and interest.

Steps of vocational rehabilitation

1. **Evaluation:** physical capacity (functional capacity evaluation), education, skills, interests, aptitude, work history, local job market.
2. **Guidance and counselling.**
3. **Training:** industrial training institutes, Vocational Rehabilitation Centres for Handicapped, NGOs; work conditioning and work hardening.
4. **Placement:** old job with changes, open employment, sheltered workshop, supported employment, self-employment.
5. **Follow-up** to solve problems at work and keep the job.

Significance

- **Matches the job to remaining abilities**, so training is not wasted and failure is avoided.
- Builds **motivation, self-esteem** and acceptance of disability; reduces depression.
- Leads to **earning and economic independence**; less dependence on family and state.
- Gives a **social role and dignity**, the final aim of rehabilitation (participation).
- Guides the **family and employer**; identifies adaptations, aids and job modifications.
- Informs about schemes: **4% reservation** in government posts (RPWD Act 2016), training centres, concessional loans (NDFDC).

Physiotherapist's part
Functional capacity evaluation of physical demands; work conditioning and work hardening; advice on workplace adaptations.

Work is the step that turns rehabilitation into full participation in society.

7 marks Training for activities of daily living, with examples

Asked: W17 (2f)

Activities of daily living (ADL) are the everyday tasks a person needs for self-care and independence.

- **Basic ADL:** feeding, grooming, bathing, dressing, toileting, transfers, walking or wheelchair mobility.
- **Instrumental ADL:** cooking, shopping, housework, money, phone, transport.

Steps of ADL training

1. **Assess** with the **Barthel Index** or **FIM**.
2. Set goals **with the person**.
3. **Analyse the task** and break it into steps.
4. Teach step by step (**forward or backward chaining**), in the real setting and at the usual time.
5. Use the easiest method first and **fade help**.
6. Add **aids** and change the environment.
7. **Train the family**; review progress.

Examples

Condition	Training and aids
Hemiplegia	**Dress the affected arm first** and undress it last; sit to dress; elastic laces
Paraplegia	**Sliding-board transfer** to the toilet; long-handled mirror for skin checks
Rheumatoid hands	**Built-up handles**, lever taps
Low vision	**Tactile marks** on the stove

Other aids: button hook, reacher, sock aid, built-up spoon, universal cuff.

Long-term conditions
Teach **energy conservation** and **joint protection**.

Who trains

- **Occupational therapist:** self-care tasks, adaptive devices, home modification.
- **Physiotherapist:** transfers, walking or wheelchair mobility, aids and their training, home programmes.
- **Nurse:** skin, bladder and bowel care; teaching the family.

Train in the real home at the usual time, so the skill carries over.

7 marks Management of the disabled hand

Asked: W17 (2d)

The hand is the main tool for self-care and work and a sense organ. **Causes:** nerve injury, leprosy, burns, crush injury, stroke, cerebral palsy, rheumatoid arthritis, amputation. Plan: **restore, protect, adapt**.

Assessment

- Deformity (claw hand, wrist drop), wasting, scars, oedema (volumeter).
- Range by goniometry; **grip dynamometer** (sitting, elbow 90°, mean of 3 trials) and pinch gauge; muscle test of little finger abduction, thumb abduction, wrist extension.
- **Sensation:** monofilaments, two-point discrimination.
- **Function:** Nine-Hole Peg Test, Jebsen–Taylor test, DASH, real tasks.

Functional grips (Napier)
Power: cylindrical (glass), spherical (ball), hook (bag).
Precision: tip pinch (pin), three-jaw (pen), lateral key pinch (key). Train each with graded objects: large to small, light to heavy.

Treatment

- Oedema and pain: elevation, retrograde massage; mobilise, stretch, strengthen; dexterity on pegboards.

- **Splints** (static or dynamic; bamboo or tubing in villages): **cock-up** for radial palsy; **knuckle-bender** for mobile claw hand; thumb web spacer for median palsy; **safe position** for burns (wrist 20–30° extension, knuckles 70–90° flexion, fingers straight).
- **Sensory re-education:** localisation, then textures and objects, eyes closed then checked with eyes open.
- **Anaesthetic hand (NLEP self-care):** inspect daily, soak about half an hour, scrape hard skin, oil while wet, protect from heat and friction.

Adaptive devices and ADL

Built-up padded handles, universal cuff, lever taps, button hook, Velcro, one-handed methods; practise eating, cooking, washing and dressing at home.

Work and vocational aspects

Job analysis, graded work simulation, modified or insulated tools, vocational guidance or job change; tendon transfer for a mobile claw hand at an NLEP centre; certify the **hand component** at maximum improvement.

Restore, protect, then adapt: if the hand cannot be restored, change the tool or the task.

7 marks Role of NGOs in the rehabilitation team

Asked: S19 (2c)

An **NGO** is a **voluntary, non-profit** organisation run independently of the government, registered as a society or trust. NGOs work alongside government services and are important members of the wider rehabilitation team, especially in CBR.

Roles

1. **Identification and awareness:** surveys and early identification of persons with disability; awareness to reduce stigma.
2. **Direct services where government does not reach:** therapy, camps, aids and appliances, special and inclusive schools.
3. **Training:** CBR workers, parents and volunteers.
4. **Livelihood:** vocational training, self-help groups.
5. **Advocacy:** rights, accessibility and policy; linking people to government schemes.
6. **Resources:** raising funds; research; disaster relief.
7. **Innovation:** flexible and close to the community, so they often pilot programmes that government later adopts.
8. **Partnership with government:** the Deendayal Disabled Rehabilitation Scheme gives grants to NGOs to run rehabilitation services.

Examples

NGO	Work
ADAPT (Spastics Society of India)	Education and therapy for children with cerebral palsy; training; inclusive education
Indian Red Cross Society	Disaster relief, blood banks, first aid, care of the sick and disabled
BMVSS, Jaipur	**Jaipur foot** and other aids, distributed free
HelpAge India	Older people

Others: Amar Seva Sangam, Mobility India, CBM. In the team, NGO workers sit alongside CBR workers, families and professionals, and help carry rehabilitation from the institution to the home.

NGOs bridge institutions and communities: they run outreach camps, refer to hospitals and support people at home.

4(c) National policies and legal rights for persons with disability

Full notes p. 144

3 marks Difficulties faced by a community-dwelling person with disability

Asked: W17 (1f)

- **Physical:** steps, narrow doors, inaccessible toilets, uneven roads, no ramps or lifts.
- **Transport:** buses and autos hard to board; cost of private transport.
- **Attitudes:** stigma, pity, discrimination, overprotection; difficulty in marriage.
- **Economic:** unemployment, poverty, cost of aids and treatment.
- **Services:** distant health and rehabilitation centres; lack of information about schemes; paperwork for certificates.
- **Education and communication:** inaccessible schools; communication barriers for deaf and blind persons.
- **Social:** isolation, dependence on family, risk of abuse.

In the social model these barriers, not the impairment, turn a disability into a handicap.

3 marks Persons with Disabilities Act, 1995

Asked: S18 (1b), S23 (1e), S24 (1b) (asked 3 times)

Full name: **Persons with Disabilities (Equal Opportunities, Protection of Rights and Full Participation) Act, 1995**.

- **7 disabilities:** blindness, low vision, leprosy-cured, hearing impairment, locomotor disability, mental retardation, mental illness.
- **Benchmark:** 40% or more.
- **3%** of government jobs reserved.
- **Free education up to 18 years**; 3% seats in government-aided institutions.
- Prevention and early detection; non-discrimination in transport and buildings; aids; unemployment allowance; Chief and State Commissioners.

Replaced by the **RPWD Act 2016**: 21 specified disabilities, **4%** job reservation, at least 5% seats in higher education, in line with the UN Convention (2006).

3 marks Three national programmes for prevention of disability

Asked: S20 (1e), W21 (1b), W24 (1e) (asked 3 times)

Programme	Prevents
Universal Immunisation Programme (1985) with Pulse Polio (1995)	**Polio** paralysis and disability from measles, rubella (congenital rubella), diphtheria, tetanus
National Programme for Control of Blindness and Visual Impairment (1976)	**Blindness** from cataract, refractive error, glaucoma; childhood blindness

Programme	Prevents
National Programme for Prevention and Control of Deafness (2006)	**Hearing loss**: early detection, treatment, hearing aids

Others: National Leprosy Eradication Programme (1983); National Iodine Deficiency Disorders Control Programme (iodised salt); Rashtriya Bal Swasthya Karyakram (2013, child screening); National Programme for Health Care of the Elderly (2010).

7 marks Architectural barriers faced by persons with disability and their modifications

Asked: S18 (2c)

Architectural barriers are features of buildings, roads and transport that stop a person with disability from entering, moving about or using a place. In the **social model** they are a main cause of handicap. The **RPWD Act 2016** gives a right to accessibility (buildings, transport, ICT), and the **Accessible India Campaign** (Sugamya Bharat Abhiyan, 2015) promotes accessible buildings, transport and websites. Measurements follow the **Harmonised Guidelines (2021)**.

Barriers and modifications

Barrier	Modification
Steps at the entrance	**Ramp** no steeper than **1:12**, clear width at least **1,200 mm**, handrails on both sides (about **760–900 mm** high) extending beyond the ends, landing at top and bottom
Narrow doors	Clear width about **900 mm**; lever handles; no high thresholds
No turning space	Clear **turning circle of 1,500 mm** in rooms, toilets and corridors
Inaccessible toilet	Western toilet at wheelchair seat height, **grab bars**, washbasin without pedestal, outward-opening door, non-slip floor
Upper floors	**Lift** with wide doors and low buttons; ground-floor rooms
High counters and switches	Lowered counters, switches and handles within reach from a wheelchair
Pavements and roads	**Kerb cuts**, even paths, **tactile paving** for blind persons, audible signals
Signs and information	Large clear signs with contrast, **Braille**, sign-language information

Role of the physiotherapist

- **Accessibility audit** of the home, school, workplace and public buildings.
- Advice on ramps, rails, toilet changes and furniture height; training in wheelchair skills for the barriers that remain.
- Linking the family to the panchayat or schemes for a ramp and toilet.

> **Draw:** A ramp with a 1:12 gradient, handrails and landings; an accessible toilet with grab bars and a 1,500 mm turning circle.

4(e, f) Community-based rehabilitation: principles, strategies and the physiotherapist

Full notes p. 150

3 marks Aspects (features) of community-based rehabilitation

Asked: S16 (1b), W24 (1d) (asked 2 times)

CBR (ILO, UNESCO and WHO, 2004): "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities".

- Takes place in the person's **own home, village or town**, using **local resources** and simple technology.
- The person and **family are active partners** and decide goals.
- **Community workers** (CBR workers, anganwadi workers, ASHAs, teachers, volunteers) do most of the work, trained and supported by professionals.
- Covers **health, education, livelihood, social life and empowerment**, not only therapy.
- Works with existing services (PHC, schools, panchayats, schemes) and **refers** to institutions when needed.
- **Low cost, wide coverage**, sustained by the community.

3 marks Principles of CBR

Asked: W21 (1a), S22 (1a), S25 (1a) (asked 3 times)

CBR is rehabilitation within the community, by the community, using local resources (ILO, UNESCO and WHO, 2004).

- **Community participation and ownership:** the village plans and runs the programme (CBR committee, gram panchayat).
- **Inclusion and equal opportunity** in ordinary schools, jobs and festivals.
- **Empowerment and self-advocacy** through self-help groups and disabled people's organisations.
- **Local resources and appropriate technology:** aids made by the village carpenter.
- **Multisectoral approach**; **integration with existing services** (PHC, anganwadi, school).
- **Two-way referral** with PHC, district hospital, DDRC.
- **Sustainability**; **gender equality** and respect for children.

3 marks Role of the anganwadi worker in CBR

Asked: S17 (1b), S22 (1d) (asked 2 times)

The anganwadi worker is the village-level worker of **ICDS** (1975), serving children under 6, pregnant women and nursing mothers.

- **Identification:** house-to-house survey, growth monitoring, milestone checks.
- **Referral:** to PHC, RBSK team and **DEIC**; helps with certificates and UDID.
- **Prevention:** immunisation, supplementary nutrition, vitamin A, iodised salt, antenatal care.
- **Home programmes** taught by the physiotherapist: exercises, positioning, stimulation.

- **Pre-school inclusion**; counselling mothers; records and links to schemes.

7 marks Principles of CBR and team members of CBR

Asked: S17 (2d)

CBR (ILO, UNESCO and WHO, 2004): "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities", carried out by persons with disabilities, their families and communities with health, education, vocational and social services.

Principles

Principle	Meaning in practice
Community participation and ownership	The village identifies needs, plans and runs the programme (CBR committee, gram panchayat)
Inclusion and equal opportunity	Ordinary schools, jobs, festivals and decisions; no discrimination
Empowerment and self-advocacy	Self-help groups and disabled people's organisations
Local resources, appropriate technology	Aids made by the village carpenter or blacksmith; exercise with household objects
Multisectoral approach	Health, education, social welfare, livelihood and panchayat together
Integration with existing services	Through primary health care, anganwadis and schools
Referral system	Two-way links with PHC, district hospital, DDRC, national institutes
Sustainability	Low cost, trained local people, community funds
Gender equality	Equal attention to women and girls; respect for children

Team members

Level	Members
Family and village	**Person with disability**, family, **CBR worker** or volunteer, **anganwadi worker**, **ASHA**, teacher, self-help group, gram panchayat, local craftsmen
Block	PHC and CHC staff (doctor, ANM), multipurpose rehabilitation worker or CBR supervisor, block resource centre, RBSK mobile teams
District	District hospital, DEIC, **DDRC**, **physiotherapist**, occupational therapist, prosthetist and orthotist, speech therapist, psychologist, social worker, NGOs
State and national	National institutes and composite regional centres; government departments

CBR teams work in the **transdisciplinary** way: one key worker (CBR worker or family) delivers the programme, trained and supported by the professionals.

7 marks Benefits of teamwork in CBR

Asked: S20 (2b)

In **CBR**, rehabilitation is carried out by persons with disability, their families and communities together with health, education, vocational and social services. No single worker can meet all needs, so CBR depends on a **team**.

The team

- **Village:** person and family, CBR worker, anganwadi worker, ASHA, teacher, self-help group, panchayat, local craftsmen.
- **Block and district:** PHC doctor and ANM, CBR supervisor, DEIC, DDRC, physiotherapist, occupational therapist, prosthetist, psychologist, social worker, NGOs.
- **National:** national institutes and composite regional centres.

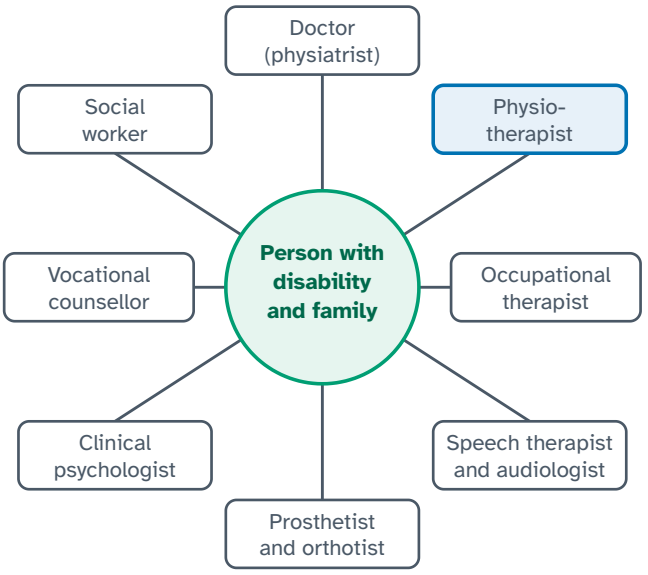
How the team works

Multidisciplinary (each works separately) → interdisciplinary (shared findings, common goals) → **transdisciplinary** (one key worker delivers the programme, trained by the others): the usual model in CBR.

Benefits

1. **Whole-person care:** disability touches health, school, work and social life, so many sectors act together.
2. **Wide reach:** few professionals train and support many local workers.
3. **One consistent plan:** shared goals avoid repeated assessments and conflicting advice.
4. **Early identification** by village workers and quick **referral** both ways; follow-up close to home.
5. **Lower cost:** local resources and schemes are pooled, so the programme lasts.
6. **Attitude change:** when many local people are involved, stigma falls.
7. **Empowerment:** the person and family are team members who set goals; the community owns the programme.

The rehabilitation team



The person and family (green) are members of the team and set the goals; the physiotherapist is shown in blue. Add the nurse, the special teacher and, in CBR, the CBR worker and community.

Teamwork makes CBR cheaper and sustainable, with continuity of care from village to institution.

7 marks Community-based rehabilitation

Asked: W20 (2e)

Definition (ILO, UNESCO and WHO, 2004): "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities", carried out by people with disabilities, their families and communities together with health, education, vocational and social services.

History

The WHO promoted CBR after **Alma-Ata (1978)**, because most disabled people in low-income countries could not reach

institutions; joint position papers (**1994, 2004**) widened it to rights and inclusion; the **WHO CBR guidelines (2010)** set out the CBR matrix.

Features

- In the person's own home and village, with **local resources** and simple technology.
- Person and **family are partners**; community workers deliver most of the work.
- Works through existing services; **refers** to institutions when needed; low cost, wide coverage.

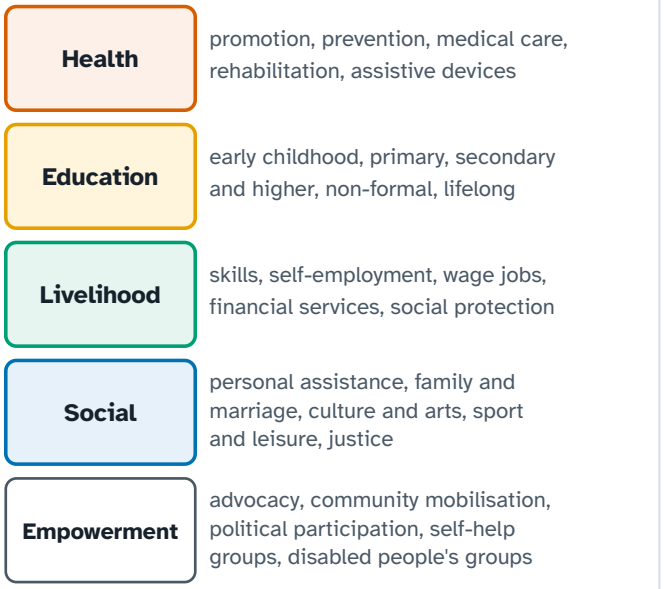
Principles

Community participation; inclusion; empowerment; local resources; multisectoral approach; integration with existing services; referral; sustainability; gender equality.

CBR matrix (WHO, 2010)

Component	Elements
Health	Promotion, prevention, medical care, rehabilitation, assistive devices
Education	Early childhood, primary, secondary and higher, non-formal, lifelong
Livelihood	Skills, self-employment, wage jobs, financial services, social protection
Social	Personal assistance, family and marriage, culture and arts, sport and leisure, justice
Empowerment	Advocacy, community mobilisation, political participation, self-help groups, disabled people's groups

Five components of CBR and their elements



In the exam, draw the five components as five columns side by side with their elements underneath; empowerment runs across all of them.

Steps of a programme

Situation analysis and household survey → plan with a CBR committee → train local workers and families → implement (home visits, camps, aids, school inclusion, livelihood) → monitor, evaluate and hand over.

Add two lines comparing CBR with IBR: low cost, wide coverage, skills learnt in the real home.

7 marks CBR strategies in rural and urban areas

Asked: W19 (2e)

CBR is rehabilitation within the community, using local resources and existing services, with the person and family as partners (ILO, UNESCO and WHO, 2004). Strategies differ because rural and urban areas have different resources.

Steps of a CBR programme (both settings)

1. **Situation analysis:** meet leaders, map resources, household survey to identify persons with disability.
2. **Plan with the community:** CBR committee; priorities from the CBR matrix.
3. **Train** local workers and families in simple assessment, home programmes and referral.
4. **Implement:** home visits, camps, school inclusion, aids, livelihood links, self-help groups.
5. **Monitor and evaluate**; hand over to local ownership.

Urban strategies

- **Urban health centres** and community centres for clinics and camps.
- **Clubs, mahila mandals** and social centres for awareness, volunteers and funds.
- **Ordinary schools** for inclusion and screening.
- **Industries** for vocational training, jobs and workplace adaptation.
- **Sports centres** for adapted sport; NGOs and **corporate social responsibility** funds.

Rural strategies

- **PHCs**, rural hospitals and the district hospital for screening, treatment and referral.
- **Anganwadi workers and ASHAs** for identification and home programmes.
- **Locomotor aids from local materials** made by village craftsmen: bamboo crutches, wooden walkers, parallel bars, standing frames, CP chairs.
- **Panchayat** support; **self-help groups** for livelihood.
- **Camps** for disability certificates and aids.

End with two-way referral links (village → PHC → DDRC → national institute) and sustainability through community ownership.

7 marks Approaches to CBR

Asked: S26 (2b)

CBR (ILO, UNESCO and WHO, 2004) is "a strategy within general community development for the rehabilitation, equalization of opportunities, poverty reduction and social inclusion of all people with disabilities". It began after Alma-Ata (1978) as service delivery through primary health care and has widened into community development. Programmes differ by **who controls them** and **where the work is done**; most mix several approaches.

A. By who controls it

Approach	How it works	For and against
Top-down (government-led)	Planned from outside; workers follow set packages (Iran, Viet Nam; DDRC camps)	Easy to run in many places; more money, reach, sustainability. But weak local commitment
Bottom-up (community, NGO or disabled people's groups)	Disabled people and families decide (PROJIMO, Mexico; Mobility India)	Flexible, fits culture, strong ownership. But fewer resources, smaller reach

B. By where the work is done

1. **Home-based:** a local supervisor teaches a family member (WHO 1989 manual). For: reaches remote homes, skills learnt where used, cheap. Against: no back-up for aids or specialist help; heavy for poor families.
2. **Community centre-based:** village centre, anganwadi or early intervention centre. For: low-cost base, aids made locally, back-up for home work. Against: travel; fewer homes reached.
3. **Outreach from institutions:** camps and mobile teams from hospitals, national institutes, DDRCs. For: specialist skill near home. Against: periodic visits only.
4. **Integrated with primary health care:** PHC staff, ASHAs, anganwadi workers, RBSK. For: existing staff, wide and cheap. Against: tends to stay medical.
5. **Community development and rights-based:** all five matrix components; self-help groups and DPOs lead; **twin-track** (mainstream inclusion plus specific support). For: tackles poverty, school, work. Against: complex; evidence limited.

Indian example: Amar Seva Sangam (rural Tamil Nadu) uses community rehabilitation workers at home or at a local centre, guided by specialists through an app.

Conclusion: CBR works best with government support that stays sensitive to local culture and disabled people's organisations (WHO, 2010); home work needs a local centre and referral behind it.

7 marks Role of the anganwadi worker in CBR

Asked: S26 (2e)

The **anganwadi worker** is the village-level worker of the **Integrated Child Development Services (ICDS, 1975)**. She runs an anganwadi centre for about **400–800 people** (300–800 in tribal and hilly areas) with one helper, and serves **children under 6**, **pregnant women** and **nursing mothers**. Being in every village, she is a key member of the CBR team.

Area	Role in CBR
Identification	House-to-house survey; **growth monitoring** and simple **milestone checks**; spotting children with delay or disability early
Referral	To the PHC, **RBSK** mobile team and **District Early Intervention Centre**; helps with disability certificates and UDID
Prevention	**Immunisation** (polio, measles), supplementary nutrition, vitamin A, iodised salt, anaemia control, antenatal care, safe delivery, birth spacing
Home programmes	Carries out simple exercises, positioning and stimulation **taught by the physiotherapist**; checks aids
Pre-school inclusion	Includes children with disability in pre-school play and learning; prepares them for school
Education and awareness	Counsels mothers, reduces stigma, mothers' meetings
Records and links	Keeps registers; links families with schemes, ASHA, ANM and the panchayat

Link with the physiotherapist

The physiotherapist trains her in simple assessment, positioning, exercises and when to refer, and reviews the children she identifies; the anganwadi can also serve as a training place.

She works at all three levels of prevention: primary (immunisation, nutrition), secondary (early detection and referral) and tertiary (home programmes, inclusion).

7 marks Role of the physiotherapist in a school for physically handicapped children

Asked: W22 (2f)

Children may have cerebral palsy, spina bifida, muscular dystrophy, limb deficiency or the after-effects of polio. The aim is to help each child **move, learn and take part** in school life, working with the special educator, teachers, parents, OT, speech therapist and doctor.

1. **Screening and assessment:** GMFCS level, tone, contractures, strength, posture and spine, balance, gait, hand function, self-care; goals in the **individual education plan**.
2. **Individual programmes:** individual and group therapy in the school day: stretching, strengthening, balance, gait, play; a written home programme.
3. **Seating and positioning:** adapted chair and table (pelvis level, feet supported); change position through the day; **standing frame 30–60 min a day** if the child cannot stand.
4. **Mobility aids:** prescribe, fit and check AFOs, walkers, crutches, wheelchairs; wheelchair skills.
5. **Accessibility:** ramps no steeper than **1:12** with handrails, accessible toilets, ground-floor classes, safe playground.
6. **Training:** teachers and attendants in safe lifting, transfers and positioning; parents in the home programme.
7. **Sports and play:** adapted physical education; WHO (2020): children 5–17, including those with disability, an average of **60 min a day** of moderate-to-vigorous activity, with vigorous and strengthening activity on at least **3 days a week**, as able.
8. **Prevention:** contractures, hip and spine deformity, obesity, pressure sores; early orthopaedic referral.
9. **Inclusion and transition:** prepare children for the neighbourhood school (seating near door and board, peer buddy, extra time); later, vocational training.

Law and schemes

- **RPWD Act 2016:** section 16, schools must admit without discrimination, make buildings accessible and give reasonable accommodation, sport and transport; section 17, survey every 5 years and free assistive devices up to 18 years; section 31, **free education 6–18 years** in a neighbourhood or special school of choice.
- **Samagra Shiksha** (pre-school to class XII): assessment camps, aids, escort and transport allowance, stipend for girls, therapy at block level, ramps and toilets, home-based education for severe disability.
- Disability certificate and UDID, ADIP aids, RBSK school screening and DEIC.

The goal is not only therapy but full participation in learning and school life.

4 Rehabilitation of an amputee in the community

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7 marks Psychosocial issues of being an amputee

Asked: W17 (2c)

Losing a limb affects not only walking but also the person's mind, body image, family, work and place in society. These issues decide whether the amputee uses the prosthesis and returns to normal life.

Psychosocial issues

1. **Grief and loss:** shock, denial, anger, sadness, then gradual acceptance; more severe after sudden trauma.
2. **Depression and anxiety**; **post-traumatic stress** after accidents.
3. **Body image:** embarrassment, hiding the limb, low self-esteem.
4. **Phantom sensation** and **phantom limb pain**; stump pain.
5. **Loss of role and income:** unable to return to work (drivers, farmers, labourers); financial strain on the family.
6. **Dependence** on others; strain on family relationships; **sexual and marital** concerns.
7. **Stigma and social isolation**; difficulty in marriage for young people; cultural beliefs (punishment, bad luck).
8. **Practical worries:** cost and repair of the prosthesis, travel to centres, inaccessible homes and toilets.

Management

- **Counselling** and psychology referral.
- **Peer support:** meeting a successful amputee; amputee groups.
- **Early prosthetic fitting** and good training, which restore mobility and confidence (Jaipur foot allows squatting and cross-legged sitting).
- **Phantom pain:** explanation, desensitisation, **mirror therapy**, TENS, medication by the doctor.
- **Involve the family**; vocational rehabilitation and adapted work.
- Information on **concessions and schemes** (UDID, free prosthesis under ADIP or from BMVSS); sport and leisure.

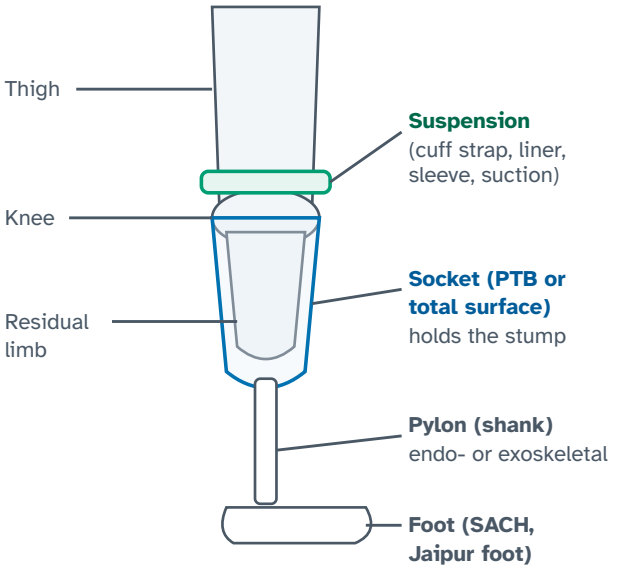
A confident amputee walks better: start counselling and peer contact in the pre-prosthetic stage, not after the limb is fitted.

7 marks Prosthetic check-out for a below-knee amputee

Asked: S18 (2d)

Check-out is the systematic examination of a new prosthesis on the amputee, to confirm **fit, comfort, function and safety** before final delivery. It has static (before donning, standing, sitting), dynamic (walking) and after-removal parts.

Parts of a below-knee prosthesis



The socket (blue) carries weight on pressure-tolerant areas; the suspension (green) keeps it on; the pylon and foot replace the leg and ankle.

1. Before donning

Prosthesis as prescribed; finish and weight; socket smooth inside; suspension and foot correct; stump skin intact.

2. Standing

- Comfortable with **equal weight** on both legs; **pelvis level** (correct length).
- Knee in **slight flexion**; foot flat with slight toe-out like the other foot.
- Socket in **total contact** without gaps; patellar tendon bar at the right level; suspension holds; pylon vertical in both views.

3. Sitting

Sits with the knee bent about **90°** comfortably; no pinching behind the knee by the posterior wall; prosthesis stays on; shoe flat on the floor.

4. Walking

Smooth gait, equal step length, no pain, **no pistoning**; stairs and slopes. Look for deviations:

Deviation	Prosthetic cause
Excess knee flexion at heel strike	Foot too dorsiflexed or too far back; socket too flexed; heel too hard
Knee stays straight	Foot too plantarflexed or too far forward; heel too soft
Lateral thrust of pylon	Foot **inset**
Medial thrust	Foot **outset**
Early drop-off	Keel too short or foot too far back
Pistoning	Poor suspension or loose socket (add socks or new socket)

5. After removal

Redness only over **pressure-tolerant areas** (patellar tendon, medial tibial flare, pretibial muscles), fading within **10–20 minutes**; no abrasions or blisters; stump sock not wrinkled.

Sensitive areas to relieve in a PTB socket: tibial crest and tubercle, fibular head, distal tibia and fibula, hamstring tendons.

Part 5: Disaster management

5 Disaster management

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7 marks Types of disaster and principles of disaster management

Asked: W19 (2a), S24 (2a) (asked 2 times)

Disaster (Disaster Management Act, 2005): a catastrophe, mishap, calamity or grave occurrence, from natural or man-made causes, or by accident or negligence, that causes substantial loss of life, human suffering, or damage to property or the environment, and is **beyond the coping capacity** of the affected community.

A disaster happens when a **hazard** (the threat) meets a **vulnerable** community: **risk = hazard × vulnerability ÷ capacity**.

Types

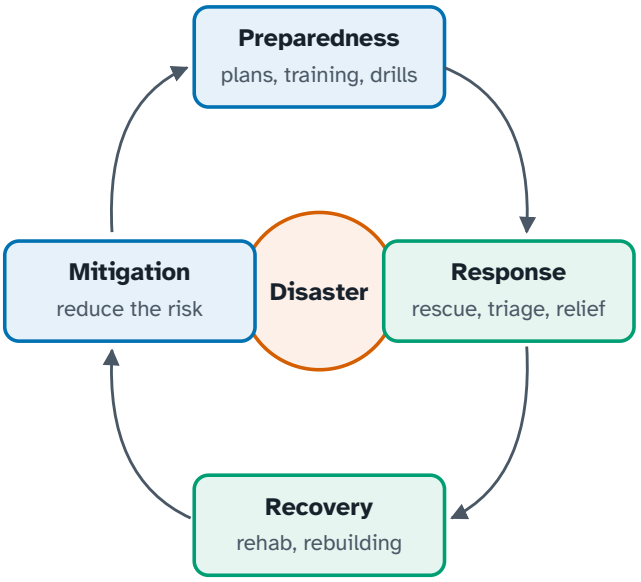
Type	Examples (India)
Natural: geophysical	Earthquakes (Latur 1993, Bhuj 2001), tsunami (2004), landslides
Natural: hydrological and weather	Floods (Kerala 2018), cyclones (Odisha 1999), cloudbursts, droughts, heat and cold waves
Natural: biological	Epidemics and pandemics (COVID-19)
Man-made: technological	Industrial and chemical accidents (**Bhopal gas leak 1984**), fires, building collapse, nuclear accidents
Man-made: transport and crowds	Rail and road accidents, stampedes
Man-made: conflict	Terrorism, bomb blasts, war, riots

Also by **onset** (sudden: earthquake; slow: drought) and by **scale** (local, state, national).

Principles of disaster management

1. **Comprehensive:** all hazards, all phases, all people affected.
2. **Progressive:** anticipate and act before disasters (preparedness, mitigation).
3. **Risk-driven:** priorities set by hazard and vulnerability assessment.
4. **Integrated and coordinated:** all levels of government, agencies, NGOs and communities under one plan and command.
5. **Collaborative:** trust and partnership among organisations and the community.
6. **Flexible:** adapt to each situation.
7. **Professional:** trained people, standards, evidence.
8. **Community participation and inclusion:** women, older people and persons with disability in planning; **"build back better"** in recovery (Sendai Framework).

The disaster management cycle



Blue phases come before a disaster, green phases after it. Recovery feeds back into mitigation ("build back better").

In India: DM Act 2005 set up the **NDMA** (chaired by the Prime Minister), State and District Authorities, the **NDRF** and the National Institute of Disaster Management; National Disaster Management Plan (2016) aligned with the Sendai Framework (2015–2030).

Link the principles to the cycle: mitigation → preparedness → response → recovery.

Part 6: Industrial health

6(a) Industrial health: models, the team and the worker care spectrum

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7 marks Role of the physiotherapist in industrial health

Asked: W19 (2c)

Occupational health (Joint ILO/WHO Committee, 1950, revised 1995) aims at the highest degree of physical, mental and social well-being of workers, prevention of ill health caused by working conditions, protection from risks, and **the adaptation of work to people and of each person to their job**.

Industrial therapy is the application of physiotherapy to workers and workplaces: preventing work injury, treating work-related disorders close to the job, and returning injured workers to safe, productive work. The physiotherapist works in the **industrial model**: at or near the workplace, focused on the worker's function in the job and on prevention, with the employer as a partner and **early return** with modified duties.

Roles

Area	What the physiotherapist does
Pre-placement	Screens new workers against **job demands**; records a baseline; advises adaptations
Job analysis and ergonomics	Job and task analysis; ergonomic evaluation (**REBA, RULA, NIOSH**); workstation redesign
Prevention and education	**Back schools**, safe lifting, posture, **micro-breaks**, workstation exercises, warm-up before heavy shifts
Health promotion	**Employee fitness programmes**; weight and NCD risk; stress management
Early intervention	**On-site clinic** to treat aches early, before they become disorders; first aid

Area	What the physiotherapist does
Disability management	**FCE**, work conditioning, work hardening, modified duties, **graded return**
Hazard control	Advice on vibration, manual handling, standing work, PPE
Records and research	Injury data, programme evaluation, cost–benefit reports to management
Inclusion	Adapts work for older workers and persons with disability; **reasonable accommodation** under the RPWD Act

Team work: with the worker, employer and supervisor, occupational health physician, occupational therapist, ergonomist and safety officer, psychologist, vocational counsellor and case manager (ESIC).

Indian context: most workers are in the informal sector (farms, construction, small units) with little protection; the Factories Act, 1948 and the ESI Act, 1948 cover health, safety and benefits.

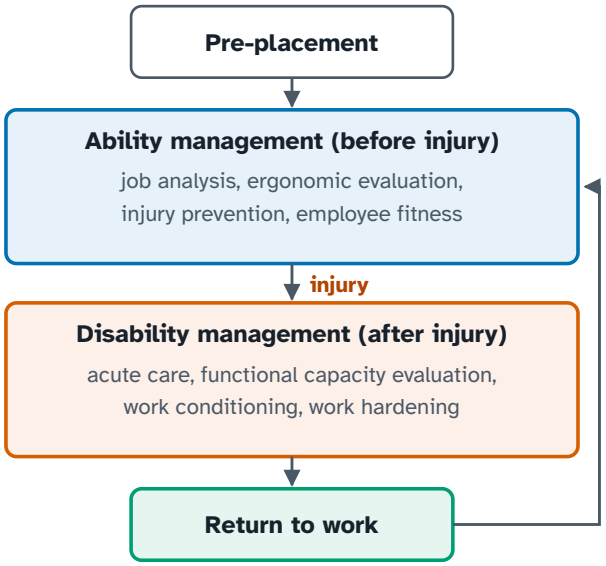
Outcome: fewer injuries, days lost and compensation costs; better productivity.

7 marks Pre-placement in the worker care spectrum

Asked: S25 (2d)

The **worker care spectrum** runs from hiring to return to work: **pre-placement → ability management** (before injury: job analysis, ergonomic evaluation, injury prevention, employee fitness) → injury → **disability management** (acute care, FCE, work conditioning, work hardening) → **return to work**, which links back to ability management.

Worker care spectrum: from hiring to return to work



This page covers the blue half; the orange half is on disability management. After return to work, the cycle starts again with ergonomics and prevention.

Pre-placement (post-offer) screening

The **first step**, long before any injury: **after a job is offered**, the worker's health and physical abilities are tested against the **essential demands of that job**.

Aims

- Place the worker in a job they can do **safely**.
- Find needs for training, aids or workplace changes.
- Record a **baseline** for comparison if the worker is later injured.
- Reduce injuries and compensation costs.

Components

1. **Job demand analysis** of the post: weights, postures, frequency, environment.
2. **Health questionnaire and medical examination:** past back or joint problems, heart and lung disease, vision and hearing.
3. **Musculoskeletal screen:** posture, range, strength, flexibility.
4. **Job-matched functional tests:** lifting and carrying to the job's weights and heights; pushing and pulling; positional tolerance (stooping, kneeling, climbing, overhead work); grip; aerobic fitness.
5. **Result:** fit; fit with adjustments (training, tools, rotation); temporarily unfit with a plan to improve. Report to the employer in **functional terms, not diagnoses**.

Rules

- Tests must be **job-related, standardised and safe**; the same tests for all candidates for a post.
- Must not exclude persons with disability unfairly: the employer must consider **reasonable accommodation** (RPWD Act, 2016).
- Results are **confidential**.

Example: a warehouse loader is tested on floor-to-waist lifts and carrying at the weights of the actual job.

6(b i) Ability management: job analysis, ergonomics and injury prevention

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3 marks Workplace health promotion

Asked: S18 (1c)

Workplace health promotion (Luxembourg Declaration, 1997): "the combined efforts of **employers, employees and society** to improve the health and well-being of people at work".

It works through:

- Better **work organisation and working environment**.
- **Active participation** of workers.
- **Personal development**.

Examples: ergonomic workplaces; fitness and health checks; healthy canteens; **tobacco-free workplace**; stress management; flexible hours.

Physiotherapy role: employee fitness programmes, back care and posture education, workstation exercises; benefits are fewer injuries and sick days and better productivity.

3 marks Ergonomic physiotherapy evaluation

Asked: S19 (1b)

A systematic assessment of the worker, job and workstation by the physiotherapist to find **musculoskeletal risk factors** and correct them.

1. Meet management and workers; injury and absence records; **Nordic Musculoskeletal Questionnaire**.
2. **Walk-through observation** and video; job and task analysis.
3. **Measure** workstation heights, reaches, seat and screen; loads, forces, frequency; light, noise, temperature.
4. **Score risk:** RULA, REBA, NIOSH lifting equation, Strain Index.
5. **Recommend changes** by the hierarchy of controls: redesign, aids, layout, breaks and rotation, training, exercise.
6. **Re-evaluate** after the changes.

3 marks Employee fitness programme

Asked: W19 (1d)

A workplace programme to improve workers' **health and fitness** and reduce injuries and absence.

- **Plan:** needs assessment and management support; a committee with workers.
- **Assess:** health risk appraisal and fitness tests (BP, BMI, waist, aerobic fitness, strength, flexibility).
- **Programme:** on-site exercise classes or gym, walking groups, **stretching breaks**, sports; education on back care, nutrition, weight, smoking, stress and sleep.
- **Support:** incentives, convenient timing, family involvement.
- **Evaluate:** participation, fitness changes, injuries, absence, productivity.

Benefits: fewer musculoskeletal injuries and sick days; better fitness, morale and productivity; lower health costs.

3 marks Workstation exercises

Asked: S20 (1b)

Short exercises done at the desk or workplace, **5–10 repetitions every 30–60 minutes**, to break static posture in sedentary workers.

- **Neck:** chin tucks; side bending and turning, holding **10 seconds**.
- **Shoulders and upper back:** shoulder rolls and shrugs; squeeze the shoulder blades back; overhead reaches.
- **Back:** seated back extension over the chair back; seated twists; standing back bends; pelvic tilts in sitting.
- **Wrists and hands:** flexor and extensor stretches; fist and spread fingers; tendon gliding.
- **Legs:** ankle pumps; seated knee straightening; heel raises; marching.
- **Eyes:** look into the distance; blink; palming.

3 marks Job demand analysis

Asked: S21 (1c)

A systematic record of the **physical, mental and environmental demands** of a job, with weights, distances and **how often** each occurs; one of the three parts of job analysis.

- **Physical:** lifting, carrying, pushing and pulling; sitting, standing and walking time; bending, kneeling, squatting, climbing, reaching above shoulder; gripping, fine finger work, repetition; vibration.
- **Mental and sensory:** attention, memory, decisions, vision, hearing, speed and deadlines.
- **Environment:** heat, cold, noise, light, dust, fumes, heights, wet floors, shift work.
- **Frequency** (Dictionary of Occupational Titles): occasionally up to **a third** of the day; frequently **a third to two-thirds**; constantly **more than two-thirds**.

Uses: physical demand level of the job; pre-placement; ergonomic change; matching with the FCE; return to work.

3 marks Job analysis

Asked: S26 (1a)

Job analysis: a systematic study of a job to record what the worker does, what it demands and where the risks lie. It has three parts:

Part	What it records
Job description	Title, purpose, main duties, tools and machines, materials, hours and shifts, breaks, supervision, pay system (piece rate or salary)
Job demand analysis	Physical, mental and environmental demands and how often each occurs (occasionally, frequently, constantly)
Task analysis	Each task in steps with posture, force, repetition, duration, tools and environment; risk factors found

Purpose: prevention and ergonomic redesign; pre-placement; matching with the FCE; designing work hardening; light duties; training; compensation.

7 marks Importance of ergonomic evaluation at the workplace

Asked: S16 (2c)

Ergonomics (International Ergonomics Association, 2000) is the scientific discipline concerned with the understanding of interactions among humans and other elements of a system; simply, **fitting the job to the person**. An **ergonomic evaluation** studies the worker, the tasks, the workstation and the environment to find risk factors and correct them.

Importance

- Identifies **risk factors before injury** occurs; prevents work-related musculoskeletal disorders, accidents and absence.
- Improves **comfort, productivity and quality**; reduces fatigue, errors and **compensation costs**.
- Gives **objective scores** to prioritise and justify changes, and to show improvement after them.
- Helps **return injured workers** to work and **include older workers and workers with disability**.
- Meets the employer's **legal duty** to provide a safe workplace.
- Forms the basis for redesign, training, rest breaks, rotation and workstation exercises.

Steps of the evaluation

1. Meet management and workers; review **injury and absence records**; symptom survey with the **Nordic Musculoskeletal Questionnaire**.
2. **Walk-through observation** and video of the job; job and task analysis.
3. **Measure** the workstation (heights, reaches, seat, screen), loads, forces, frequency and environment (light, noise, temperature).
4. **Score risk** with a tool: **RULA, REBA, NIOSH lifting equation, Strain Index**.
5. **Recommend changes** from the hierarchy of controls: redesign, tools and aids, layout, breaks and rotation, training, exercise.
6. **Re-evaluate** after the changes.

Example: a REBA score of 11–15 for a loading task means very high risk, action needed now; a lower score after redesign shows the change worked.

7 marks Importance of task analysis

Asked: S19 (2e)

Job analysis has three parts: **job description**, **job demand analysis** and **task analysis**. **Task analysis** breaks each task into steps and records, for each step, the **posture, force, repetition, duration, tools and environment**, with the risk factors found.

How it is done

Walk-through observation and video of the worker; list the tasks; break each into steps; note risk factors; score with REBA, RULA or the NIOSH lifting equation.

Importance

1. Finds the **exact steps and postures** that cause pain or injury, so the fix is specific.
2. Basis for **ergonomic redesign** of tools, workstation and methods.
3. **Matches the job to the worker** in pre-placement and return to work (compared with the functional capacity evaluation).
4. Designs job-specific **work conditioning and work hardening** (job simulation).
5. Shows what **light duties** are possible during recovery; guides **job rotation**.
6. Basis for **training safe methods** and setting **fair production rates**.
7. Supports **compensation** decisions.

Example

A labourer lifting **50 kg bags** into a truck at about **3 feet**: steps of bending to the floor, lifting, twisting and loading. The load is far above the NIOSH recommended limit of **23 kg**, so the task must change: 25 kg bags or two-person lifts, trolleys or a ramp, loading from waist height, rotating tasks.

Task analysis tells you what to change; FCE tells you what the worker can do. Compare the two before return to work.

7 marks Define ergonomics; describe one ergonomic scale for musculoskeletal risk (REBA)

Asked: W19 (2b), S23 (2b) (asked 2 times)

Ergonomics

International Ergonomics Association (2000): "the scientific discipline concerned with the understanding of interactions among humans and other elements of a system", and the profession that applies theory and methods to design in order to optimise **human well-being** and **overall system performance**. Simply, **fitting the job to the person**. Domains: **physical, cognitive, organisational**.

Rapid Entire Body Assessment (REBA)

Developed by **Hignett and McAtamney (2000)** to score **whole-body postures** in tasks that are unpredictable or varied (health care, service, industry).

1. **Group A:** trunk, neck and legs, plus a **load or force** score → score A.
2. **Group B:** upper arm, lower arm and wrist, plus a **coupling (grip)** score → score B.
3. Scores A and B give **score C** from a table.
4. An **activity score** is added (static posture held over a minute, repeated small actions, or rapid large posture changes) → **REBA score 1–15**.

REBA score	Risk	Action
1	Negligible	None needed
2–3	Low	May be needed
4–7	Medium	Needed
8–10	High	Needed soon
11–15	Very high	Needed now

Use

- Scores lifting, loading and patient-handling postures quickly by observation.
- Ranks jobs for change; re-scoring after redesign shows improvement.
- Related tools: **RULA** for upper-limb seated or standing work (score 1–7); **NIOSH lifting equation** (23 kg recommended limit in ideal conditions).

Viva: group A is trunk, neck and legs; a score of 11–15 means act now.

7 marks Cognitive ergonomics

Asked: S21 (2f)

Ergonomics (International Ergonomics Association, 2000): the scientific discipline concerned with understanding interactions among humans and other elements of a system, applied to optimise **human well-being** and **system performance**. It has three domains:

Domain	Concerned with
Physical	Anatomy, anthropometry, physiology and biomechanics: postures, manual handling, repetition, musculoskeletal disorders, workplace layout
Cognitive	Mental processes (perception, memory, reasoning, motor response): mental workload, decision-making, skilled performance, human error, work stress, training
Organisational	Work systems: shift design, teamwork, communication, work pace, participation

Areas of cognitive ergonomics and solutions

- **Mental workload:** overload causes errors and stress; underload causes boredom and lapses. Balance tasks and breaks.
- **Attention and alertness:** monotonous monitoring, **night shifts** and fatigue reduce vigilance; breaks and task variety.
- **Display and control design:** clear labels, logical layout, warnings that stand out, controls that move the expected way.
- **Human error:** design so mistakes are hard to make and easy to correct (**checklists**, forcing functions).
- **Human–computer interaction:** simple screens and software, readable fonts, fewer interruptions.
- **Decision-making and skilled performance:** training and decision support for complex tasks.
- **Work stress:** stress management.

Examples

- **Control rooms:** clear displays and warnings that stand out.
- **Drivers and pilots:** alertness, fatigue and breaks.
- **Surgeons and health-care workers:** checklists against error; night-shift fatigue.
- **Call centres:** workload, monotony, task variety.

Stress and mental overload raise muscle tension and the chance that pain persists, so combine cognitive changes with physical ergonomics, breaks and relaxation.

7 marks Purpose and principles of ergonomics

Asked: S22 (2c)

Ergonomics (International Ergonomics Association, 2000): "the scientific discipline concerned with the understanding of interactions among humans and other elements of a system", applied to design to optimise **human well-being** and **overall system performance**. Simply, **fitting the job to the person**. Domains: physical, cognitive, organisational.

Purpose

- Prevent **work-related musculoskeletal disorders** and accidents.
- Reduce **fatigue and errors**.
- Improve **comfort, productivity and quality**.
- Include **older workers and persons with disability** by adapting work.
- Reduce absence and **compensation costs**; help injured workers return to work.

Principles

1. Work in **neutral postures**: spine in its curves, elbows by the side, wrists straight, neck upright.
2. **Reduce excessive force:** lighter loads, mechanical aids, good grips.
3. Keep everything within **easy reach**.
4. Work at the **right height**: around elbow height; lower for heavy work, higher for fine work.
5. Reduce excessive **repetition and static loading**; vary tasks.
6. Minimise **fatigue** with rest breaks and job rotation.
7. Minimise **contact stress and vibration**.
8. Provide **clearance and space**; allow change of posture (sit–stand).
9. Keep a comfortable **environment**: light, temperature, noise.
10. Make displays and controls easy to understand; involve workers in changes (**participatory ergonomics**).

Applied example: at a computer workstation the screen is about an arm's length away with its top at or slightly below eye level, the keyboard is at elbow height with wrists straight, and the worker takes a movement break every 30–60 minutes.

> **Draw:** Work heights: precision work above elbow height, light work at elbow height, heavy work below elbow height.

7 marks Workstation exercises for a sedentary worker

Asked: S25 (2b)

Sedentary desk workers (clerks, bank staff, computer operators) develop **neck and shoulder pain** with a forward head and rounded shoulders, thoracic stiffness, **low back pain** from slumped sitting, wrist and hand symptoms, eye strain, weight gain and low fitness. **Workstation exercises** break static posture during the working day.

Dose

5–10 repetitions, every 30–60 minutes; hold stretches about **10 seconds**.

Region	Exercises
Neck	Chin tucks; side bending and turning, holding 10 seconds
Shoulders and upper back	Shoulder rolls and shrugs; squeeze the shoulder blades back; arm reaches overhead
Mid back	Seated back extension over the chair back; seated twists; side bends
Wrists and hands	Wrist flexor and extensor stretches; fist and spread fingers; tendon gliding
Low back and hips	Standing back bends with hands on hips; pelvic tilts in sitting; stand up and walk
Legs	Ankle pumps and circles; seated knee straightening; heel raises in standing; marching
Eyes	Look into the distance regularly; blink; palming

Pair with

- **Workstation set-up:** screen about an arm's length away (50–100 cm), top at or slightly below eye level; feet flat, lumbar support; keyboard and mouse at elbow height with wrists straight.

- **Outside work hours:** deep neck flexor, scapular retractor and lower trapezius strengthening, thoracic mobility, pectoral stretching and general fitness; strengthening and fitness reduced neck pain in office workers (Chen et al., 2018).

Teach 2–3 exercises per micro-break so the worker actually does them.

6(b ii) Disability management: functional capacity evaluation, work conditioning and work hardening

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3 marks Job simulation

Asked: S16 (1f)

Job simulation: reproducing the real tasks of a job, with the **same tools, weights, heights, postures and pace**, in the clinic or at a mock workstation.

- **To train:** builds job-specific tolerance in **work hardening**.
- **To test:** shows whether the worker can meet the job demands in a **functional capacity evaluation**.
- Tasks come from the **job analysis** and are graded in weight, height, distance, repetitions and time.
- **Examples:** work simulators with interchangeable tool handles; work sample batteries (such as **Valpar** work samples); mock construction, assembly or office stations; a mock truck bed for a loader.

It leads to a safe, graded return to the real job.

3 marks Work conditioning

Asked: S17 (1d), S22 (1b) (asked 2 times)

Work conditioning: an **intensive, work-related, goal-oriented** conditioning programme to restore **systemic, neuromusculoskeletal and cardiopulmonary function** (strength, endurance, mobility, motor control, aerobic capacity) so the person can return to work.

- Usually **one discipline** (physiotherapist); part of a day, several days a week; starts after acute care when there is a physical deficit.
- **Aerobic training** (treadmill, cycle, walking) **20–40 minutes**.
- **Progressive strengthening** of the injured area and the whole body (trunk, legs, grip).
- Flexibility, mobility, motor control and core stability.
- Basic **body mechanics** and lifting practice; education.

It is followed by **work hardening** (interdisciplinary, job-specific) when physical capacity is nearly enough.

3 marks Define work hardening and work conditioning

Asked: W21 (1e), W22 (1d) (asked 2 times)

Work conditioning: an intensive, work-related, goal-oriented conditioning programme to restore **systemic, neuromusculoskeletal and cardiopulmonary function** (strength, endurance, mobility, motor control, aerobic capacity) so the person can return to work.

Work hardening: a highly structured, goal-oriented, individualised, **interdisciplinary** programme using **real or simulated work tasks** to restore **physical, behavioural and vocational functions** for return to a specific job.

Feature	Work conditioning	Work hardening
Team	Usually one discipline (PT)	PT, OT, psychologist, vocational counsellor, case manager
Focus	General physical capacity	Job-specific tasks, work tolerance, work behaviours
Hours	Part of a day	Builds up to a full working day

3 marks Define functional capacity assessment

Asked: W22 (1e), W25 (1c) (asked 2 times)

Functional capacity assessment (evaluation): a **standardised** set of tests that measures a person's **safe maximum ability** to perform work-related physical tasks, and compares it with the **demands of a job** or a **physical demand level**.

- **Components:** intake; musculoskeletal screen; material handling (lifting floor to waist, waist to shoulder, overhead; carrying; pushing and pulling); positional tolerances; hand function (grip, pinch, dexterity); aerobic test; effort and consistency; report with an 8-hour projection.
- **Uses:** return-to-work decisions and job matching; planning work conditioning and hardening; modified duties; disability rating and compensation; pre-placement.
- **Example:** Isernhagen Work System (kinesiophysical, 28 tests).

7 marks Functional capacity evaluation (assessment)

Asked: W16 (2e), S17 (2f), S19 (2b) (asked 3 times)

Functional capacity evaluation (FCE): a **standardised** set of tests that measures a person's **safe maximum ability** to perform work-related physical tasks, and compares it with the **demands of a job** or a **physical demand level**. It is a step of disability management.

Uses

Return-to-work decisions and job matching; planning work conditioning and hardening; deciding modified duties; disability rating and compensation; pre-placement screening; measuring progress.

Components

Part	What is tested
Intake	History, job description, symptoms, self-rated abilities, medical clearance, consent
Musculoskeletal screen	Posture, ROM, strength, sensation, HR and BP
Material handling	Lifting **floor to waist, waist to shoulder, overhead**; carrying; pushing and pulling, weights increased to a safe maximum
Positional tolerances	Sitting, standing, walking, stooping, kneeling, crouching, reaching, climbing stairs and ladders, balance
Hand function	**Grip dynamometer**, pinch, dexterity tests

Part	What is tested
Aerobic capacity	Step or treadmill test
Effort and consistency	Repeated trials, HR response, observed body mechanics, consistency between tests
Report	Safe abilities, physical demand level, comparison with the job, **projection to an 8-hour day**, recommendations

Approaches

- **Kinesiophysical** (**Isernhagen Work System**, 28 work-related tests): the evaluator watches body mechanics to decide the safe maximum.
- **Psychophysical:** uses the worker's own judgement of an acceptable load.
- Duration: usually several hours, often over **1–2 days**.

Limits

Affected by pain, fear and motivation; predicts return to work only modestly; needs a trained evaluator and a standard protocol.

Report in functional terms, for example "medium work: up to 50 lb (23 kg) occasionally, 25 lb (11 kg) frequently".

7 marks Work hardening, with two examples

Asked: W18 (2e)

Work hardening: a highly structured, goal-oriented, individualised, **interdisciplinary** programme using **real or simulated work tasks** to restore **physical, behavioural and vocational functions** for return to a specific job.

Features

- **Team:** physiotherapist, occupational therapist, psychologist, vocational counsellor, case manager.
- **Focus:** job-specific tasks, productivity, safety, work tolerance, work behaviours (punctuality, pacing, attitude), pain coping.
- **Hours:** build up towards a **full working day**, several days a week.
- **Entry:** when physical capacity is nearly enough but job tolerance, confidence or behaviour limit return; usually after work conditioning.

Content

- **Job simulation:** real or simulated tasks from the job analysis, graded in weight, height, distance, repetitions and time.
- Physical conditioning continued alongside.
- **Body mechanics and pacing** during the tasks; ergonomic changes for the real job.
- Pain management, stress and **fear-of-movement** counselling; work habits (timekeeping, following instructions).
- Regular **re-measurement** against the job demands; plan for **graded return**.

Two examples

1. **Warehouse loader after low back strain:** lifting crates floor → waist → shoulder with gradually heavier loads (for example **10 kg to 25 kg**); carrying 20 m; pushing a loaded trolley; loading a mock truck bed; building up to the full shift's number of lifts with correct technique.
2. **Data-entry clerk after carpal tunnel release or wrist fracture:** graded typing and mouse sessions from **15 to 60 minutes**; filing and lifting files; dexterity tasks; workstation set-up and micro-breaks; building up to a full working day.

Work conditioning restores general physical capacity (one discipline); work hardening is job-specific and interdisciplinary.

6(b iii) Environmental stress at work: physical, chemical and mechanical hazards

Full notes p. 193

3 marks Causative factors of cumulative trauma disorders

Asked: W16 (1f)

Cumulative trauma disorders (work-related musculoskeletal disorders): injuries of muscles, tendons, nerves and joints from **repeated small strains without enough recovery**.

Factors	Examples
Work	**Repetition, force**, awkward or **static posture**, **vibration**, contact pressure from tool edges, cold, long hours without breaks, piece-rate pressure
Personal	Age, female sex, pregnancy and menopause, **diabetes, hypothyroidism, rheumatoid arthritis**, obesity, previous injury, low fitness, smoking
Psychosocial	Stress; high demands with little control, tight deadlines, monotony

Examples: carpal tunnel syndrome, de Quervain's tenosynovitis, lateral epicondylitis, tension neck syndrome.

3 marks Byssinosis

Asked: S18 (1e)

Byssinosis: occupational lung disease from inhaling **cotton, flax and hemp dust** in **textile mills** (carding and blowing rooms).

- **Feature:** chest tightness and breathlessness on the **first day back at work** after a break (**"Monday chest tightness"**); later **chronic airflow obstruction**.
- **Schilling grades:** ½ occasional symptoms on the first day of the week; 1 on every first day; 2 on the first and other days; 3 grade 2 with permanent incapacity.
- **Prevention:** dust control, ventilation, respirators, periodic spirometry, job transfer.
- **Physiotherapy:** breathing exercises, airway clearance, exercise training.

3 marks Asbestosis

Asked: W18 (1f)

Asbestosis: a pneumoconiosis; lung fibrosis from inhaling **asbestos fibres**.

- **Jobs:** asbestos mining, cement sheets, insulation, brake linings, ship-breaking.
- **Features: lower-lobe fibrosis**, **pleural plaques** and thickening, clubbing; restrictive lung disease.
- **Sequelae: lung cancer** (more with smoking) and **mesothelioma**, decades later; cor pulmonale, respiratory failure.
- **Prevention:** substitution (asbestos bans), wet methods, enclosure, exhaust ventilation, respirators; periodic chest X-ray and spirometry; notifiable under the Factories Act.
- **Physiotherapy:** breathing control, pursed-lip breathing, thoracic mobility, graded exercise training, energy conservation, smoking cessation.

3 marks Three hazards of exposure to chemical agents

Asked: S19 (1d)

Chemical agents (dusts, gases, fumes, solvents, metals, pesticides) harm workers by three routes:

1. **Local action on skin and eyes:** dermatitis, burns, ulcers, skin cancer.
2. **Inhalation** of dusts, gases, fumes and vapours: **pneumoconiosis** (silicosis, asbestosis, byssinosis), asthma, poisoning by carbon monoxide, lead and solvents.
3. **Ingestion** from contaminated hands, food and smoking at work: **lead and mercury poisoning**.

Prevention: substitute safer materials; wet methods, enclosure, exhaust ventilation; correctly fitted respirators; pre-placement and periodic examination.

3 marks Sequelae of pneumoconiosis

Asked: W20 (1d)

Pneumoconiosis: lung disease from inhaling mineral or organic dust, causing **fibrosis** (silicosis, coal workers' pneumoconiosis, asbestosis, byssinosis).

- Progressive fibrosis and **restrictive lung disease**; **progressive massive fibrosis**.
- **Tuberculosis**, especially with silica (silico-tuberculosis); chronic bronchitis and airflow obstruction.
- **Pulmonary hypertension and cor pulmonale**; respiratory failure.
- **Lung cancer** (asbestos, silica) and **mesothelioma** (asbestos); pneumothorax.
- Disability, loss of work and income, early death.

Physiotherapy: breathing control, airway clearance, graded exercise training as pulmonary rehabilitation, energy conservation.

3 marks Types of occupational hazards

Asked: W25 (1b)

Type	Examples
Physical	Heat, cold, poor lighting and glare, **noise, vibration**, ultraviolet and ionising radiation, high or low air pressure
Chemical	Dusts (silica, coal, asbestos, cotton), gases and fumes, solvents, lead, mercury, pesticides; by inhalation, skin contact or ingestion
Biological	Tuberculosis and hepatitis B in health workers; anthrax and brucellosis in animal handlers; tetanus, leptospirosis
Mechanical and ergonomic	Machinery accidents, falls, lifting, repetitive and forceful work, awkward postures
Psychosocial	Job stress, shift work, monotony, harassment

Control them from the top of the hierarchy: elimination, substitution, engineering, administrative, then PPE.

7 marks Workplace hazards and risk assessment tools

Asked: S18 (2b)

Workplace hazards

A **hazard** is anything at work that can cause harm; the **risk** is how likely the harm is and how severe it would be.

Type	Examples
Physical	Heat, cold, poor lighting, **noise, vibration**, radiation, air pressure
Chemical	Dusts (silica, coal, asbestos, cotton), gases, fumes, solvents, lead, mercury, pesticides
Biological	Tuberculosis and hepatitis B in health workers; anthrax, brucellosis, tetanus, leptospirosis
Mechanical and ergonomic	Machinery accidents, falls, lifting, repetitive forceful work, awkward postures
Psychosocial	Job stress, shift work, monotony, harassment

Risk assessment: steps

1. **Identify hazards:** walk-through, job analysis, injury records, worker interviews.
2. Decide **who might be harmed** and how.
3. **Evaluate the risk** (**likelihood × severity** on a risk matrix) and decide controls.
4. **Record** findings and act.
5. **Review** regularly and after any change or injury.

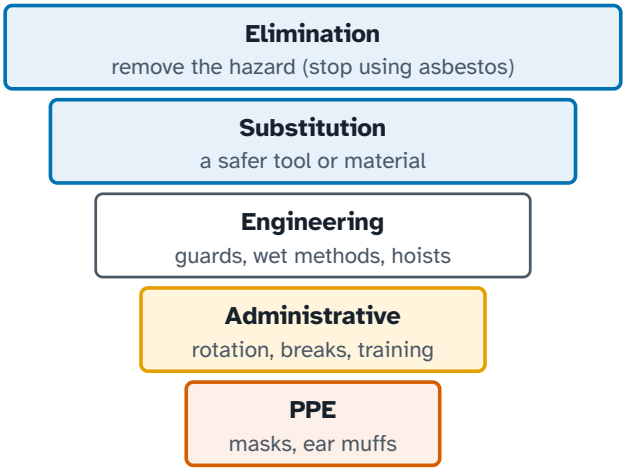
Ergonomic risk assessment tools

Tool	What it assesses
RULA	Neck, trunk and upper-limb posture with force and repetition; score **1–7** (7 change immediately)
REBA	Whole-body posture, load and coupling; score **1–15** (11 or more very high risk)
NIOSH lifting equation	Recommended weight limit (**23 kg** in ideal conditions) and lifting index
Strain Index, OCRA	Distal upper-limb risk in repetitive hand work
OWAS, QEC	Working postures; quick exposure check
Nordic Musculoskeletal Questionnaire	Symptoms by body region in the past 12 months and 7 days

Control

Hierarchy of controls, most to least effective: **elimination → substitution → engineering → administrative → PPE**.

Hierarchy of controls



Most effective at the top (blue), least effective at the bottom (orange). Answer every prevention question from the top down; PPE is the last line, not the first.

PPE is the last line, not the first.

7 marks Influence of physical agents on industrial workers' health

Asked: W20 (2d)

Physical agents are one of the five types of occupational hazard (physical, chemical, biological, mechanical, psychosocial).

Agent	Effects on health	Prevention
Heat (furnaces, glass, steel, kitchens)	Heat rash, cramps, syncope, heat exhaustion, **heat stroke** (an emergency); dehydration; more accidents	Shielding, ventilation, acclimatisation, water and salt, rest in shade, work in cooler hours
Cold (cold stores, high altitude)	Chilblains, frostbite, trench foot, **hypothermia**; stiffness and injuries	Insulated clothing and gloves, warm breaks, limit exposure
Poor or glaring light	Eye strain, headache, accidents, **miner's nystagmus**	Even lighting without glare; task lights
Noise (looms, presses, drilling)	**Noise-induced hearing loss** (first around **4 kHz**), tinnitus, stress, accidents	Quieter machines, enclosures; limit exposure (**8 hours at 90 dBA** under Indian factory rules); ear plugs and muffs; hearing tests
Vibration	Hand-arm: **vibration white finger**, numbness, carpal tunnel syndrome, weak grip. Whole-body: low back pain	Low-vibration tools, anti-vibration handles, suspended seats; EU limits (hand-arm action 2.5, limit 5 m/s^2; whole-body action 0.5, limit 1.15 m/s^2)
Ultraviolet (welding, sun)	**Welder's flash** (photokeratitis), sunburn, cataract, skin cancer	Welding shields and goggles, screens, covering clothing
Ionising radiation (X-ray, nuclear)	Skin burns, radiation sickness, cataract, **cancer and leukaemia**, infertility, birth defects	Time, distance, **shielding** (lead aprons), dosimeter badges; **20 mSv a year** averaged over 5 years (ICRP)

Prevention follows the hierarchy of controls: engineer the hazard out first; PPE (ear muffs, goggles, lead aprons) is the last line.

7 marks Contributing factors for industrial injury

Asked: S22 (2e)

Industrial injuries arise from sudden accidents (machinery, falls) or from **cumulative microtrauma** (repeated small strains). They rarely have one cause: **worker, machine, environment, organisation and ergonomic** factors act together. For example, a tired worker on a long night shift, hurrying to meet a production target on a wet floor beside an unguarded machine.

Factor	Examples
Worker	**Fatigue**, lack of training or experience, haste, **alcohol**, illness, poor vision or hearing, stress, not using protection
Machine and tools	**Unguarded moving parts**, faulty or poorly maintained equipment, unsuitable tools
Environment	Poor lighting, noise, heat, **slippery or cluttered floors**, poor ventilation, work at height
Organisation	**Long hours, night shifts**, production pressure, poor supervision, no safety rules or PPE, poor maintenance
Ergonomic	**Heavy lifting**, awkward postures, repetition, badly designed workstations

Prevention

- **Risk assessment:** identify hazards, decide who may be harmed, rate likelihood × severity, act, review after any change or injury.
- **Hierarchy of controls:** elimination and substitution; engineering (guards on moving parts, hoists, good lighting and ventilation, clean non-slip floors); administrative (safety rules, supervision, maintenance, limits on hours and night shifts, rotation, breaks, training); PPE last.
- **Ergonomic evaluation** of jobs (REBA, RULA, NIOSH) and redesign.
- **Physiotherapy:** safe lifting and posture education, warm-up before heavy shifts, micro-breaks, fitness programmes, early reporting and early care of aches.

Injury records and worker interviews show where the factors cluster; correct the job before blaming the worker.

7 marks Cumulative trauma disorder; explain one common CTD

Asked: W22 (2b), S24 (2b) (asked 2 times)

Cumulative trauma disorders (work-related musculoskeletal disorders): injuries of muscles, tendons, nerves and joints from **repeated small strains without enough recovery**.

Causes

- **Work:** repetition, force, awkward or static posture, vibration, contact pressure, cold, long hours without breaks, piece-rate pressure.
- **Personal:** age, female sex, pregnancy, diabetes, hypothyroidism, rheumatoid arthritis, obesity, previous injury, low fitness, smoking; stress.

Stages

Stage	Features
1	Ache during work, settles overnight; work not affected; **reversible**
2	Pain early in the shift, lasts overnight; sleep disturbed; work capacity reduced
3	Pain at rest and with light tasks; cannot do even light duties; months to years

Examples: carpal tunnel syndrome, de Quervain's, trigger finger, lateral epicondylitis, rotator cuff tendinopathy, tension neck syndrome, hand-arm vibration syndrome.

Carpal tunnel syndrome

- **What:** compression of the **median nerve** under the **flexor retinaculum** at the wrist.
- **Features:** tingling and numbness of thumb, index, middle and half the ring finger, **worse at night** and with gripping; later weakness and **thenar wasting** (abductor pollicis brevis).
- **Tests: Phalen's** (wrist flexion 60 seconds), **Tinel's**, carpal compression; nerve conduction studies confirm.
- **Treatment: night splint** in neutral; nerve and tendon gliding; ergonomic changes and breaks; steroid injection; carpal tunnel release for thenar wasting, constant numbness or failed conservative care.

Prevention

Ergonomic redesign, breaks, job rotation, **early reporting** while still at stage 1.

> **Draw:** Carpal tunnel in cross-section: flexor retinaculum, median nerve, flexor tendons.

6(b iv) Mechanical and psychological stress at work

Full notes p. 203

3 marks Ergonomic advice to an air hostess

Asked: W18 (1a)

Main stresses: lifting bags into **overhead bins**, pushing and pulling **service trolleys**, prolonged standing, cramped postures, **turbulence** (falls) and long, irregular duty days. Back injuries are linked most often with turbulence, overhead bins and trolleys.

- **Avoid lifting heavy bags overhead**; passengers lift their own; heavy bags go under the seat.
- Push trolleys with **both hands, close to the body**; report broken wheels.
- **Turbulence:** stop service, secure trolleys, sit with the belt fastened; use handholds.
- Store galley items so they need **no overhead reach** or deep bending.
- **Supportive shoes** without high heels, **calf raises**, stockings if the legs swell; a **sleep plan**.

Back, core and upper-body strengthening off duty.

3 marks Effects of shift work on health

Asked: W21 (1c)

Shift work, especially at night, disturbs the **body clock** (circadian rhythm).

- **Poor sleep and fatigue**; reduced vigilance in monotonous night work; more **errors and accidents**.
- **Digestive problems**, weight gain, metabolic and heart risk.
- Night shift work is classed as **"probably carcinogenic"** (IARC, 2019).
- Strain on **family and social life**; stress.

Advice: forward-rotating shifts; protected sleep in a dark, quiet room; regular meals; exercise; avoid long runs of night shifts.

3 marks Mechanical stresses faced by truck drivers

Asked: S24 (1d)

- **Prolonged sitting** in a fixed posture: low back and neck pain, stiffness, poor leg circulation.
- **Whole-body vibration** through the seat: low back pain and disc degeneration.
- **Poor seat and reach:** twisting, reaching, neck strain on mirrors and turning.
- **Loading, unloading, climbing in and out** of the cab: back and knee strain, falls.
- **Long hours and night driving:** fatigue, sleep loss, obesity, accidents.

Advice: break every **2 hours** to walk and stretch; suspended seat with lumbar support; mirrors set to avoid neck twisting; use handholds and steps; planned sleep.

7 marks Ergonomics for low back ache in a housemaid

Asked: S16 (2a)

A housemaid's low back ache builds up from many small strains (**cumulative microtrauma**): hours of **stooping**, floor-level work and **lifting** in several houses a day. In a Bengaluru survey of 408 women domestic workers, the working day was about **6 h**, most worked **7 days a week**, and pain in the past 12 months was commonest in the **knees (38.2%)**, upper back (31.9%) and **low back (27.5%)**.

Risk factors

- **More than 10 years** in the job and **psychological distress** each about doubled the odds of low back pain.
- Sustained **stooping and squatting**; lifting from the floor; load far from the body; bending with twisting.
- Washing-platform and **seat height** are linked with symptoms; prolonged squatting or kneeling nearly doubles the odds of **knee osteoarthritis**.
- Low fitness, obesity, previous injury.

Task analysis and ergonomic changes

Task and strain	Ergonomic change
Sweeping with a short jhadu: trunk bent, reaching	**Long-handled broom**; step with it and sweep close to the body
Mopping by hand: squatting or kneeling, twisting	**Long-handled mop** and wringer bucket; if by hand, a **knee pad**, change sides and leading hand
Washing utensils and clothes: deep squat, back bent	**Low stool (paat)** of suitable height, or a **raised platform** or sink; work close; wring small loads
Carrying water: load on one side, twisting	**Half-filled buckets**; two smaller loads close to the body; a wheeled bucket; **turn with the feet**
Several houses: long hours, few breaks	**Alternate stooped and upright tasks**; short breaks

Physiotherapy

- **Assess** pain, mobility, core endurance, hamstrings and knees; watch each chore; score the worst posture with **REBA**.
- **Educate:** back pain is rarely serious; **stay active** and keep working with the changes.
- **Strengthen** (**8–12 reps, 2–3 days a week**): abdominal bracing, bridging, four-point arm and leg lifts, sit-to-stand or wall squats.
- **Stretch** hamstrings, hip flexors and calves (**10–30 s** holds); practise bending from the hips.
- Regular walking, **weight control** (protects the knees too), relaxation for stress.

Current practice (NICE NG59): no lumbar belts, TENS, ultrasound or interferential therapy for low back pain; change the task first, then exercise.

7 marks Job stress and psychological hazards at work: causes, prevention and management

Asked: W17 (2a), S18 (2a), W18 (2c) (asked 3 times)

Psychological (psychosocial) hazards are features of work that cause strain: **job stress**, shift work, monotony, harassment. Job stress arises when demands are high and control, reward or support are low.

Causes (aetiology)

- **Demand–control model:** high demands with little control over the work.
- **Effort–reward imbalance:** great effort with little reward.
- **Monotony** and job dissatisfaction.
- Anxiety about finishing work on time and to quality; tight deadlines.
- **Long hours and shift work**.
- Poor relationships, **harassment**, poor support, job insecurity.

Effects

Muscle tension and pain, headache, fatigue, poor sleep, **raised BP**, anxiety, depression, more **errors and accidents**, absence. Psychosocial factors raise muscle tension and the chance that pain persists. In long-route drivers: irritability, anxiety, poor sleep and loneliness.

Prevention at work

- Reasonable workload and deadlines; more **control and variety**.
- Breaks; support from supervisors; clear roles; safe workplaces.
- Shift planning: forward-rotating shifts, no long runs of night shifts.

Management

- **Relaxation:** Jacobson's progressive muscle relaxation; Mitchell's physiological relaxation; diaphragmatic breathing.
- **Exercise:** regular aerobic exercise, stretching programmes at work, **yoga**.
- **Ergonomic and task changes** that remove physical strain; education on posture, sleep and pacing.
- **Referral** to a counsellor or doctor for anxiety or depression.
- **Evaluate** with pain scores, absenteeism and stress questionnaires.

Example: a long-route driver with stress from traffic and deadlines needs planned rest, relaxation and breathing, fitness and family time.

7 marks Ergonomic care for physiotherapists

Asked: S19 (2d)

Physiotherapists get the same **work-related musculoskeletal disorders** they treat. Lifetime prevalence is **55–91%**; the **low back** is commonest, then the neck, upper back and shoulders. In a Delhi survey about **9 in 10** had pain that began after starting work, and in Australia about **1 in 6** changed jobs or left the profession.

Risk factors

- **Lifting and transferring dependent patients**: rated the top cause.
- **Manual therapy**: mobilisation and manipulation are linked with **thumb** symptoms.
- Repetitive movements; awkward and static postures; **many patients a day**; **working while injured**.
- Commoner in women, younger therapists and rehabilitation settings.

High-risk tasks and prevention

Task	Prevention
Transfers	Manual limit about **16 kg (35 lb)** even for a cooperative patient; above it use **hoists, slide sheets**, transfer boards or sit-to-stand aids; let the patient help; plan the move
Manual therapy	Plinth at the right height; body weight with **straight wrists**; vary the technique and contact point (heel of the hand, not the thumb)
Low plinths, mats, beds	**Height-adjustable plinth**: **5–10 cm below elbow** for light work, **20–40 cm below** for heavy pressure; raise the bed; knee pad for floor work
Long, busy days	Fair caseload; **scheduled breaks**; rotate heavy and light tasks

Other measures

- **Equipment first:** knowing body mechanics alone does not prevent injury.
- **Body mechanics:** patient close to the waist, stable wide base, slight hip and knee bend, **no twisting**, head up.
- Regular **ergonomics training** and yearly health checks; **report pain early**; do not work through injury.
- **Own fitness:** daily activity; back, core, neck, shoulder and grip strengthening; stretching between patients.
- **Self-check:** Nordic questionnaire survey of the team; REBA scores of transfer and treatment postures.

Follow the hierarchy of controls: equipment and layout first, then work organisation, then technique and fitness.

7 marks Role of the physiotherapist in stress management in industry

Asked: S20 (2e), W21 (2e), W24 (2e) (asked 3 times)

Work stress arises when job demands are high and control or reward is low (**demand–control** and **effort–reward** models), with monotony, deadlines, long hours, shift work, poor relationships, harassment and job insecurity. It causes **muscle tension and pain**, headache, fatigue, poor sleep, raised BP, anxiety, depression, errors, accidents and absence.

Role of the physiotherapist

1. **Screening and education:** identify workers with stress-related pain and fatigue; teach posture, sleep and pacing.
2. **Jacobson's progressive muscle relaxation:** tense, then release, each muscle group in turn.

3. **Mitchell's physiological relaxation:** move into the opposite of the stress posture.
4. **Diaphragmatic breathing** practised daily and during stressful moments.
5. **Exercise:** regular **aerobic exercise**, stretching programmes at work, **yoga**; employee fitness programmes with group classes and walking groups.
6. **Ergonomic and task changes** that remove physical strain; workstation exercises and micro-breaks.
7. **Shift-work advice:** protected sleep, regular meals, exercise.
8. **Referral** to a counsellor or doctor for anxiety or depression.

At the organisation level

Advise management on reasonable workload and deadlines, more control and variety, breaks, supervisor support, clear roles and forward-rotating shifts; build stress management into workplace health promotion.

Health promotion

Walking or yoga, sleep routine, weight and posture awareness.

Evaluation

Pain, absenteeism and stress questionnaires before and after the programme.

Stress raises muscle tension and the chance that pain persists, so relaxation and exercise belong in every industrial back and neck programme.

7 marks Musculoskeletal hazards of prolonged standing

Asked: S21 (2a)

Many workers stand for most of the shift: **watchmen, traffic police, surgeons, teachers, air hostesses**. Standing still is **static work**: the same muscles stay contracted, and the legs get little movement to pump blood back to the heart.

Hazards and their causes

Effects reported consistently (Waters and Dick, 2015):

Hazard	Why it happens
Low back pain	**Static loading** of the back muscles and joints; no change of posture
Leg pain and swelling	Little **calf-muscle pumping**, so fluid collects in the legs
Varicose veins	**Pooling of blood** in the leg veins over years
Fatigue and muscle pain	Sustained contraction of back and leg muscles without rest
Foot pain, plantar heel pain	**Hard floors** and poor footwear

Prevention and physiotherapy

- **Change position often and walk** when possible: the most effective measure.
- **Sit–stand options** or a leaning stool; rest breaks.
- **Anti-fatigue mats**.
- Supportive, cushioned **footwear and insoles**.
- A **footrest** to alternate the feet and shift weight.
- **Compression stockings** for those with swelling.
- **Calf raises and ankle pumps** during the shift.
- Back and leg strengthening and general fitness outside work.

Job examples: a teacher alternates sitting and standing; a surgeon uses an anti-fatigue mat and supportive shoes; an air hostess does calf exercises and wears supportive shoes.

Viva: movement beats every other measure; mats, sit–stand options, footwear and stockings also help.

7 marks Factors causing low back pain in industry and its prevention

Asked: W22 (2d), W25 (2a) (asked 2 times)

Industrial low back pain usually builds up from **cumulative microtrauma** rather than one injury.

Factors

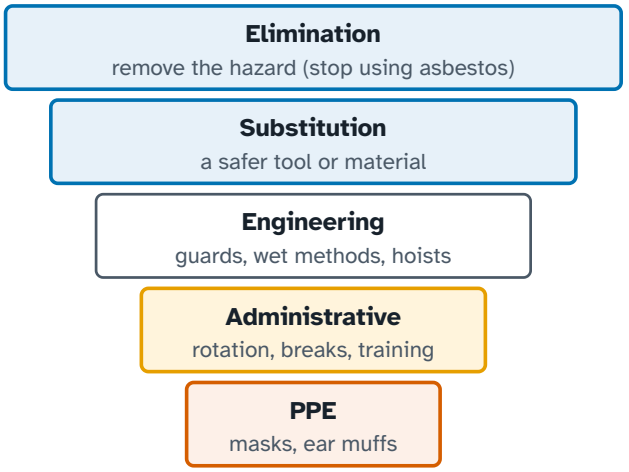
Group	Factors
Manual handling	**Heavy, frequent or awkward lifting** (bending and twisting); lifting **far from the body** or from the floor
Posture	Prolonged sitting or standing; stooping and squatting work
Vibration	**Whole-body vibration** (drivers)
Accidents	Slips and falls
Personal	Low fitness, smoking, obesity
Psychosocial	Job dissatisfaction and stress

Squatting and stooping work (automobile workers, farm labour) adds back strain and knee and ankle pain.

Revised NIOSH lifting equation: 23 kg is the recommended limit under ideal conditions, reduced for distance from the body, height, travel, twisting, frequency and poor grip. A 50 kg bag lifted to 3 feet is well beyond it, so the task itself must change.

Prevention

Hierarchy of controls



Most effective at the top (blue), least effective at the bottom (orange). Answer every prevention question from the top down; PPE is the last line, not the first.

- **Redesign first:** lighter bags, trolleys, hoists, conveyors, loads at waist height, team lifts. For a labourer loading 50 kg bags: 25 kg bags or two-person lifts, a trolley or ramp, loading from waist height.
- **Squatting work:** low stools or kneeling pads; rotate tasks.
- **Ergonomic evaluation** and task analysis (NIOSH, REBA).
- **Administrative:** job rotation, rest breaks.
- **Vibration:** suspended, well-maintained seats; limit driving hours.

- **Posture:** change position often; sit–stand options.
- **Fitness:** core and hip strengthening, hamstring flexibility, aerobic fitness, weight control, smoking cessation.
- **Early reporting** and early physiotherapy: education and reassurance, graded activity.

Training in lifting technique alone did not prevent back pain (Verbeek et al., 2011): change the task, then train.

7 marks Ergonomic evaluation and modifications for a surgeon

Asked: S26 (2a)

Surgeons hold one posture for hours, with the **neck bent** over the field and the arms held still. The pooled prevalence of pain is **35–60%**, and **12%** of those with a disorder need leave, a change in practice or early retirement.

Ergonomic evaluation

1. **Symptom survey:** Nordic Musculoskeletal Questionnaire (past 12 months and 7 days).
2. **Observe and video** operations: posture, time each posture is held, breaks.
3. **Measure:** table against elbow height, monitor, loupes or microscope, pedals, floor, footwear.
4. **Score posture: RULA** (neck and arms) or **REBA** (whole body).
5. **Recommend** changes, equipment first; then **re-evaluate**.

Problems and modifications by type

Type and problems	Modifications
Open (disorders in 66–94%): neck flexion with loupes and headlamps, standing, twisting	Hands near **elbow height, elbows about 90°**; light loupes angled downwards; lights between surgeon and assistant; a step for shorter staff
Laparoscopic (73–100%): monitor to the side or high, long instruments, bent wrists, uneven stance at pedals	Table lower, at **0.7–0.8 × elbow height**, elbows 90–110°; **monitor straight ahead**, top edge at eye level; handles for a straight wrist; pedals in front
Microsurgery: neck flexion at the eyepieces, long static posture, unsupported arms	**Sit** with feet and **forearms supported**; eyepieces set for a neutral neck

For every surgeon

- **Micro-breaks:** stretching for **1.5–2 min every 20–40 min** reduced pain without lengthening the operation (American College of Surgeons: every 30–60 min).
- **Anti-fatigue mat**, weight on both legs, supportive shoes, stockings for leg swelling.
- **Instruments** that fit the hand; ratchets so the hand does not squeeze for long; power grip; powered tools.
- **Plan** table height, patient position and monitors before the case.
- **Exercise:** deep neck flexor nods (**10 s, 10 times**); trapezius, levator scapulae and pectoral stretches (**30 s, twice**); general fitness.

A lower RULA or REBA score after the changes shows they worked.

Index by paper

Each paper's short questions (Q1 at 3 marks, Q2 at 7 marks) with the page of the model answer. Sit a paper in 100 minutes (five of each), then check.

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Abbreviations

1RM: One-repetition maximum
AAM: Ayushman Arogya Mandir
AB: Ayushman Bharat
ACOG: American College of Obstetricians and Gynecologists
ACSM: American College of Sports Medicine
ADAPT: Able Disabled All People Together (formerly the Spastics Society of India)
ADIP: Assistance to Disabled Persons for purchase of aids and appliances
ADL: Activities of daily living
AIIMS: All India Institute of Medical Sciences
ANM: Auxiliary nurse midwife
ASHA: Accredited Social Health Activist
ATP: Adenosine triphosphate
BIA: Bioelectrical impedance analysis
BMI: Body mass index
BMR: Basal metabolic rate
BMVSS: Bhagwan Mahaveer Viklang Sahayata Samiti (Jaipur foot)
BP: Blood pressure
CBM: CBM Global (an international disability charity)
CBR: Community-based rehabilitation
CHC: Community health centre
COPD: Chronic obstructive pulmonary disease
CP: Cerebral palsy
CT: Computed tomography
CTD: Cumulative trauma disorder
DDRC: District Disability Rehabilitation Centre
DEIC: District Early Intervention Centre
DEXA: Dual-energy X-ray absorptiometry
DM: Diabetes mellitus
DNA: Deoxyribonucleic acid
DVT: Deep vein thrombosis
ECT: Every contraction timed
EMG: Electromyography
ESI: Employees' State Insurance
ESIC: Employees' State Insurance Corporation
EU: European Union
FCE: Functional capacity evaluation
FES-I: Falls Efficacy Scale–International
FIM: Functional Independence Measure
FITT: Frequency, intensity, time and type
FITT-VP: Frequency, intensity, time, type, volume and progression
FRC: Functional residual capacity
FSH: Follicle-stimulating hormone
GDS: Geriatric Depression Scale
GMFCS: Gross Motor Function Classification System
HDL: High-density lipoprotein (good cholesterol)
HR: Heart rate
HRR: Heart-rate reserve
IADL: Instrumental activities of daily living
IAP: Indian Academy of Pediatrics
IARC: International Agency for Research on Cancer
IBR: Institution-based rehabilitation
ICDS: Integrated Child Development Services
ICF: International Classification of Functioning, Disability and Health
ICIDH: International Classification of Impairments, Disabilities and Handicaps (WHO, 1980)
ICIQ-UI: ICIQ urinary incontinence short form
ICRP: International Commission on Radiological Protection
ICS: International Continence Society
ICT: Information and communication technology
ICU: Intensive care unit
IGNOAPS: Indira Gandhi National Old Age Pension Scheme
ILO: International Labour Organization
JSSK: Janani Shishu Suraksha Karyakram
JSY: Janani Suraksha Yojana
LH: Luteinising hormone
LPG: Liquefied petroleum gas (cooking gas)
MET: Metabolic equivalent
METs: Metabolic equivalents (1 MET = resting energy use)
MMSE: Mini-Mental State Examination
MNA: Mini Nutritional Assessment
MRC: Medical Research Council
MRI: Magnetic resonance imaging
NCD: Non-communicable disease
NDFDC: National Divyangjan Finance and Development Corporation
NDMA: National Disaster Management Authority
NDRF: National Disaster Response Force
NFHS-5: National Family Health Survey, fifth round (2019–21)
NGO: Non-governmental organisation
NICE: National Institute for Health and Care Excellence (UK)
NIOSH: National Institute for Occupational Safety and Health (USA)
NLEP: National Leprosy Eradication Programme
NPHCE: National Programme for Health Care of the Elderly
OA: Osteoarthritis
OCRA: Occupational Repetitive Actions index
OT: Occupational therapy
OWAS: Ovako Working Posture Analysis System
$PaCO_2$: Arterial carbon dioxide pressure
PaO_2: Arterial oxygen pressure
PC: Phosphocreatine (creatine phosphate)
PCOS: Polycystic ovary syndrome
PE: Pulmonary embolism
PERFECT: Power, endurance, repetitions, fast contractions, every contraction timed (pelvic floor grading)
PHC: Primary health centre
PM-JAY: Pradhan Mantri Jan Arogya Yojana
PMMVY: Pradhan Mantri Matru Vandana Yojana
PMSMA: Pradhan Mantri Surakshit Matritva Abhiyan
POMA: Performance-Oriented Mobility Assessment (Tinetti)
POP-Q: Pelvic Organ Prolapse Quantification
PPE: Personal protective equipment
PT: Physiotherapy
PTB: Patellar tendon bearing
QEC: Quick Exposure Check
RBSK: Rashtriya Bal Swasthya Karyakram
REBA: Rapid Entire Body Assessment
ROM: Range of motion
RPE: Rating of perceived exertion (Borg scale)
RPWD: Rights of Persons with Disabilities
RULA: Rapid Upper Limb Assessment
SPPB: Short Physical Performance Battery
SUI: Stress urinary incontinence
TENS: Transcutaneous electrical nerve stimulation
TLC: Total lung capacity
TUG: Timed Up and Go test
TV: Tidal volume
UDID: Unique Disability ID
UN: United Nations
UNESCO: United Nations Educational, Scientific and Cultural Organization
UTI: Urinary tract infection
VC: Vital capacity
VO_2max: Maximal oxygen uptake
WHO: World Health Organization
YMCA: Young Men's Christian Association